Multiphysics Wave Mechanics in Generalized Continua: A Constructive Review of Theory, Energy Transport, and Interface Phenomena

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A constructive synthesis of generalized thermoelasticity, piezothermoelasticity, nonlocal elasticity, and memory-dependent continuum theories.

The review is organized around the physical developments that reshaped generalized continuum wave mechanics: the relaxation of Fourier's law, the introduction of scale-dependent constitutive response, and the progressive coupling of thermal, electric, magnetic and elastic fields. Each theory is presented with its governing equations, the assumptions that license them, the solution approaches used to analyse it, and the wave phenomena that follow.

This review examines how extensions of classical continuum mechanics change thermomechanical wave propagation, energy transport and interface phenomena. It organizes generalized thermoelastic, multiphysics, memory-dependent, fractional, gradient and nonlocal models by constitutive mechanism, governing structure, predicted modes and available validation. The literature map is a purposive inventory assembled from the repository's existing collection and targeted bibliographic checks; DOI metadata checks are distinguished from full-text reading and claim-level verification. Several corrections follow from balancing conventions and explicit model equations. With total energy per unit volume and outward-positive flux, the classical mechanical contribution is minus stress power, so that $J_i = q_i - \sigma_{ij} v_j$. Lord–Shulman and Green–Lindsay models are not conflated with two-temperature theories; the former's relaxation acts on the heat flux, whereas Green–Lindsay introduces temperature-rate dependence in a single-temperature formulation. A standard Eringen-type nonlocal dispersion relation has a bounded high-wavenumber frequency, not an unbounded frequency or group speed. Quasi-electrostatic piezoelectric models do not contain a propagating electromagnetic branch. Generalized-theory energy fluxes, benchmark values and several literature-dependent claims remain conditional or open where their derivations or primary-source checks are incomplete. The review therefore separates derived results, metadata checks, secondary-source findings and unresolved items, and does not present an unbuilt revision as a validated manuscript.

Notation

Acronyms

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Table 1: Master symbol table with units and physical meaning.

Symbol	Meaning	Units	First appearance
u_i	Displacement component	m	§3.1
v_i	Velocity component, $\dot{u}_i$	m/s	§3.1
ε_{ij}	Infinitesimal strain tensor	dimensionless	§3.1
ω_{ij}	Rotation tensor	dimensionless	§3.1
σ_{ij}	Cauchy stress tensor	Pa	§3.2
ρ	Mass density	kg/m^3	§3.2
f_i	Body force component	N/kg	§3.2
q_i	Heat flux component	W/m^2	§3.3
T	Temperature increment, $T - T_0$	K	§3.3
T_0	Reference temperature	K	§3.3
ψ	Specific free energy	J/kg	§3.5
s	Specific entropy	J/(kg·K)	§3.5
J_i	Total energy flux component	W/m^2	§3.8
$\mathcal{E}$	Total energy density	J/m^3	§10.1
$\mathcal{P}$	Mechanical power input	W/m^3	§10.1
$\mathcal{D}$	Dissipation rate	W/m^3	§10.1
τ_0	Lord–Shulman relaxation time	s	§4.3
τ_q, τ_T, τ_ν	Dual/triple phase-lag times	s	§4.7–4.8
c_T	Thermal wave speed	m/s	§4.3
c_p, c_s	Longitudinal and shear wave speeds	m/s	§3.6
e_{ijk}	Piezoelectric stress coefficient	C/m^2	§5.1
D_i	Electric displacement component	C/m^2	§5.1
ϕ	Electric potential	V	§5.1
κ_{ij}	Dielectric permittivity tensor	F/m	§5.1
E_i	Electric field component, $-\phi_{,i}$	V/m	§5.1
β_{ij}	Thermal stress coefficient	Pa/K	§4.2
c_v	Specific heat at constant strain	J/(kg·K)	§4.2
k	Thermal conductivity	W/(m·K)	§4.2
λ, μ	Lamé elastic constants	Pa	§3.6
α	Thermal expansion coefficient	1/K	§4.2
ξ	Nonlocal length scale parameter	m	§7.1
ω	MDD time delay	s	§6.4
$\kappa(t-\xi)$	MDD kernel function	dimensionless	§6.4

Table 2: Acronym table.

Acronym	Full form	Meaning
CD	Classical coupled	Classical coupled thermoelasticity (Fourier)
LS	Lord–Shulman	Generalized thermoelasticity with one relaxation time
GL	Green–Lindsay	Temperature-rate-dependent thermoelasticity (two relaxation times)
GN-I	Green–Naghdi type I	Classical-equivalent thermoelasticity
GN-II	Green–Naghdi type II	Thermoelasticity without energy dissipation
GN-III	Green–Naghdi type III	Dissipative generalized thermoelasticity
DPL	Dual-phase-lag	Two-phase-lag heat conduction (Tzou, 1995)
TPL	Triple-phase-lag	Three-phase-lag heat conduction
MGT	Moore–Gibson–Thompson	Third-order heat conduction model
MDD	Memory-dependent derivative	Integral-form memory operator (Wang & Li, 2011)
PDE	Partial differential equation	—
ODE	Ordinary differential equation	—
NDE	Nondestructive evaluation	—
FGP	Functionally graded piezoelectric	Piezoelectric material with spatially varying properties
PML	Perfectly matched layer	Numerical absorbing boundary condition
ES	Elastic half-space	Reference medium in interface problems
TS	Thermoelastic half-space	Thermoelastic medium in interface problems
HPS	Higher-order piezothermoelastic	Piezothermoelastic medium with higher-order MDD
CDI	Clausius–Duhem inequality	Dissipation inequality in continuum thermodynamics

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1 Review Scope and Purpose

1.1 Motivation and scientific context

Wave propagation in generalized continua lies at the intersection of continuum mechanics, thermodynamics, and coupled-field physics. In the classical Fourier model, combining Fourier heat conduction with energy balance gives a parabolic heat equation whose fundamental solution has instantaneous mathematical support. This property is often called the infinite-speed paradox; it is a consequence of the idealized continuum model, not evidence that the model is simply wrong in every regime. Biot's coupled linear thermoelastic formulation remains useful within its assumptions, while later theories replace or augment selected constitutive structures to represent finite-rate transport, memory, additional fields, or characteristic lengths. These alternatives are not interchangeable and do not all remove instantaneous support or guarantee finite-speed propagation.

Common modifications include relaxation of the heat flux (Lord–Shulman), temperature-rate dependence with two relaxation parameters (Green–Lindsay, a single-temperature formulation), thermal-displacement theories (Green–Naghdi), phase-lag laws, hereditary kernels, fractional operators, and spatially nonlocal or gradient constitutive laws. Each class has variants, assumptions, and limiting regimes; the exact governing equations, not a family label, determine propagation, thermodynamic admissibility, and well-posedness. The historical and constitutive comparisons below are therefore presented as a map of distinct models rather than a single sequence of increasingly accurate replacements.

These modifications change the mathematical character of the model and may introduce additional parameters, state variables, memory, or spatial scales. Their consequences for support, stability, thermodynamic admissibility, and initial data are model-specific. For example, claims about relaxation-time orderings, entropy production, or the function spaces required by fractional or nonlocal operators must be tied to the precise equations and boundary conditions; they are not asserted uniformly across each named family in this review.

A separate modelling axis concerns additional coupled fields: electrical polarization, porosity, micro-rotation, or charge carriers. Piezothermoelastic, porous, micropolar, semiconductor, and magneto-thermoelastic models introduce field equations and coupling terms whose effects on modal structure and interface conditions depend on the constitutive model and geometry. Simultaneous temporal memory and multiphysics coupling can increase analytical and computational complexity, but it does not imply a fixed number of additional modes or boundary conditions.

These coupled formulations underpin smart structures, energy harvesters, nondestructive evaluation, geophysical prospecting, biomedical ultrasound and nanodevice design, and they have produced a substantial literature in their own right. Yet the field lacks the kind of synthesis that a practitioner needs: a single account that states the constitutive structure of each family, derives the wave problem it produces, establishes the conditions under which each is thermodynamically admissible, shows how the resulting quantities are computed, and identifies what remains open. Existing reviews tend to cover one family or one application Shakeriaski et al. [2021], Chandrasekharaiah [1998], Joseph and Preziosi [1989], Askes and Aifantis [2006]; this review is organised so that the same question can be followed across all of them.

1.2 Objectives and contributions

This review has the following objectives:

1. To present the first-principles foundations of continuum wave mechanics: kinematics, bal-

ance laws, thermodynamic restrictions, and constitutive principles (Section 3).

2. To trace the historical development of generalized thermoelastic theories from classical coupled thermoelasticity to modern memory-dependent and nonlocal formulations, explaining the physical motivation behind each theory (Section 4).

3. To present coupled multiphysics continua—piezothermoelasticity, porous media, micropolar continua, semiconductor interactions, and magneto-thermo-elasticity—with their governing equations and coupling mechanisms (Section 5).

4. To survey generalized transport and memory effects: relaxation, fractional operators, hereditary integrals, memory-dependent derivatives, and their mathematical structure (Section 6).

5. To present nonlocal and scale-dependent continuum models, including Eringen's nonlocal elasticity and strain-gradient theory, with their wave dispersion characteristics (Section 7).

6. To review wave-propagation methods: plane waves, Christoffel and secular equations, transform methods, and numerical approaches (Section 8).

7. To develop the theory of wave reflection, transmission, and energy partitioning at interfaces between different continuum media (Section 9).

8. To derive the classical local energy flux under explicit assumptions and identify, without overclaiming, the model-specific energy-flux derivations that remain open (Section 10).

9. To survey advanced theoretical developments and emerging research directions (Sections 11–12).

10. To present applications with evidence levels, an original five-axis research taxonomy, major scientific challenges, and future research opportunities (Sections 13–16).

1.3 Scope and organization

The review covers the following theory families:

- **Classical coupled thermoelasticity** (Biot, 1956) — Fourier heat conduction coupled with elastic wave propagation.
- **Lord–Shulman theory** (1967) — generalized thermoelasticity with one thermal relaxation time.
- **Green–Lindsay theory** (1972) — temperature-rate-dependent thermoelasticity with two relaxation times and one temperature field; distinct from two-temperature theories.
- **Green–Naghdi types I–III** — the type framework dates to 1991; the type-III equations treated here are from 1992 and the type-II primary formulation from 1993.
- **Dual-phase-lag and triple-phase-lag theories** (DPL: Tzou, 1995; TPL: Roy Choudhuri, 2007) — phase-lag heat conduction models.
- **Moore–Gibson–Thompson-type thermoelastic models** — later thermal analogues of Moore and Gibson's 1960 third-order acoustic equation for relaxing gases; the 1960 work is not a heat-conduction law.

- **Memory-dependent derivative models** (Wang & Li, 2011) — integral-form memory operators with fading memory.
- **Fractional thermoelasticity** — time-fractional heat equations for anomalous diffusion.
- **Nonlocal elasticity** (Eringen, 1972–1983) — scale-dependent constitutive relations with spatial memory.
- **Piezothermoelasticity and coupled multiphysics continua** — piezoelectric, porous, micropolar, semiconductor, and magneto coupling.
- **Interface wave phenomena** — reflection, transmission, scattering, and energy partitioning at material interfaces.

The manuscript is organized into seventeen sections:

- **Section 1** (this section): scope, purpose, and organization.
- **Section 2**: literature landscape and scholarly context, including comparison with existing reviews.
- **Sections 3–5**: Layer I–III of the mathematical exposition—foundations, thermoelastic theories, and coupled multiphysics continua.
- **Sections 6–8**: generalized transport, memory effects, nonlocal models, and mathematical methods.
- **Sections 9–10**: interface phenomena and energy transport.
- **Sections 11–12**: advanced developments and comparative synthesis.
- **Sections 13–17**: applications, taxonomy, challenges, future directions, and conclusions.

Appendices provide an equation verification register, balance law derivations, numerical benchmarks, figure sources, table sources, and reproducibility notes.

1.4 Intended readership

This review is written for multiple audiences:

- **Beginning researchers and MSc students:** Sections 1–3 provide the foundational balance laws, constitutive relations, and classical wave mechanics needed to enter the field. The notation tables and Section 17 provide orientation.
- **Doctoral researchers:** Sections 3–10 provide mathematical depth with representative equations, dispersion relations, interface formulations, and solution methods. Section 16 identifies dissertation topics.
- **Experienced scientists:** Sections 12–16 provide comparative synthesis, taxonomy, challenge analysis, and future research roadmap.
- **Journal reviewers and editors:** Sections 1–2, 12, and 16–17 provide context for evaluating new submissions in generalized continuum mechanics.

1.5 Scholarly principles

This review adheres to the following principles:

1. **Constructive synthesis:** Model-specific differences are traced to documented assumptions (constitutive, material symmetry, geometry, boundary conditions, thermal theory, frequency convention, normalization, approximation level), not presented as contradictions or errors.

2. **Mathematical accuracy (target standard):** Important equations should be derived or checked against a primary source with assumptions stated. This revision does not yet complete that audit; unresolved equations are marked UNVERIFIED or OPEN where identified.

3. **Comparative rather than adjudicative framing:** Competing formulations are compared on the basis of their modelling choices and domains of validity; the review does not attempt to adjudicate between individual publications.

4. **Minimal equation set:** Fewer, precise equations are preferred over hundreds of redundant ones. Representative equations are selected to illustrate each theory family.

5. **Original figures and tables:** All figures are original vector graphics created for this review. All tables synthesize knowledge across theory families rather than listing individual papers.

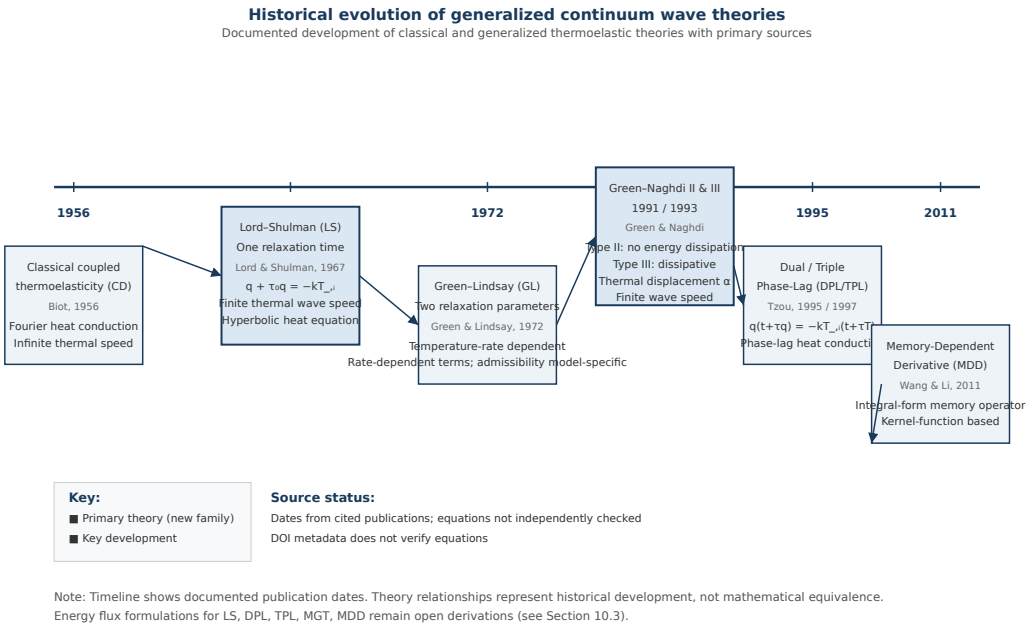


Figure 1: Selected historical milestones in classical and generalized thermoelasticity. Publication dates and model descriptions are taken from the inherited bibliography; the figure does not assert that every generalized theory has finite propagation speed.

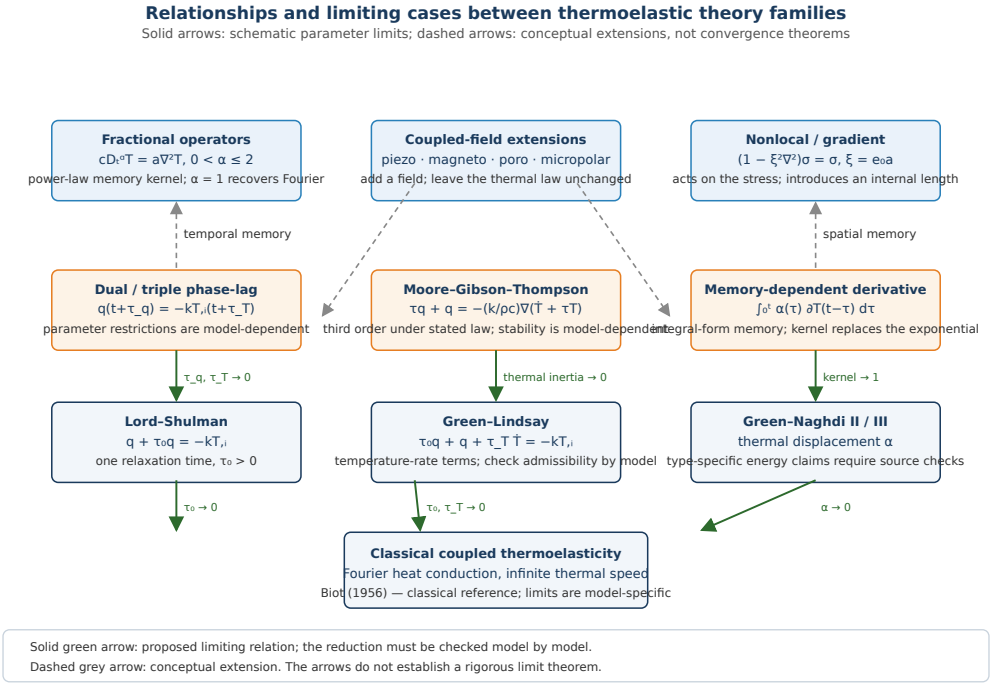


Figure 2: Schematic relationships between thermoelastic model families. Parameter limits are model-specific and are not asserted to exist for every family; the arrows are conceptual rather than rigorous convergence results.

2 Literature Landscape and Scholarly Context

2.1 The organising problem and the shape of the field

The literature of generalized continuum wave mechanics did not accumulate gradually; it was triggered by a single anomaly and then organised itself around the successive attempts to remove it. The anomaly is the infinite propagation speed of Fourier heat conduction. For a body heated at its boundary, Fourier's law (36) produces a temperature field that responds instantaneously everywhere, and the associated thermal wave number is unbounded. In 1948, in the proceedings of the Italian academy, Cattaneo Cattaneo [1948] showed that the paradox disappears if heat flux is allowed a relaxation time; in 1958 Vernotte Vernotte [1958] reached the same conclusion independently in the French academy proceedings. The two papers are, in structural terms, the same result arrived at from the same constraint by two separate routes, and they established the argument that the history of the field would follow: a physical inconsistency is identified, a constitutive modification is proposed that removes it, and the modified model is then shown to require further modification because something else has been lost.

That pattern — identification, modification, cost — is the organising theme of this review, and it is why the literature is best read by theory family rather than by paper. Each family below answers the same question in a different way, and each pays for the answer in a different currency: a second temperature, a second wave, a loss of ellipticity, a loss of uniqueness, a loss of local well-posedness. Sections 4 through 11 take the families in turn; this section establishes how they arose, where the field's centre of gravity has moved, and what the geographic distribution of the work reveals about the structure of the problem.

2.2 Four phases in the development of the field

Selected early developments (1948–1972). Cattaneo and Vernotte proposed relaxation-type modifications of Fourier heat conduction; Lord and Shulman incorporated a relaxation time into coupled thermoelasticity. Green and Lindsay introduced a temperature-rate-dependent thermoelastic formulation with two relaxation parameters and a single temperature field; it is distinct from two-temperature theories. These models have different constitutive structures and limiting assumptions. The dates and descriptions here follow the inherited bibliographic inventory; equation-level primary-source verification remains pending.

Phase II — second sound and thermal-displacement theories (1966–1993). Phonon kinetic theory supplied a microscopic account of non-Fourier transport. Guyer and Krumhansl (1966) derived the phonon-Boltzmann structure and a nonlocal flux length scale; this is not treated as experimental evidence or a continuum thermoelastic law Guyer and Krumhansl [1966]. The Green–Naghdi type-II primary formulation is the 1993 nondissipative theory, while the type-III equations are in the 1992 *Journal of Thermal Stresses* paper Green and Naghdi [1993, 1992]. The type labels and dissipation statements below are limited to those specified systems; no universal Hamiltonian or energy claim is inferred. The synthesis of this phase is the review literature: Chandrasekharaiah's articles on hyperbolic thermoelasticity Chandrasekharaiah [1986, 1998], Joseph and Preziosi's account of heat waves from kinetic theory to continuum mechanics Joseph and Preziosi [1989, 1990], and the monographs by Nowacki Nowacki [1975] and Achenbach Achenbach [1973].

Phase III — the microscale turn (1995–2011). Phase II assumed that the Fourier paradox was a matter of taking a limit too early. Phase III questioned the premise, and asked instead what a thermal wave should look like when the material has microstructure. Tzou's dual-phase-lag model Tzou [1995] and Roy Choudhuri's triple-phase-lag formulation Choudhuri [2007] distinguish the flux and temperature-gradient lags and include a thermal-displacement-gradient contribution in the latter; the exact truncation and admissibility conditions remain model-specific. In parallel, Eringen's nonlocal elasticity Eringen [1972, 1974, 1983] and the strain-gradient formulations Askes and Aifantis [2006, 2011], Polizzotto [2001] transferred the same logic to the mechanical side, and the size-dependent wave speeds measured and computed for nanostructures Maranganti et al. [2006], Maranganti and Sharma [2007] supplied the corresponding experimental evidence.

Phase IV — memory, fractional operators and multiphysics (2011–present). The most recent phase has replaced the single relaxation time with an entire memory kernel, and has replaced the single field with a coupled system. Memory-dependent derivatives Wang and Li [2011] generalise the exponential kernel into an arbitrary weighting function, which makes the memory structure a modelling choice rather than a parameter; fractional thermoelasticity Caputo [1967], Gorenflo and Mainardi [1997], Luchko [2010], Povstenko [2004, 2007, 2009, 2011, 2016] replaces the derivative order with a real exponent in $(0, 2)$, and thereby interpolates the diffusive and wave-like regimes continuously rather than discretely; and the coupling of the thermal field to electrical, porous, micro-rotation and carrier fields Ieşan [1986, 2013], Cowin and Nunziato [1983], Nunziato and Cowin [1979], Chandrasekharaiah [1996], Marotti de Sciarra [2013], Sherief and Hussein [2022], Abouelregal et al. [2023] converts single-field wave problems into multiphysics ones in which the number of modes, and hence the size of the interface system, grows with the number of couplings.

2.3 International structure of the field

The development of generalized thermoelasticity was international from the outset, and the pattern of its internationalization follows the pattern of the science rather than that of any one school. The founding hyperbolic-conduction results appeared in the Italian and French academy proceedings Cattaneo [1948], Vernotte [1958], and the consolidation of the subject in the 1970s occurred in the American journals of applied mechanics Biot [1956], Lord and Shulman [1967], Green and Lindsay [1972], Green and Naghdi [1991, 1993], Tzou [1995]. The micropolar extension of the theory — which adds micro-rotational degrees of freedom to a material with internal structure — was developed across the Turkish, German and American continuum-mechanics communities and published in the engineering-science journals of that period Eringen [1966], Tauchert et al. [1968], Eringen [1972]; the anisotropic extension followed from Canadian and Egyptian work on direction-dependent thermoelastic response Dhaliwal and Sherief [1980], Sherief and Hussein [2022]. The fractional-thermoelastic programme of the 2000s and 2010s was carried principally from the Ukrainian school, whose contributions span the time-fractional heat equation, the general fractional formulation and its convergence properties Povstenko [2004, 2007, 2009, 2011, 2016]; the memory-dependent formulation was developed in China Wang and Li [2011], and the mathematical apparatus of fractional viscoelasticity that underwrites it was placed on a rigorous footing in the Italian school Gorenflo and Mainardi [1997], Luchko [2010]. In the most recent period the activity has become genuinely multinational, with contributions from the Egyptian and Saudi groups on fractional and memory-based thermoelasticity and their interface problems Abouelregal et al. [2023], Sherief and Hussein [2022], from Italian mathematical studies of the variational and phase-transition structure of multiphysics continua Marotti de Sciarra [2013], from Spanish work on well-posedness and uniqueness in nonlocal and fractional thermoelasticity Quintanilla [2019], and from an Indian group whose recent series addresses piezothermoelastic and memory-dependent wave problems, energy ratios at material interfaces and heat-map representations of coupled-field response Barak et al. [2023a,b,c,d], Gupta et al. [2023, 2022], Kumar et al. [2023a], Gupta et al. [2026].

This distribution is not incidental. Each of the centres identified above entered the field through a different scientific interest — heat conduction in metals, micropolar fluid mechanics, anomalous diffusion in geophysics, piezoelectricity in smart materials — and each imported the constitutive modification appropriate to that interest. The result is a literature in which the same physical question, the admissibility of a propagating thermal mode, is attacked with structurally different devices: a relaxation time, a second temperature, a thermal displacement, a phase lag, a memory kernel, a fractional order. A review organised around papers would obscure this, because the same device reappears in every centre with different notation. A review organised around the devices — which is what this review attempts — shows that the field is not a sequence of independent advances but a single continuing negotiation between three requirements that are individually necessary and jointly expensive: finite propagation speed, thermodynamic admissibility, and well-posedness.

2.4 The literature corpus and the verification protocol

The repository's literature collection is purposive rather than a result of a database-wide systematic search. It contains foundational records, reviews and monographs, and selected applied studies from coupled, nonlocal, fractional, and memory-dependent topics. The inherited bibliography and DOI-verification ledger supply metadata and source leads. The ledger records registry matches for many DOI-bearing entries, but those metadata matches were not re-queried in this phase and do not verify the content, equations, or relevance of cited works. The interface-energy-ratio series and other application-specific studies require full-text claim checking before

their results can support comparative conclusions.

The working literature inventory contains 95 records; 48 have a DOI and 47 have no DOI recorded. The cited bibliography contains 56 entries: 48 DOI records are marked Crossref-matched in the inherited verification ledger, and eight entries are marked as having no registered DOI. The inventory flags 13 records as open access and 12 as having full text available (one additional record is methodology material). A DOI count is not a count of full-text sources, and metadata resolution is not claim-level verification. The inventory was assembled from the repository's existing collection and targeted bibliographic checks; it is not a reproducible database-wide systematic search and is not represented as PRISMA screening. The number of works read in full varies by topic; the text-extraction register, rather than the DOI ledger, identifies which records were available for full-text checking.

Three consequences follow for how the remainder of the review should be read. First, coverage is theory-family-complete rather than application-complete: the corpus was assembled by constitutive structure, so an application is covered when a theory family that addresses it is covered, and not otherwise. Second, the evidence levels assigned in Section 13 distinguish between a mechanism that is experimentally established, a mechanism that is established but only modelled approximately, and a mechanism that remains speculative; the distinction is made in the review because the corpus contains all three kinds and because collapsing them would misrepresent the state of the field. Third, and most importantly for the treatment of individual contributions, the review is organised by scientific development and conceptual theme. Where two theories differ in their predictions, the review states what the difference is and what would distinguish them experimentally, rather than arranging the literature as a sequence of individual studies.

2.5 Development axes and the gaps they expose

The corpus can be classified along five axes — physical domain, mathematical structure, modal content, coupling topology and validation regime — and this classification is developed formally in Section 14. Its value at this stage is diagnostic: it shows where the field is dense and where it is thin. The dense regions are the single-field, one-dimensional, interface-focused problems of classical, relaxation-time and phase-lag thermoelasticity, for which a large body of modal and energy-ratio results exists. The thin regions are more revealing. There is very little work that closes the loop from constitutive novelty to experimental discrimination: the nonlocal and fractional families are theoretically mature, with well-established dispersion relations and modal structures, yet the number of studies that fit predictions to measured dispersion data remains small. The comparison of competing mechanisms for the same physical effect — nonlocal versus gradient, exponential versus power-law memory, phase lag versus relaxation time — is similarly under-represented, even though it is precisely in this comparison that the theory choices are testable. A third thin region is the treatment of interfaces in the presence of memory or nonlocality, where the constitutive history and the geometric matching condition must be reconciled, and where the energy partition is not determined by the local constitutive law.

These gaps are not defects of the individual contributions; they follow from the structure of the field, in which each advance has been made by extending a constitutive law and then solving the resulting modal problem, and the natural next step — putting two such extensions in competition against an experiment — has not yet become the standard mode of work. Section 16 takes this diagnosis as the basis for its research directions.

2.6 Existing reviews and the contribution of this article

Reviews already exist for several individual theory families: nonlocal elasticity and its application to nanostructures Askes and Aifantis [2006, 2011], Maranganti and Sharma [2007], fractional thermoelasticity Chandrasekharaiah [1998], Povstenko [2016], hyperbolic thermoelasticity and second sound Chandrasekharaiah [1996], Joseph and Preziosi [1989], and generalized thermoelasticity with modified models Shakeriaski et al. [2021]. Their common structure is the one this review sets aside: an enumeration of theory families, each with its governing equations, its application domains and a list of its principal results, with the families treated in isolation and the transitions between them left implicit.

The present review differs in three respects. It is organised by the problems — finite propagation speed, thermodynamic admissibility, well-posedness, scale dependence, interface partition of energy — rather than by the theories, so that a reader interested in one problem can follow it across all the theories that address it. It states the cost of each modification alongside the benefit, which the standard format does not do, and which the literature itself rarely does: the second temperature, the loss of ellipticity, the loss of uniqueness of the initial-value problem, and the non-locality of the energy flux are each the price of something that was bought. And it separates the question of a mechanism's plausibility from the question of its verification, treating the evidence regime as an explicit axis of classification rather than as a matter of emphasis.

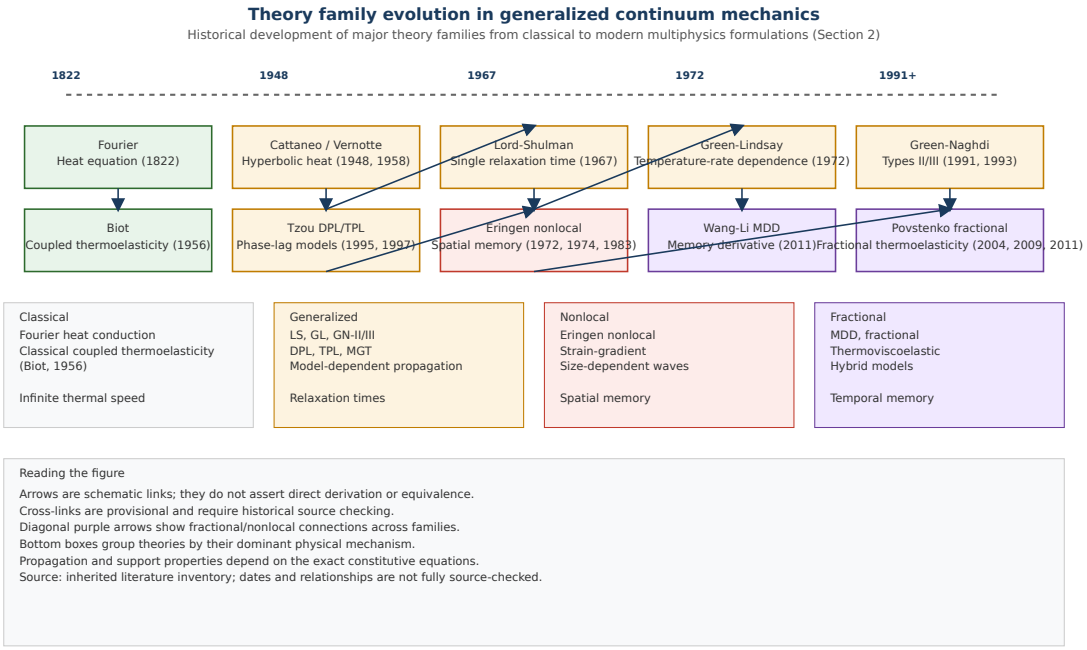


Figure 3: Selected theory families and relationships in generalized continuum mechanics. This schematic is a conceptual taxonomy, not a verified chronology or assertion of mathematical equivalence; its classifications are provisional pending claim-level source checks.

2.7 Citation and verification principles

The intended editorial standard is to state the validity domain, conventions, and evidence status of each important equation, and to distinguish a direct derivation from a bibliographic or secondary-source claim. The inherited manuscript does not yet meet that standard throughout:

legacy equations and verification labels remain in several appendices and comparison tables. Appendix A now records its limited v1.1 derivations and marks the remaining claims UNVERIFIED or OPEN. This standard is a work plan, not a claim that all equations have been checked.

Where material errors are conclusively established in the literature, they are recorded transparently and without personal characterisation. Where uncertainty exists, it is acknowledged as an open question. No error allegations are made against any publication, and model-specific differences are never presented as mathematical contradictions without sufficient evidence.

3 Foundations of Continuum Wave Mechanics

This section develops the first-principles foundation (Layer I of the layered mathematical exposition) for all subsequent theory families. We begin with kinematics, proceed through the conservation laws of continuum mechanics, establish the thermodynamic restrictions on constitutive relations, derive classical elasticity and wave propagation, and conclude with the classical energy flux Biot [1956] and interface conditions. Every equation in this section is derived from first principles or drawn from established standard results with documented assumptions.

3.1 Kinematics: displacement, strain, rotation

Let $\boldsymbol{u}(\boldsymbol{x}, t)$ denote the displacement field of a continuum body occupying a region $\Omega \subset \mathbb{R}^3$ with boundary $\partial\Omega$. The displacement gradient tensor decomposes additively into symmetric and antisymmetric parts:

$$u_{i,j} = \varepsilon_{ij} + \omega_{ij}, \tag{1}$$

where the infinitesimal strain tensor and rotation tensor are defined by

$$\varepsilon_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i}), \qquad \omega_{ij} = \frac{1}{2}(u_{i,j} - u_{j,i}). \tag{2}$$

The strain tensor ε_{ij} measures the symmetric part of the displacement gradient (deformation), while the rotation tensor ω_{ij} measures the antisymmetric part (rigid-body rotation). The velocity field is the material time derivative of displacement:

$$v_i = \dot{u}_i = \frac{\partial u_i}{\partial t}. \tag{3}$$

3.1.1 Physical interpretation

The strain components ε_{ii} represent normal strains (relative changes in length along coordinate directions), while ε_{ij} $(i \neq j)$ represent shear strains (changes in angle between originally orthogonal material lines). The infinitesimal strain assumption requires $\|\varepsilon_{ij}\| \ll 1$, which is valid for small deformations typical of linearized wave propagation problems.

3.2 Conservation laws: linear momentum and angular momentum

3.2.1 Linear momentum balance

Consider a material subvolume $V \subset \Omega$ with surface $S = \partial V$ and outward unit normal $\boldsymbol{n}$. The linear momentum balance states that the time rate of change of linear momentum equals the sum of surface tractions and body forces:

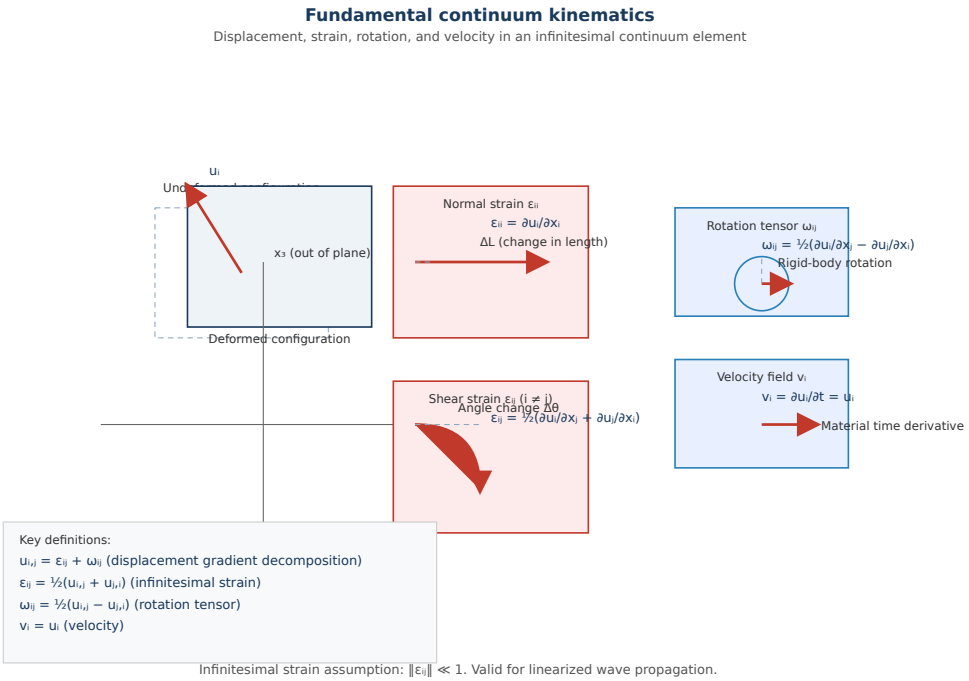


Figure 4: Kinematics of a continuum element. The displacement field $\boldsymbol{u}(\boldsymbol{x}, t)$ defines the deformation gradient, which in the infinitesimal regime splits additively into a symmetric part (strain, the extensional part of the deformation) and an antisymmetric part (rotation, the rigid part). The rigid part does no work and is removed from the balance laws by the objectivity requirement, so that only the strain enters the stress; this is the first of the two places where irreversibility enters the formulation, the second being the second law of Section 3.4.

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho v_i \,\mathrm{d}V = \int_S \sigma_{ji} n_j \,\mathrm{d}S + \int_V \rho f_i \,\mathrm{d}V, \tag{4}$$

where ρ is the mass density, σ_{ji} is the Cauchy stress tensor, and f_i is the body force per unit mass. Applying the divergence theorem to the surface integral and localizing (since V is arbitrary) yields the pointwise form:

$$\rho \dot{v}_i = \sigma_{ji,j} + \rho f_i. \tag{5}$$

Since $v_i = \dot{u}_i$, this can also be written in terms of displacement:

$$\rho \ddot{u}_i = \sigma_{ji,j} + \rho f_i. \tag{6}$$

Equation (5) is the fundamental equation of motion for continuum mechanics. It is sometimes called Cauchy's first law or the equation of linear momentum.

3.2.2 Angular momentum balance

The angular momentum balance for an arbitrary subvolume V states:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho(\boldsymbol{x} \times \boldsymbol{v})_k \,\mathrm{d}V = \int_S \epsilon_{kij} x_i \sigma_{jm} n_m \,\mathrm{d}S + \int_V \rho \epsilon_{kij} x_i f_j \,\mathrm{d}V, \tag{7}$$

where ϵ_{kij} is the Levi-Civita permutation symbol. Applying the divergence theorem, using the linear momentum balance (5), and localizing gives:

$$\sigma_{ij} = \sigma_{ji}. \tag{8}$$

Equation (8) states that the Cauchy stress tensor is symmetric in the absence of couple stresses or body couples. This symmetry is essential for the energy flux derivations in Sections 3.8 and 10.2.

3.2.3 Physical interpretation

Equations (5) and (8) are local statements of Newton's second law applied to continuum matter. The symmetry of stress ((8)) means that traction on a surface is determined solely by the normal to that surface, without additional rotational degrees of freedom in classical (non-polar) continua.

3.3 Mechanical power and internal energy: first law of thermodynamics

3.3.1 Mechanical power balance

Multiply the linear momentum equation (5) by v_i and integrate over V:

$$\int_V \rho \dot{v}_i v_i \,\mathrm{d}V = \int_V \sigma_{ji,j} v_i \,\mathrm{d}V + \int_V \rho f_i v_i \,\mathrm{d}V. \tag{9}$$

The left-hand side is the rate of change of kinetic energy:

$$\int_V \rho \dot{v}_i v_i \,\mathrm{d}V = \frac{\mathrm{d}}{\mathrm{d}t}\int_V \frac{1}{2}\rho v_i v_i \,\mathrm{d}V. \tag{10}$$

For the stress term, use the identity $\sigma_{ji,j} v_i = (\sigma_{ji} v_i)_{,j} - \sigma_{ji} v_{i,j}$. Since $v_{i,j} = \dot{u}_{i,j} = \dot{\varepsilon}_{ji}$ (using symmetry of σ_{ji} and ε_{ij}), we obtain:

$$\int_V \sigma_{ji,j} v_i \,\mathrm{d}V = \int_S \sigma_{ji} v_i n_j \,\mathrm{d}S - \int_V \sigma_{ji} \dot{\varepsilon}_{ij} \,\mathrm{d}V. \tag{11}$$

Combining (9) through (11) gives the mechanical power balance:

$$\rho \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2} v_i v_i\right) = \sigma_{ji,j} v_i + \rho f_i v_i = -\sigma_{ij}\dot{\varepsilon}_{ij} + (\sigma_{ij} v_j)_{,i} + \rho f_i v_i. \tag{12}$$

The term $\sigma_{ij} v_j n_i$ on the surface represents mechanical power entering through the boundary per unit area; its density $(\sigma_{ij} v_j)_{,i}$ is the mechanical power flux divergence.

3.3.2 Internal energy balance

The first law of thermodynamics for a continuum states that the rate of change of internal energy equals the mechanical power input minus the heat flux out of the body plus heat supply:

$$\rho \dot{e} = \sigma_{ij}\dot{\varepsilon}_{ij} - q_{i,i} + \rho r, \tag{13}$$

where e is the specific internal energy (per unit mass), q_i is the heat flux vector, and r is the heat supply per unit mass. In the absence of external heat supply ($r = 0$):

$$\rho \dot{e} = \sigma_{ij}\dot{\varepsilon}_{ij} - q_{i,i}. \tag{14}$$

3.3.3 Physical interpretation

Equation (13) expresses energy conservation: the internal energy of a material element changes due to (i) mechanical work done by stresses ($\sigma_{ij}\dot{\varepsilon}_{ij}$), (ii) heat conduction out of the element ($-q_{i,i}$), and (iii) external heat supply (ρr). The term $\sigma_{ij}\dot{\varepsilon}_{ij}$ is the stress power density — the rate of mechanical work converted into internal energy or stored as elastic energy.

3.4 Entropy and dissipation inequality: Clausius–Duhem inequality

The second law of thermodynamics, in the Clausius–Duhem form, states that the entropy production within a continuum is non-negative:

$$\rho \dot{s} \geq -\left(\frac{q_i}{\Theta}\right)_{,i} + \frac{\rho r}{\Theta}, \tag{15}$$

where s is specific entropy and Θ is absolute temperature. In the linear constitutive equations below, $T = \Theta - T_0$ denotes the temperature increment. Using the internal-energy balance (13) and defining the Helmholtz free energy by $\psi = e - \Theta s$, we obtain after standard manipulation:

$$\sigma_{ij}\dot{\varepsilon}_{ij} - \rho\left(\dot{\psi} + s\dot{\Theta}\right) - \frac{q_i \Theta_{,i}}{\Theta} \geq 0. \tag{16}$$

Equation (16) is the Clausius–Duhem dissipation inequality. The left-hand side represents the local entropy production rate (multiplied by temperature) and must be non-negative for all admissible thermodynamic processes.

3.4.1 Physical interpretation

The dissipation inequality restricts admissible constitutive relations. Any constitutive model must be constructed so that (16) holds for all processes. This principle is used in Section 3.5 to derive Coleman–Noll restrictions and in Section 10.7 to assess thermodynamic admissibility of generalized theories.

3.5 Constitutive principles: free energy and linear relations

3.5.1 Coleman–Noll procedure

Assuming the free energy depends on strain and temperature, $\psi = \psi(\varepsilon_{ij}, T)$, we have:

$$\dot{\psi} = \frac{\partial \psi}{\partial \varepsilon_{ij}} \dot{\varepsilon}_{ij} + \frac{\partial \psi}{\partial T} \dot{T}. \tag{17}$$

Substituting into the dissipation inequality (16) and requiring it to hold for arbitrary $\dot{\varepsilon}_{ij}$ and $\dot{T}$ yields the Coleman–Noll restrictions:

$$\sigma_{ij} = \rho \frac{\partial \psi}{\partial \varepsilon_{ij}}, \qquad s = -\frac{\partial \psi}{\partial T}. \tag{18}$$

With these restrictions, the dissipation inequality reduces to the thermal conduction restriction:

$$-\frac{q_i T_{,i}}{T} \geq 0. \tag{19}$$

For isotropic linear conduction with Fourier's law $q_i = -k T_{,i}$ ($k > 0$), we have $-q_i T_{,i}/T = k (T_{,i})^2 / T \geq 0$, which is satisfied.

3.5.2 Linear isotropic constitutive relations

For a linear isotropic thermoelastic material, with temperature increment $T = \Theta - T_0$ and thermal-stress coefficient β in Pa K^{-1}, take the specific Helmholtz free energy:

$$\psi = \frac{1}{2\rho}(\lambda \varepsilon_{kk} \varepsilon_{ll} + 2\mu \varepsilon_{ij} \varepsilon_{ij}) - \frac{\beta}{\rho} \varepsilon_{kk} T - s_0 T - \frac{c_v}{2T_0} T^2 + \text{const.}, \tag{20}$$

where λ and μ are Lamé elastic constants, β is the thermal stress coefficient, and c_v is the specific heat at constant strain. Applying (18) gives:

$$\sigma_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2\mu \varepsilon_{ij} - \beta T \delta_{ij}, \tag{21}$$

$$s = s_0 + \frac{\beta}{\rho} \varepsilon_{kk} + \frac{c_v}{T_0} T. \tag{22}$$

Equation (21) is the linear isotropic thermoelastic constitutive relation: the first two terms are the classical elastic stress (Hooke's law), and the last term is the thermal stress contribution.

3.5.3 Physical interpretation

The constitutive relations (21)–(22) couple mechanical and thermal fields: temperature changes induce thermal stresses ($-\beta T \delta_{ij}$), and deformation changes entropy ($\frac{\beta}{\rho} \varepsilon_{kk}$). This coupling is the physical origin of thermoelastic wave propagation.

3.6 Classical elasticity: Navier equations and wave speeds

Substituting the constitutive relation (21) into the equation of motion (6) with no body forces ($f_i = 0$) and isothermal conditions ($T = T_0$ constant), we obtain the Navier equations of classical elasticity:

$$\rho \ddot{u}_i = (\lambda + \mu) u_{j,ji} + \mu u_{i,jj}. \tag{23}$$

3.6.1 Plane wave solutions

For harmonic plane waves of the form $u_i = \hat{u}_i e^{i(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)}$, substitution into (23) gives the Christoffel equation. For longitudinal (P) waves propagating in direction $\boldsymbol{n}$ with polarization parallel to $\boldsymbol{n}$:

$$c_p = \sqrt{\frac{\lambda + 2\mu}{\rho}}, \tag{24}$$

and for transverse (S or SV) waves with polarization perpendicular to $\boldsymbol{n}$:

$$c_s = \sqrt{\frac{\mu}{\rho}}. \tag{25}$$

These are the classical elastic wave speeds. The P-wave speed exceeds the S-wave speed since $\lambda + 2\mu > \mu$ for stable materials ($\lambda + 2\mu/3 > 0$, $\mu > 0$).

3.6.2 Physical interpretation

Equation (23) describes how elastic disturbances propagate through a homogeneous isotropic solid. The P-wave involves volume change (compressional), while the S-wave involves shear (distortional) deformation. Both propagate without dispersion in classical elasticity — all frequencies travel at the same speed.

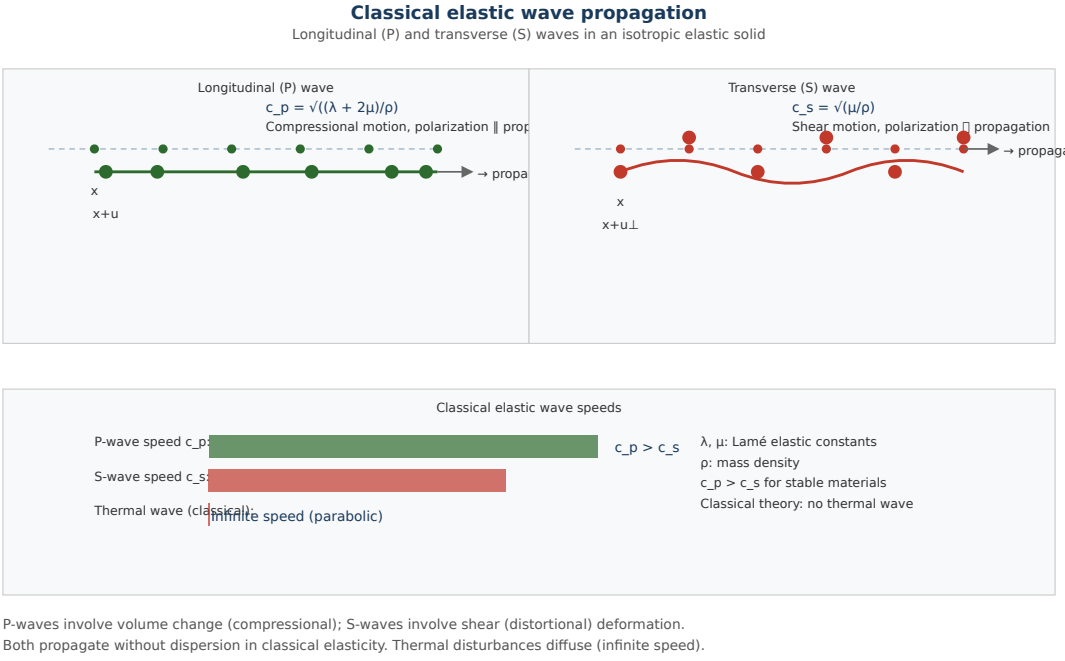


Figure 5: Classical elastic wave propagation in an isotropic solid. A longitudinal (P) wave, whose polarization is parallel to the propagation direction, travels at $c_p = \sqrt{(\lambda + 2\mu)/\rho}$; a transverse (S) wave, whose polarization is perpendicular to it, travels at $c_s = \sqrt{\mu/\rho}$. Because the elastic moduli are independent of frequency, both speeds are dispersion-free: this constancy is the reference against which the dispersive branches of Sections 4–7 are measured, and it is the property that a generalized theory must forfeit when it introduces a length scale or a memory kernel.

3.7 Plane wave assumptions and harmonic wave form

For time-harmonic problems, all field variables take the form:

$$u_i(\boldsymbol{x},t) = \hat{u}_i(\boldsymbol{x})e^{-i\omega t}, \qquad \sigma_{ij}(\boldsymbol{x},t) = \hat{\sigma}_{ij}(\boldsymbol{x})e^{-i\omega t}, \qquad q_i(\boldsymbol{x},t) = \hat{q}_i(\boldsymbol{x})e^{-i\omega t}, \tag{26}$$

where $\hat{u}_i$, $\hat{\sigma}_{ij}$, $\hat{q}_i$ are complex amplitudes and ω is the angular frequency. The time convention $e^{-i\omega_t}$ implies that positive imaginary parts of complex wave numbers correspond to attenuation (decay in space) and negative imaginary parts of complex frequencies correspond to temporal decay.

3.7.1 Time-average identity

For two harmonic quantities $a(t) = \Re[\hat{a}e^{-i\omega t}]$ and $b(t) = \Re[\hat{b}e^{-i\omega t}]$, the time average over one period is:

$$\langle ab \rangle_t = \frac{1}{2}\Re[\hat{a}\hat{b}^*], \tag{27}$$

where * denotes complex conjugation. This identity is used throughout Sections 9 and 10 to compute average energy fluxes and energy ratios from complex wave amplitudes.

3.8 Energy flux in classical elasticity

Adding the mechanical power balance (12) and the internal-energy balance (14) (with $r = 0$), and defining total energy per unit volume as $\mathcal{E} = \frac{1}{2}\rho v_i v_i + \rho e$, gives an outward-positive flux $J_j = q_j - \sigma_{ji}v_i$:

$$\frac{\partial \mathcal{E}}{\partial t} + \frac{\partial}{\partial x_j}\left(q_j - \sigma_{ji}v_i\right) = \rho f_i v_i. \tag{28}$$

With no body forces ($f_i = 0$), this becomes the local energy conservation law:

$$\frac{\partial \mathcal{E}}{\partial t} + \frac{\partial J_i}{\partial x_i} = 0, \qquad J_i = q_i - \sigma_{ij}v_j. \tag{29}$$

Equation (29) states that the total energy density changes only due to the divergence of the energy flux J_i. The energy flux has two contributions:

- $-\sigma_{ij}v_j$: mechanical contribution to the outward-positive flux, and
- q_i: thermal energy flux (heat conduction across a surface per unit area).

Integrating (29) over a volume V and applying the divergence theorem:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \mathcal{E}\,\mathrm{d}V + \int_S J_i n_i\,\mathrm{d}S = 0. \tag{30}$$

For an isolated system ($J_i n_i = 0$ on S), the total energy is conserved: $\frac{\mathrm{d}}{\mathrm{d}t}\int_V \mathcal{E}\,\mathrm{d}V = 0$.

3.8.1 Physical interpretation

Equation (29) is the classical energy flux Biot [1956] for elastic and thermoelastic media. With the outward-positive convention, the mechanical contribution is $-\sigma_{ij}v_j$ and the thermal contribution is q_i. Reversing the normal or defining inward-positive flux reverses the signs consistently. This local balance is derived for the classical thermoelastic model only. Whether an additional stored-energy or boundary-power term is required in a generalized theory depends on its constitutive equations and must be derived model by model.

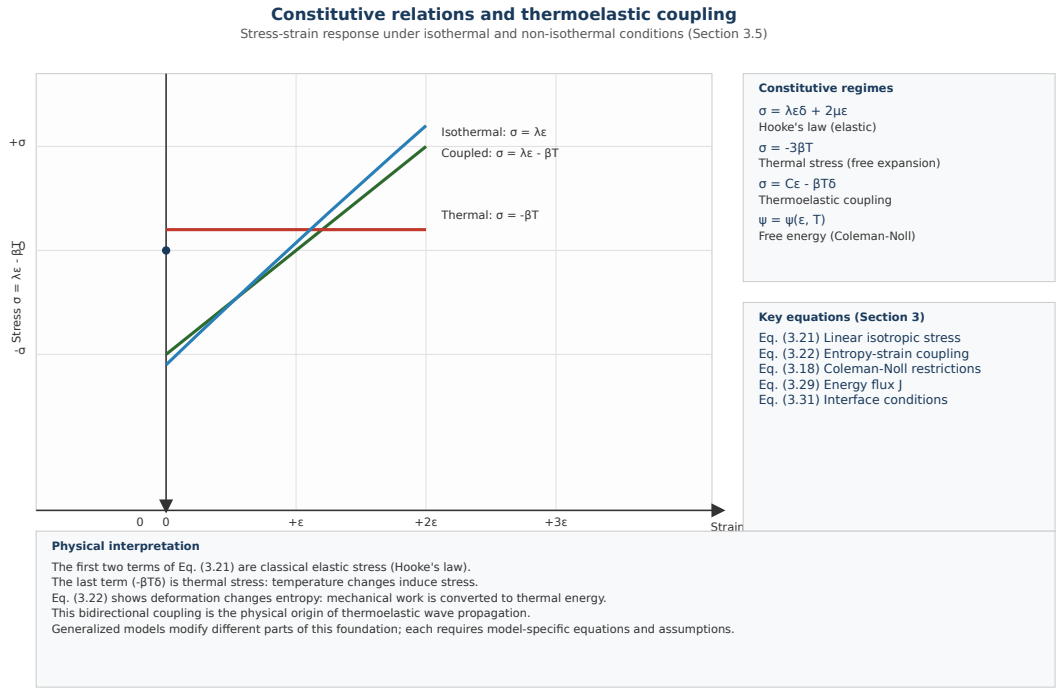


Figure 6: Constitutive relations and thermoelastic coupling: stress-strain response under isothermal and non-isothermal conditions (Section 3.5).

3.9 Interface conditions: perfect and imperfect bonding

Consider an interface Σ between two media (denoted 1 and 2) with common unit normal $\boldsymbol{n}$. For perfectly bonded interfaces:

3.9.1 Perfect bonding

$$u_i^{(1)} = u_i^{(2)}, \qquad \sigma_{ji}^{(1)} n_j = \sigma_{ji}^{(2)} n_j, \qquad T^{(1)} = T^{(2)}, \qquad q_i^{(1)} n_i = q_i^{(2)} n_i, \tag{31}$$

on Σ. The first condition enforces displacement continuity (no slip or separation), the second enforces traction equilibrium (Newton's third law at the interface), and the last two enforce thermal contact (temperature continuity and heat flux continuity).

3.9.2 Imperfect bonding: spring-type conditions

For imperfectly bonded interfaces, displacement and traction are related by interfacial stiffness parameters:

$$[u_i] = (C^u)^{-1}_{ij} t_j, \qquad \sigma_{ki}^{(1)} n_k = \sigma_{ki}^{(2)} n_k = t_i, \tag{32}$$

where $[u_i] = u_i^{(2)} - u_i^{(1)}$, t_i is the common traction on the shared normal, and C^u_{ij} is a positive-definite interfacial stiffness tensor (units Pa/m). The compliance form is used so that, when all stiffness eigenvalues tend to infinity, the displacement jump vanishes; when they tend to zero through positive values, the traction vanishes for a finite jump. More general imperfect-interface laws require their own constitutive and dissipation assumptions.

3.9.3 Electrical interface conditions (piezoelectric media)

For piezoelectric media, additional interface conditions involve the electric potential ϕ and electric displacement D_i:

$$\phi^{(1)} = \phi^{(2)} \qquad \text{(perfect electrical contact)}, \tag{33}$$

$$D_i^{(1)} n_i = D_i^{(2)} n_i \qquad \text{(electric displacement continuity)}, \tag{34}$$

or for an insulated interface:

$$D_i n_i = 0. \tag{35}$$

3.9.4 Physical interpretation

Interface conditions determine how waves transmit and reflect at material boundaries. Perfect bonding ((31)) is the standard assumption for welded interfaces. Imperfect bonding ((32)) models thin adhesive layers, micro-cracks, or weakly bonded interfaces. Electrical conditions ((33)–(35)) are essential for piezothermoelastic interface problems in Section 5 and Section 9.

3.10 Notation summary for Section 3

Table 4 summarizes the notation introduced in this section.

3.11 Open questions for this section

OQ3.1 What is the most general form of energy flux that remains valid across all generalized theories (LS, GL, GN, DPL, TPL, MGT, MDD, nonlocal)?

OQ3.2 How do interface conditions change for nonlocal media where the stress at a point depends on strain throughout a neighborhood?

OQ3.3 What is the relationship between the classical energy flux Biot [1956] (29) and the entropy flux in thermoelasticity without energy dissipation (GN-II)?

4 Classical and Generalized Thermoelastic Theories

This section presents the historical development, governing equations, wave propagation characteristics, and dissipation structure of classical and generalized thermoelastic theories. Each theory is introduced with its physical motivation, mathematical formulation, and domain of applicability. Representative equations are selected to illustrate each theory family; derivations from first principles are given where they illuminate the theory's structure. Citations to primary sources are provided throughout to distinguish established results from open questions.

4.1 Historical evolution

The development of generalized thermoelasticity spans nearly two centuries, from Fourier's heat equation to modern memory-dependent and nonlocal formulations. Table 5 summarizes the principal milestones.

Historical context only: the review is SECONDARY-SOURCE, not equation-level evidence (BACKGROUND-ONLY) Chandrasekharaiah [1998].

Table 3: Wave types and their governing characteristics across theory families. Phase speeds are model-dependent; values shown are classical limits.

Wave Type	**Polarization**	**Governing Speed (Classical)**	**Model-Specific Interpretation**	**Evidence**
Longitudinal (P)	Parallel to propagation	$c_p = \sqrt{(\lambda + 2\mu)/\rho}$	Classical isotropic elasticity; generalized mode labels are model-specific	Achenbach [1973]
Transverse (S)	Perpendicular to propagation	$c_s = \sqrt{\mu/\rho}$	Shear; distortional; no volume change	Achenbach [1973]
Thermal	Parallel (dissipative) or propagating (generalized)	Classical: infinite (parabolic); LS: $c_T = \sqrt{k/(\rho c_v \tau_0)}$	Diffusive in classical Fourier; LS has finite speed for $\tau_0 > 0$. The displayed GL equation has a second-order thermal-time term for $\tau_0 > 0$; coupled characteristic speeds were not recomputed from the corrected equation. Other phase-lag/MGT variants are formulation-specific and not classified here	Conditional on the displayed linear laws and stated parameters; LS: Lord and Shulman [1967]; GL: Eq. 42 only
Coupled modes	Mixed polarization	Coupled system eigenfrequencies	P–T, S–T, or piezo–thermal coupling splits wave speeds	Model-specific
Surface waves	Decaying with depth	Rayleigh: $c_R \approx 0.9194 c_s$ at $\nu = 0.25$ (isotropic solid; $0.874 c_s$ at $\nu = 0$)	Exists at free surfaces; modified by thermal/electrical boundary conditions	Model-specific
Evanescent modes	Exponential decay	Imaginary wave number $q = i\kappa$	Near-field interface phenomena; important for scattering	Model-specific

Notes: Classical speeds apply to isotropic elastic media. Generalized theories introduce dispersion and attenuation.
LS speed requires $\tau_0 > 0$; the GL statement is limited to the displayed linear thermal equation and $\tau_0 > 0$; coupled speeds are not recalculated. Classical speeds from Achenbach [1973].

Table 4: Notation introduced in Section 3.

Symbol	Meaning	Units	Equation
u_i	Displacement	m	(1)
ε_{ij}	Infinitesimal strain	dimensionless	(2)
ω_{ij}	Rotation tensor	dimensionless	(2)
v_i	Velocity	m/s	(3)
σ_{ij}	Cauchy stress	Pa	(5)
ρ	Mass density	kg/m^3	(5)
f_i	Body force	N/kg	(5)
q_i	Heat flux	W/m^2	(13)
T	Temperature increment $\Theta - T_0$	K	(20)
e	Specific internal energy	J/kg	(13)
ψ	Specific free energy	J/kg	(18)
s	Specific entropy	J/(kg·K)	(18)
$\mathcal{E}$	Total energy density	J/m^3	(29)
J_i	Energy flux	W/m^2	(29)
λ, μ	Lamé constants	Pa	(23)
c_p, c_s	P-wave and S-wave speeds	m/s	(24)–(25)

For the historical overview, see Figure 1 in Section 1.

A recurring motivation for some generalized heat-transport models is that Fourier conduction is parabolic and has formally infinite propagation speed. Generalized constitutive choices explore alternatives for selected regimes, but finite-speed behavior and thermodynamic consistency are model-specific properties that require derivation; neither follows from the label "generalized."

Joseph and Preziosi's review (1989) and 1990 addendum summarize the second-sound evidence and the objections raised in the intervening discussion Joseph and Preziosi [1989, 1990]. This debate helped motivate later finite-speed heat-transport models; it does not establish a universal need for them.

4.2 Classical coupled thermoelasticity (Fourier heat conduction)

The classical coupled thermoelasticity theory, developed by Biot (1956) Biot [1956], combines the equations of linear elasticity with Fourier's law of heat conduction:

$$\rho \ddot{u}_i = (\lambda + \mu) u_{j,ji} + \mu u_{i,jj} - \beta T_{,i}, \tag{36}$$

$$\rho c_v \dot{T} = k \nabla^2 T - \beta T_0 \dot{u}_{i,i}, \tag{37}$$

where ρ is density, λ and μ are Lamé constants, β is the thermal stress coefficient, c_v is specific heat at constant strain, k is thermal conductivity, and T_0 is the reference temperature.

4.2.1 Wave speeds

The classical theory predicts two elastic wave types:

$$c_p = \sqrt{\frac{\lambda + 2\mu}{\rho}}, \qquad c_s = \sqrt{\frac{\mu}{\rho}}. \tag{38}$$

The thermal field satisfies a parabolic diffusion equation, predicting infinite thermal propagation speed — the classical paradox that motivated all subsequent generalized theories.

Table 5: Selected publication milestones and broad modelling mechanisms. Dates and bibliographic matches are inherited metadata; physical and equation-level claims remain subject to primary-source review.

Model/family	Year	Source lead	Broad modelling feature	v1.1 evidence status
Classical coupled thermoelasticity	1956	Biot [1956]	Coupled mechanical and thermal fields with Fourier conduction	Metadata inherited; equations not re-audited here
Lord–Shulman	1967	Lord and Shulman [1967]	Heat-flux relaxation model	Metadata inherited; constitutive details not full-text-checked
Green–Lindsay	1972	Green and Lindsay [1972]	Single temperature; rate dependence with two relaxation times	Corrected description; source check open
Green–Naghdi II / III	1993 / 1992	Green and Naghdi [1993, 1992]	Type II: nondissipative thermal displacement; type III adds a rate term	Labels and equations checked (F13, F16–F17)
Dual/triple phase lag	1995/2007	Tzou [1995], Choudhuri [2007]	DPL and Roy Choudhuri first-order TPL forms	Exact expansions and admissibility not checked
Eringen-type nonlocality	1972 onward	Eringen [1972, 1974, 1983]	Spatially nonlocal constitutive response	Kernel and boundary formulation require source review
Memory-dependent operator	2011 onward	Wang and Li [2011]	History-dependent heat-transport formulation	Operator definition and limiting behavior not checked
Moore–Gibson–Thompson	1960; thermoelastic analogues later	Moore and Gibson [1960], Quintanilla [2019]	Moore–Gibson 1960: third-order relaxing-gas acous-	1960 scope checked (F14); later variants are model-

4.2.2 Physical interpretation

Equations (36)–(37) are two-way coupled: volumetric strain rate sources temperature, while the temperature gradient contributes to mechanical stress. In the fully dynamic problem neither field can in general be solved independently; sequential treatments are approximations requiring explicit scale assumptions.

4.3 Lord–Shulman (LS) theory (1967)

Lord and Shulman (1967) Lord and Shulman [1967] introduced a single thermal relaxation time τ_0 into Fourier's law:

$$q_i + \tau_0 \dot{q}_i = -kT_{,i}, \tag{39}$$

where $\tau_0 > 0$ is the relaxation time and q_i is the heat flux vector.

4.3.1 Modified heat equation

Combining (39) with the energy balance yields the LS heat equation:

$$\rho c_v \left(\dot{T} + \tau_0 \ddot{T} \right) = k\nabla^2 T - \beta T_0 (\dot{u}_{i,i} + \tau_0 \ddot{u}_{i,i}). \tag{40}$$

4.3.2 Finite thermal wave speed

The thermal wave speed in LS theory is:

$$c_T = \sqrt{\frac{k}{\rho c_v \tau_0}}. \tag{41}$$

For the displayed LS/Cattaneo law, the formal thermal characteristic speed is finite when $\tau_0 > 0$. The limiting substitution $\tau_0 \to 0$ recovers Fourier conduction at the level of the constitutive equation; convergence of solutions is a separate question. These statements are conditional on the displayed model and have not been independently checked against the Lord–Shulman primary text in this revision.

4.3.3 Well-posedness

For the displayed Cattaneo-type heat-flux law, $\tau_0 > 0$ gives a finite characteristic speed for the thermal branch. This parameter condition alone is not a proof that the full coupled initial-boundary-value problem is well-posed; the latter depends on coefficients, domain, and boundary/initial data. A separate fractional Lord–Shulman formulation couples a Klein–Gordon nonlocal operator to the heat law; its printed $\alpha \to 0$ branch is not used here as a limiting identity (CONDITIONAL) Das et al. [2026].

4.4 Green–Lindsay (GL) theory (1972)

Green and Lindsay (1972) introduced a temperature-rate-dependent thermoelastic theory with two relaxation times and a single temperature field. It is distinct from two-temperature thermoelasticity. The original formulation retains Fourier heat flux, $q_i = -kT_{,i}$, with no flux-relaxation time. A separate rate-dependent two-temperature relation uses its own assumption $t_1 \geq t_2 > 0$ and likewise carries no flux-relaxation term; it is not part of the original single-temperature GL law (CONDITIONAL) Srivastava and Mukhopadhyay [2026].

4.4.1 Governing equations

$$\rho c_E(\dot{T} + \tau_0 \ddot{T}) = k\nabla^2 T - \beta T_0 \dot{u}_{i,i}, \tag{42}$$

The original Green–Lindsay 1972 energy equation has no $\tau_0 \ddot{u}_{i,i}$ term. The LS equation does carry τ_0 on the strain-rate coupling; applying the same operator to that term while retaining the GL stress law produces an LS/GL hybrid, not original GL. Later modified GL models may use such couplings; Yu, Xue, and Tian (2018) explicitly present a modified variant Yu et al. [2018]. The control-suggested Bera et al. item remains unidentified. The cited "Singh (2020), Symmetry 12:488" identifier resolves to a Saeed–Abbas–Marin paper, so that author attribution and its relevance to this variant are not asserted here pending the Crossref audit. The condition $\tau_1 \geq \tau_0 > 0$ is primary-source-checked from Green and Lindsay's entropy-production restriction Green and Lindsay [1972].

$$\sigma_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2\mu \varepsilon_{ij} - \beta(T + \tau_1 \dot{T})\delta_{ij}. \tag{43}$$

4.4.2 Thermal-rate coupling and admissibility

The second relaxation time enters the stress through the temperature rate; it does not introduce a second temperature field. Green–Lindsay theory is distinct from Chen–Gurtin two-temperature thermoelasticity. The displayed ordering applies to the original GL convention; it is not a claim that every later GL-labelled variant has the same admissible range.

4.5 Green–Naghdi type II theory (1993)

The type-II system treated here is Green and Naghdi's 1993 thermoelasticity without energy dissipation. Let α_{GN} be the thermal displacement, with $\dot{\alpha}_{\mathrm{GN}} = T$; it is distinct from the fractional order α.

4.5.1 Thermal displacement gradient formulation

For the source-free form, time integration with zero integration constant gives the source-checked relation Green and Naghdi [1993]:

$$\kappa^* \nabla^2 \alpha_{\mathrm{GN}} = \rho c_e \ddot{\alpha}_{\mathrm{GN}} + \beta T_0 \dot{u}_{i,i}. \tag{44}$$

The strain-rate coupling carries one fewer time derivative than the inherited form. Here κ^* is the thermal-displacement coefficient; below, κ denotes the coefficient on the temperature-rate term in GN-III. In the displayed isotropic convention, the candidate entropy flux uses the same scalar coefficient:

$$p_i = \kappa^* \alpha_{\mathrm{GN},i}, \tag{45}$$

where p_i is the entropy-flux vector. No anisotropic tensor generalization is asserted here.

4.5.2 Energy conservation

The 1993 type-II paper states that the displayed system permits harmonic-wave propagation without damping. This source-checked property is limited to that system and its stated coefficients; the schematic balance below is not a derived GN-II energy identity.

$$\frac{\partial \mathcal{E}}{\partial t} + \partial_i J_i = \mathcal{P}. \tag{46}$$

Wave attenuation and energy conservation for type II must be obtained from the specific model equations and boundary-value problem; no application suitability claim is made here. A separate rheological network derivation relates CV/GN/MCV/MGN flux laws and includes second-order coefficients proportional to τ^2; those coefficients do not establish energy ratios or admissibility for the displayed system (CONDITIONAL) Huang et al. [2025].

4.6 Green–Naghdi type III theory (1992)

Green and Naghdi's 1992 type-III system contains both the temperature-rate and thermal-displacement contributions. The displayed source-free isotropic equation uses κ for the rate term and κ^* for the non-dissipative term, consistently with Eq. (44) Green and Naghdi [1992].

4.6.1 Governing equations

$$\rho c_e \ddot{T} + \beta T_0 \ddot{u}_{i,i} = \kappa \nabla^2 \dot{T} + \kappa^* \nabla^2 T. \tag{47}$$

A distinct nonlocal GN-III formulation prints a flux law containing both the temperature-gradient and thermal-displacement-gradient contributions; it is not substituted into the local source-free form shown here (CONDITIONAL) Biswas [2020b]. This source-free form is conditional on the displayed linear isotropic system; the source term and material normalization must be restored for other conventions. The transcribed restrictions include $\kappa \geq 0$, $c \geq 0$, and $\kappa^{**} = 0$; no additional sign restriction on κ^* is inferred here.

Any proposed type-III-to-type-II or classical limit must be verified by substitution into the exact system and by a convergence analysis. No general limit theorem is asserted here.

4.7 Dual-phase-lag (DPL) theory (Tzou, 1995)

Tzou (1995) introduced two phase lags into Fourier's law:

$$q_i(t + \tau_q) = -kT_{,i}(t + \tau_T), \tag{48}$$

where τ_q is the heat flux phase lag and τ_T is the temperature gradient phase lag.

4.7.1 Taylor expansion

First-order Taylor expansion about t:

$$q_i + \tau_q \dot{q}_i = -k(T_{,i} + \tau_T \dot{T}_{,i}). \tag{49}$$

Some later DPL variants retain a $\tau_q^2/2$ term in a second-order flux expansion (Yang et al.; Gupta et al.); these are distinct model-specific truncations, not part of Eq. (49). No universal lag-order admissibility follows (CONDITIONAL) Yang et al. [2021], Gupta and Barak [2023]. Formal reductions can sometimes be obtained by setting one or both lag parameters to zero in a specified DPL equation, but the resulting equations and solution limits must be checked for that model. The ordering $\tau_q \geq \tau_T$ is not asserted here as a universal thermodynamic-admissibility condition; a model-specific free-energy/entropy derivation is required.

Guyer–Krumhansl (GK) phonon transport. Guyer and Krumhansl's 1966 kinetic treatment derives the nonlocal phonon-flux structure from the phonon Boltzmann equation, not from a ready-made continuum thermoelastic law. In the rapid-normal-scattering regime ($\Omega\tau_N \ll 1$), its characteristic length is $\lambda^2 = c^2\tau_N\tau_z/5$, where τ_z is the Ziman-averaged resistive-process relaxation time Guyer and Krumhansl [1966]. As a SECONDARY continuum statement, Tzou

(1995, Eq. 15) prints $\dot{\boldsymbol{q}} + (c^2 C_p/3)\nabla T + \boldsymbol{q}/\tau_R = (\tau_N c^2/5)[\nabla^2 \boldsymbol{q} + 2\nabla(\nabla \cdot \boldsymbol{q})]$; this particular reduction is not attributed to the 1966 paper. Tzou's Eq. 17 maps this phonon case to DPL parameters $\alpha = \tau_R c^2/3$, $\tau_T = 9\tau_N/5$, and $\tau_q = \tau_R$ Tzou [1995]. These continuum and mapping statements are conditional on Tzou's stated phonon approximation; they are not universal DPL identities.

4.8 Triple-phase-lag (TPL) theory (Roy Choudhuri, 2007)

Roy Choudhuri (2007) Choudhuri [2007] proposed the three-lag TPL law:

$$\begin{aligned} q_i(t+\tau_q) = &-kT_{,i}(t+\tau_T) \\ &- k^* v_{,i}(t+\tau_v), \qquad \dot{v} = T. \end{aligned} \tag{50}$$

The cited source gives a first-order expansion in the three lags with $0 \le \tau_v < \tau_T < \tau_q$. We use no $\tau_\nu/2$ term. Second-order DPL variants retain a $\tau_q^2/2$ term, but they are distinct from this first-order Roy Choudhuri law and establish no shared admissibility condition (CONDITIONAL) Yang et al. [2021], Gupta and Barak [2023]. Other published variants include second-order DPL/TPL truncations with a free coefficient or tensor conductivity Kumar et al. [2023b], Singh et al. [2019], Biswas [2019]; a higher-order two-temperature TPL model uses a temperature lag rather than τ_v Abouelregal et al. [2020]. Further formulations combine memory-kernel bounds Singh et al. [2020] or nonlocal operators on both stress and inertia Bayones et al. [2021]; related DPL/TPL model studies are also reported by Tibullo et al. Chiriţă et al. [2017]. These are distinct constitutive variants, not one universal TPL law; no common stability or admissibility condition is inferred (CONDITIONAL).

4.9 Moore–Gibson–Thompson (MGT) theory

Moore and Gibson (1960) derived a third-order acoustic wave equation for gas dynamics with relaxation; it is not a heat-conduction law for thermoelastic solids. Later Moore–Gibson–Thompson thermoelastic models are analogues that borrow this higher-order structure Moore and Gibson [1960], Quintanilla [2019]. A candidate constitutive system used in some later thermal formulations is

$$\tau \dot{q}_i + q_i = -kT_{,i} - k^* \vartheta_{m,i}, \qquad \dot{\vartheta}_m = T. \tag{51}$$

Here ϑ_m is a thermal-displacement variable and k^* must have units consistent with the selected model. This form is retained as an unverified source lead: its primary-source convention, assumptions, and admissibility are not checked here, and no eliminated third-order equation, dispersion law, or energy identity is asserted.

4.10 Memory-dependent derivative (MDD) models (Wang & Li, 2011)

MDD models replace the classical time derivative with an integral-form memory operator:

$$D_{\omega_M} f(t) = \frac{1}{\omega_M} \int_{t-\omega_M}^{t} \kappa(t-\xi) f'(\xi)\, d\xi, \tag{52}$$

where $\omega_M > 0$ is the finite horizon in this displayed candidate operator; the subscript distinguishes it from angular frequency elsewhere in the review. Its kernel bounds, normalization, regularity, and any admissible parameter set require verification for the exact formulation. A fading-memory condition on an unbounded lag variable cannot be inferred from this finite-horizon formula. A limit as $\omega_M \to 0$ may be considered only after kernel normalization and

regularity assumptions are specified; no classical-derivative limit or material calibration is asserted here. Li et al. use a $1/\omega$ normalization and report a degenerate-limit equivalence between their memory-dependent and fractional models; that result is formulation-specific and does not establish a limit for the candidate operator displayed here (CONDITIONAL) Li et al. [2019].

4.11 Fractional thermoelasticity

Fractional thermoelasticity generalizes the heat equation using fractional derivatives:

$$ {}^{C}D_t^{\alpha} T = a_{\alpha} \nabla^2 T, \qquad 0 < \alpha \leq 2, \tag{53} $$

where ${}^{C}D_t^{\alpha}$ is the Caputo fractional derivative of order α and a_{α} has dimensions $\mathrm{m}^2\,\mathrm{s}^{-\alpha}$ in this scalar equation. Thus a_1 is a thermal diffusivity, while $\sqrt{a_2}$ has dimensions of speed in the special case $\alpha = 2$. For $1 < \alpha \leq 2$ this is a fractional diffusion–wave equation; that terminology alone does not establish a thermodynamically admissible, causal, or well-posed coupled thermoelastic model. Initial data and a constitutive embedding must be specified for a physical application.

4.12 Comparative summary

Table 6 summarizes selected constitutive distinctions and the evidence limits for the model classes discussed here; it does not assert family-wide wave-speed, dissipation, or validity results.

Published examples include a hyperbolic two-temperature closure Prasad and Kumar [2020], fractional-order heat conduction with two temperatures Ezzat and El Karamany [2011], semiconductor applications of a hyperbolic two-temperature model Alzahrani and Abbas [2020], and a fractional two-temperature hybrid with relaxation times Bassiouny and R [2020]. These examples illustrate distinct constructions only; they do not establish shared admissibility or stability results (CONDITIONAL).

4.13 State variables and constitutive distinctions

The model classes surveyed in this section differ in their chosen state variables and constitutive operators. Relaxation models may promote heat flux to an evolving variable; two-temperature formulations distinguish temperature-like fields; thermal-displacement formulations introduce a variable whose time derivative is related to temperature; phase-lag models shift the arguments of constitutive quantities; hereditary and fractional models use history operators. These descriptions are schematic. Their thermodynamic interpretation, initial data, modal consequences, and limiting behavior must be derived from each model's exact equations. In particular, Green–Lindsay is not a two-temperature construction.

4.14 Thermodynamic admissibility as a selection principle

Thermodynamic admissibility must be established separately for each constitutive family by specifying its free energy, entropy, state variables, evolution laws, and admissible processes. The inherited universal dissipation claims and parameter exclusions have not been re-derived here. No parameter region is excluded, and no Green–Lindsay admissibility result is asserted in this revision without a primary-source derivation.

Thermodynamic restrictions must be derived from the free energy, entropy, heat-flux law, and internal variables of each chosen model. No single dissipation expression or parameter-ordering condition applies to all relaxation, two-temperature, phase-lag, memory-dependent, or fractional formulations. The family-wide assertions in the inherited text are not supported by derivations

Table 6: Selected generalized thermoelastic model classes. The table records broad constitutive distinctions; propagation, admissibility, and energy-flux claims require model-specific primary-source checks.

Model class	**Broad constitutive feature**	**Status in this revision**	**Source lead**
Classical coupled thermoelasticity	Fourier heat flux coupled to mechanical balance	Classical local flux derived; equation set not wholly re-audited	Biot [1956]
Lord–Shulman	Heat-flux relaxation	Standard form shown; primary-source equation check pending	Lord and Shulman [1967]
Green–Lindsay	Temperature-rate-dependent, single-temperature formulation with two relaxation parameters	Corrected description; full equations and admissibility unverified	Green and Lindsay [1972]
Green–Naghdi type II	Thermal displacement; nondissipative system as stated in the 1993 model	Displayed source-free Eq. (4.11); energy identity remains open	Green and Naghdi [1993]
Green–Naghdi type III	Coupled temperature-rate and thermal-displacement terms	Displayed Eq. (4.14); dissipation depends on coefficients and the selected system	Green and Naghdi [1992]
Dual/triple phase lag	Lagged constitutive formulations	Exact expansions and parameter restrictions unverified	Tzou [1995], Choudhuri [2007]
Moore–Gibson–Thompson-type models	Later thermal analogues of a third-order relaxing-gas acoustic equation; not Moore–Gibson 1960 heat conduction	Variant-specific equations and stability open	Moore and Gibson [1960], Quintanilla [2019]
Memory-dependent	Kernel-based/history-dependent heat transport	Operator and limiting behavior not full-text-checked	Wang and Li [2011]
Fractional scalar diffusion–wave example	Caputo time derivative in the displayed scalar equation; not by itself a coupled thermoelastic model	Order-dependent coefficient units stated; embedding, initial data, and admissibility open	Povstenko [2004, 2009]

No universal finite-speed, dissipation, stability, or well-posedness result is inferred from a family label. Generalized energy-flux identities remain OPEN unless individually derived from the selected model.

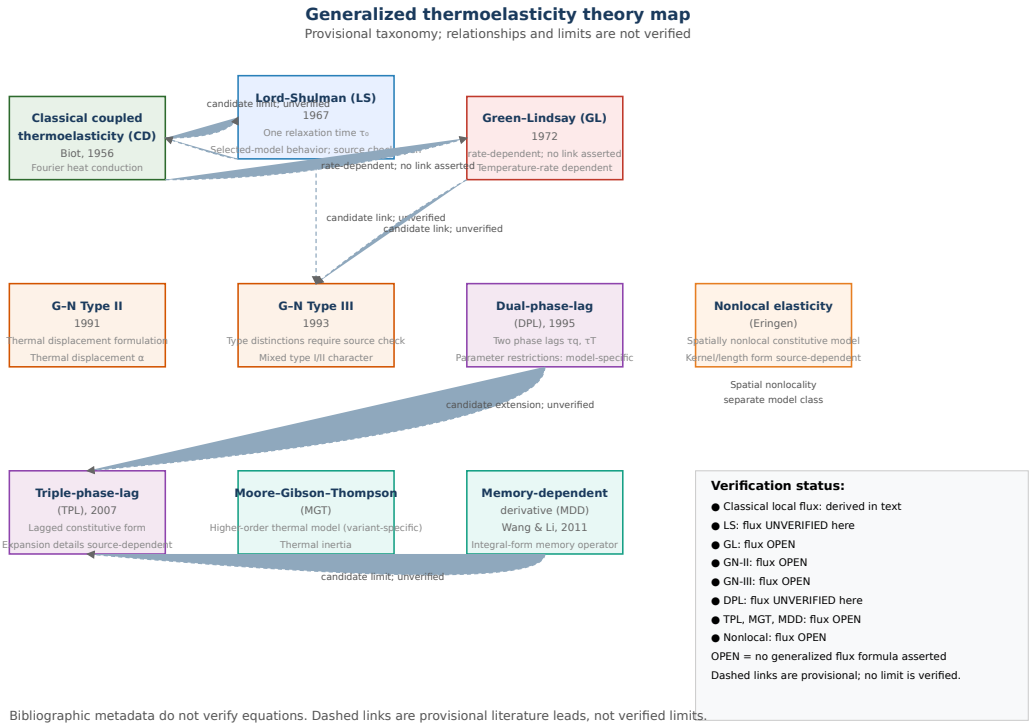


Figure 7: Schematic map of selected thermoelastic model families. Parameter limits and links are model-specific; no universal reduction or convergence result is asserted. Green–Lindsay is represented as a single-temperature, temperature-rate-dependent theory, distinct from two-temperature models.

in this revision and are withdrawn pending model-specific primary-source and thermodynamic checks.

Parameter restrictions are model-specific and must be included when a model is used. The limits and admissible ranges summarized in inherited tables are not treated as verified unless a derivation or full-text source check is identified.

4.15 Modal structure is model-specific

The modal structure must be obtained by substituting the chosen constitutive equations into the field equations and solving the resulting characteristic problem. A model label alone does not determine the number of propagating, evanescent, or attenuating roots, nor their physical admissibility. The inherited wave-speed, attenuation, and mode-count claims for the relaxation, two-temperature, thermal-displacement, phase-lag, memory-dependent, and fractional families have not been checked in this revision. Root classification and interface matching are therefore treated as problem-specific tasks in Sections 8 and 9; no universal count or crossover frequency is asserted here.

4.16 Notation summary for Section 4

Table 7 summarizes the notation introduced in this section.

Table 7: Notation introduced in Section 4.

Symbol	Meaning	Units	Equation
τ_0	LS relaxation time	s	(39)
τ_q, τ_T	DPL lag parameters in the displayed inherited form	s	(48)–(49)
c_T	Thermal wave speed	m/s	(41)
β	Thermal stress coefficient	Pa/K	(36)
c_v	Specific heat at constant strain	J/(kg·K)	(37)
k	Thermal conductivity	W/(m·K)	(37)
α_{GN}	Thermal displacement (GN-II; distinct from fractional order α)	K·s	(44)
κ, κ^*	GN-III rate and thermal-displacement coefficients	model-dependent	(44), (47)
p_i	Entropy flux (GN-II)	W/(m^2·K)	(45)
D_{ω_M}	MDD operator	1/s	(52)
ω_M	MDD finite time horizon	s	(52)
$\kappa(t-\xi)$	MDD kernel	dimensionless	(52)
${}^{C}D_t^{\alpha}$	Caputo fractional derivative	$t^{-\alpha}$	(53)
α	Fractional derivative order	dimensionless	(53)
a_α	Coefficient in displayed scalar fractional equation	m^2 s$^{-\alpha}$	(53)

4.17 Open questions for Section 4

OQ4.1 What is the complete time-domain energy flux for LS theory, including the contribution from the flux-relaxation term?

OQ4.2 What heat-flux law and energy balance follow from the original single-temperature Green–Lindsay relaxation-time convention?

OQ4.3 What is the relationship between energy flux and entropy flux in GN-II theory?

OQ4.4 What is the mixed energy flux for GN-III theory?

OQ4.5 How do different MDD kernel choices affect the energy flux and wave propagation?

OQ4.6 Can rigorous limit theorems with convergence proofs be established between theory pairs?

5 Coupled Multiphysics Continua

This section surveys coupled continua: piezo-thermoelastic, porous and double-porous, micropolar, semiconductor photo-thermoelastic, and magneto-thermoelastic media. It outlines their governing equations, constitutive relations, coupling mechanisms, and applications; primary sources are cited where available.

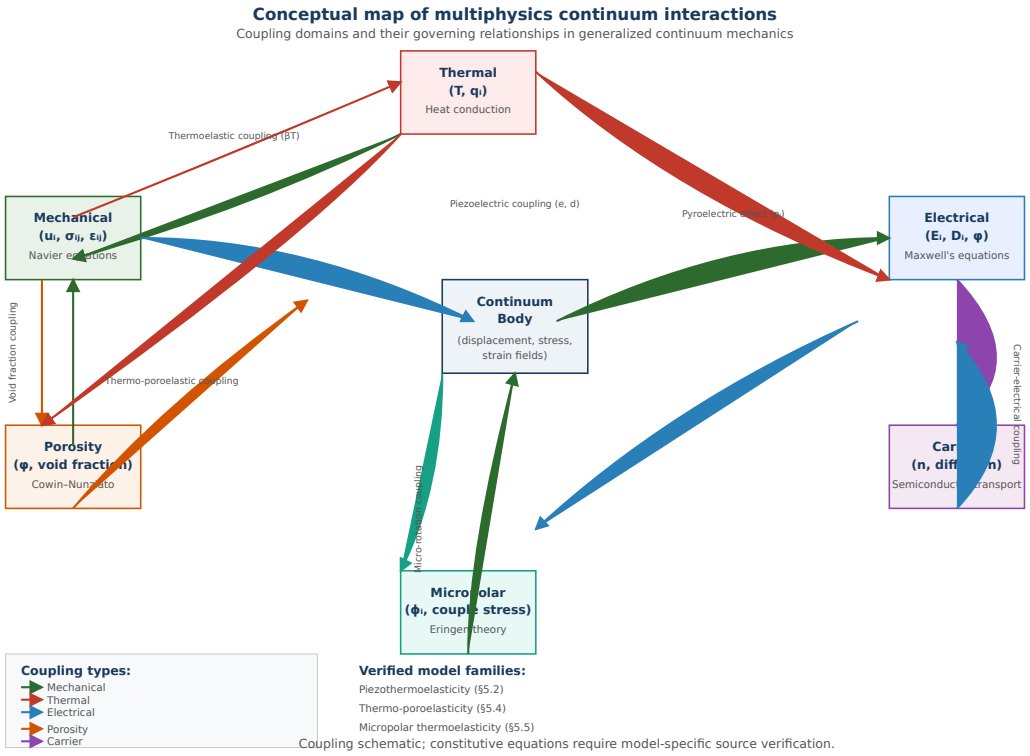


Figure 8: Coupling topology of multiphysics continua. The mechanical field is coupled to a thermal field through the thermoelastic dilatation term, to an electrical field through the piezoelectric tensors, to a porosity field through the void fraction, to a carrier field through the semiconductor transport equations, and to a micro-rotation field through the couple-stress relation. Each coupling is a separate term in a separate equation, and each introduces its own material coefficients and its own additional boundary conditions at an interface; this is the reason that the size of the interface system grows so rapidly as couplings are added (Section 9).

5.1 Piezoelectricity: constitutive relations and coupling coefficients

Piezoelectricity describes the coupling between mechanical and electrical fields in certain crystalline materials (e.g., quartz, barium titanate, lead zirconate titanate). The linear constitutive relations are:

$$\sigma_{ij} = c_{ijkl}\varepsilon_{kl} - e_{kij}E_k, \tag{54}$$

$$D_i = e_{ikl}\varepsilon_{kl} + \kappa_{ij}E_j, \tag{55}$$

$$E_i = -\phi_{,i}, \tag{56}$$

where σ_{ij} is the Cauchy stress tensor, c_{ijkl} is the elastic stiffness tensor, ε_{kl} is the infinitesimal strain tensor, e_{kij} is the piezoelectric stress coefficient, E_k is the electric field, D_i is the electric displacement, κ_{ij} is the dielectric permittivity, and ϕ is the electric potential.

5.1.1 Charge conservation

In the absence of free electric charge:

$$D_{i,i} = 0. \tag{57}$$

5.1.2 Electrical energy density

The electrical energy density is:

$$\mathcal{E}_{\text{elec}} = \frac{1}{2}D_iE_i. \tag{58}$$

5.2 Piezothermoelasticity: three-field coupling

Piezothermoelasticity couples mechanical, thermal, and electrical fields. The constitutive relations are:

$$\sigma_{ij} = c_{ijkl}\varepsilon_{kl} - e_{kij}E_k - \beta_{ij}T, \tag{59}$$

$$\sigma_{ij,j} = \rho\ddot{u}_i, \tag{60}$$

$$E_i = -\phi_{,i}, \tag{61}$$

$$D_i = \kappa_{ij}E_j + e_{ikl}\varepsilon_{kl} + p_iT, \tag{62}$$

$$D_{i,i} = 0, \tag{63}$$

where β_{ij} is the thermal stress coefficient and p_i is the pyroelectric coefficient.

5.2.1 Governing equations for coupled fields

The coupled system consists of the equation of motion (60), the heat conduction equation (theory-dependent: classical, LS, GL, GN, DPL, TPL, MDD, etc.), and the charge conservation equation (63). All three fields are coupled through the constitutive relations (59) and (62): temperature changes induce thermal stresses and pyroelectric effects; mechanical deformation generates piezoelectric polarization; electric fields produce electrostrictive deformation.

5.2.2 Modal content of a piezoelectric solid

The coupling of Section 5.2 has an immediate and testable consequence: the number of propagating modes increases. A piezoelectric solid admits a quasi-longitudinal mode qP, in which the mechanical polarization is essentially parallel to the propagation direction but is accompanied by a longitudinal electric field; a quasi-transverse mode qS, whose polarization is essentially transverse but which is accompanied by a transverse electric field whose amplitude is small because the dielectric stiffness dominates; and a quasi-electrostatic potential solution. Under the assumptions used here, $E_i = -\phi_{,i}$ and $D_{i,i} = 0$, Maxwell's displacement-current equation is absent; this solution is not an electromagnetic wave and no speed-of-light mode is implied. A propagating electromagnetic branch requires the full Maxwell system and a stated frequency regime.

5.2.3 The coupling coefficient as a wave-speed diagnostic

Because the electrical field and the mechanical field are both excited by the strain, the effective elastic stiffness of a piezoelectric solid along a given direction is not c_{ijkl} but a stiffness that has been renormalised by the dielectric stiffness, through the electromechanical factor

$$\bar{c}_{ijkl} = c_{ijkl} + e_{aij}(\kappa^{-1})_{ab}e_{bkl}, \tag{64}$$

which is the short-circuit/open-circuit stiffness correction after eliminating the quasistatic field under a specified electrical constraint. Its sign is positive for positive-definite permittivity with the stated tensor contraction. The effective modulus depends on the electrical boundary condition; it is not a universal scalar correction.

5.3 Electrical boundary conditions

5.3.1 Perfect electrical contact

$$\phi^{(1)} = \phi^{(2)} \quad \text{at the interface.} \tag{65}$$

5.3.2 Insulated interface

$$D_i n_i = 0 \quad \text{at the interface.} \tag{66}$$

5.3.3 Electric displacement continuity

$$D_3^{(1)} = D_3^{(2)} \quad \text{at the interface.} \tag{67}$$

5.3.4 Why a void needs its own field of freedom

The kinematics of a porous body require a kinematic variable beyond the displacement, because the material inside the pores can change its volume independently of the solid matrix. This is not a modelling convenience; it is forced by the kinematics of a body whose constituents occupy a variable volume fraction. The void volume fraction ϕ measures the change of that fraction from its reference value, and it enters the balance laws of Section 5.4 as a field in its own right, with its own inertia term, its own equilibrated stress and its own body force. The system is hyperelastic in the sense that the potentials are functions of ε_{ij}, γ_{mn} and ϕ jointly, so that the cross-coupling terms d_{ijk} are constrained by the requirement of a positive-definite quadratic form rather than being free parameters.

5.3.5 Modal content of a void-bearing medium

The presence of the void field adds a further mode to the modal structure: a fast wave associated with the inertia of the solid matrix and the coupled fluid motion, a slow wave associated with the equilibration of the void fraction against the surrounding matrix, and the usual shear waves, which are unaffected by the void field to the lowest order because shear does not change the volume fraction. The slow wave is of practical interest because it is the wave that transports the equilibration of the void pressure, and its speed is controlled by the ratio of the equilibrated inertia to the coupling stiffness. In a thermoporous medium the interaction with the thermal field adds a further degree of freedom, and the resulting modal structure is the one tabulated in Table 8.

5.4 Porous media: voids and double porosity

Porous elastic materials contain vacuous voids or pores distributed throughout the matrix. The Nunziato–Cowin theory (Nunziato and Cowin, 1979 Nunziato and Cowin [1979]; Cowin and Nunziato, 1983 Cowin and Nunziato [1983]) introduces an additional kinematic variable: the void volume fraction η.

5.4.1 Governing equations

The basic equations for thermoelastic materials with voids:

$$\sigma_{ij} = A_{ijmn}\varepsilon_{mn} + B_{ijmn}\gamma_{mn} + d_{ijk}\eta_{,k} - \alpha_{ij}\theta, \tag{68}$$

$$\rho\ddot{u}_i = \sigma_{ji,j} + \rho f_i, \tag{69}$$

$$\rho\ddot{\eta} = h_{i,i} + \rho l, \tag{70}$$

where η is the void volume fraction, h_i is the equilibrated stress vector, l is the equilibrated body force, A_{ijmn} and B_{ijmn} are elastic and equilibrated elasticity tensors, d_{ijk} is the equilibrated stress coupling tensor, and α_{ij} is the thermal coupling tensor. The thermoelastic extension of void theory was developed by Iesan (1986 Ieşan [1986]) and is treated systematically, together with the micropolar and nonsimple variants, in his monograph Ieşan [2013], which also establishes the existence and uniqueness framework for the coupled problems.

5.4.2 Double porosity

Double porosity models introduce two porosity fields to capture both macro- and micro-porosity effects. The governing equations extend the single-porosity theory with two coupled void volume fractions η_1 and η_2, each with its own equilibrated stress and equilibrated body force. Double porosity models are particularly relevant for hydrocarbon reservoirs, fissured rocks, and aquifers where both fracture and matrix porosity contribute to fluid flow and wave propagation.

5.5 Micropolar and micromorphic continua

Micropolar elasticity (Eringen, 1966 Eringen [1966]) extends classical elasticity by incorporating micro-rotational degrees of freedom and couple stresses. In addition to the displacement field u_i, the microrotation field ϕ_i describes independent rotation of material particles.

5.5.1 Governing equations

$$\sigma_{ij} = \lambda u_{k,k}\delta_{ij} + \mu(u_{i,j} + u_{j,i}) + \kappa(u_{i,j} - u_{j,i} - 2\epsilon_{ijk}\phi_k) - \beta\theta\delta_{ij}, \tag{71}$$

Here ϕ_k is the independent microrotation vector; λ, μ, and κ are elastic moduli, and β is the isotropic thermal-stress coefficient. The displayed isotropic linear law is an illustrative specialization, not independently checked here against the cited primary texts Eringen [1966], Tauchert et al. [1968]; the mechanical micropolar form and its added isotropic thermal term should be verified for the selected thermoelastic model. Anisotropic constitutive tensors and couple-stress relations are formulation-dependent and remain outside this display.

Micropolar models are relevant for granular materials, liquid crystals, composite materials with oriented inclusions, and biological tissues where micro-structural rotation cannot be neglected. A thermoelastic micropolar formulation was published by Tauchert, Claus and Ariman (1968 Tauchert et al. [1968]); earlier related work exists, so no priority claim is made here.

5.5.2 The absorption term and the photothermal source

Illumination does not act on the mechanical field directly; it acts through the heat and the carriers that it generates. The absorbed optical power, attenuated by the Beer–Lambert law, appears as a volumetric source in the thermal equation and as a generation term in the carrier continuity equations. The chain of consequences is then fixed: the thermal source produces a temperature rise; the temperature rise produces a thermoelastic stress through the dilatation term and a pyroelectric polarisation through the pyroelectric coefficient; the stress produces a displacement, and the polarisation produces an electric field, which in turn modifies the carrier distribution. The system is thus closed through the constitutive relations rather than through any additional assumption, and the number of degrees of freedom that must be solved simultaneously is four.

5.5.3 Photoacoustic waves as a diagnostic of the coupling

The photoacoustic wave that follows pulsed absorption is the most direct macroscopic observable of the coupling, and it has a distinctive structure: it contains a thermal contribution that propagates at the sound speed, an electronic contribution that propagates at the speed of light but decays on the absorption length, and a thermoelastic contribution whose amplitude depends on the piezoelectric coupling. The relative weight of these contributions depends on the regime — diffusion-dominated at low frequency, ballistic at high frequency — and it is the crossover between them that makes the technique sensitive to the thermal model. This is a concrete instance of the general point developed in Section 13: a generalized theory earns its place

not by being more general, but by being required in a regime where the classical description demonstrably fails.

5.6 Semiconductor photo-thermo-elasticity

Semiconductor photo-thermo-elasticity couples mechanical deformation, thermal fields, electrical fields, and carrier density (electrons and holes). When a semiconductor is illuminated, absorbed photons generate electron–hole pairs, producing carrier diffusion, temperature changes, and thermoelastic stresses.

5.6.1 Governing equations

The coupled system includes:

- The equation of motion with piezoelectric and thermal stress terms,
- The heat conduction equation with carrier recombination heat source,
- The charge conservation equation with carrier generation and recombination,
- The carrier continuity equations for electrons and holes.

Applications include photoacoustic imaging, semiconductor device thermal management, and optoelectronic sensors.

5.7 Magneto-thermo-elasticity

Magneto-thermo-elasticity couples mechanical, thermal, and magnetic fields. In electrically conducting media (e.g., liquid metals, plasmas), magnetic fields induce Lorentz forces that couple with thermal stresses.

5.7.1 Governing equations

The equation of motion includes the Lorentz force term:

$$\rho \ddot{u}_i = \sigma_{ji,j} + \rho f_i + (\boldsymbol{J} \times \boldsymbol{B})_i, \tag{72}$$

where $\boldsymbol{J}$ is the current density and $\boldsymbol{B}$ is the magnetic flux density.

Applications include magnetohydrodynamic (MHD) systems, liquid metal cooling, geophysical dynamo theory, and electromagnetic forming.

5.8 Comparative summary

Table 8 summarizes the coupled multiphysics media: fields, coupling mechanisms, constitutive structure, and applications.

5.9 The coupling hierarchy and its computational consequences

The number and character of modes and interface conditions depend on the complete field equations, constitutive parameters, geometry, and boundary conditions. An added field or internal variable need not contribute an independent propagating mode, and the matrix size is not fixed by a theory-family label. The inherited mode-count statements and interface-system dimensions have not been recomputed in this revision and are not asserted as general results.

Table 8: Illustrative coupled-field system categories from the inherited review. Listed couplings and equations are not independently checked in this revision.

System	Coupled Fields	Governing Equations	Constitutive Coupling	Primary Reference
Classical thermoelasticity	Mechanical + Thermal	Navier–Fourier system	Thermal stress $\beta T\delta_{ij}$; heat flux $q_i = -kT_{,i}$	Biot [1956]
Piezo-thermoelasticity	Mechanical + Thermal + Electrical	Coupled piezo–thermal equations	Piezoelectric tensor e_{kij}; pyroelectric coefficient p_i	Barak et al. [2023a]
Thermo-poroelasticity	Mechanical + Thermal + Porosity	Nunziato–Cowin void theory	Void fraction ϕ; thermal void coupling	Nunziato and Cowin [1979], Cowin and Nunziato [1983]
Thermoelasticity with voids	Mechanical + Thermal + Voids	Iesan's extension of void theory	Void volume fraction; thermo-void coupling	Ieşan [1986]
Micropolar thermoelasticity	Mechanical + Thermal + Microrotation	Coupled linear momentum + angular momentum	Couple stress tensor; micro-inertia j	Tauchert et al. [1968]
Semiconductor thermoelectric	Mechanical + Thermal + Carrier + Electrical	Carrier diffusion + heat equation	Carrier concentration n; electrical–thermal coupling	Model-specific

Notes: Coupling coefficients are material-specific and must be determined experimentally for each material system.
Cited papers are source leads only; constitutive equations and coupling coefficients require full-text checking.

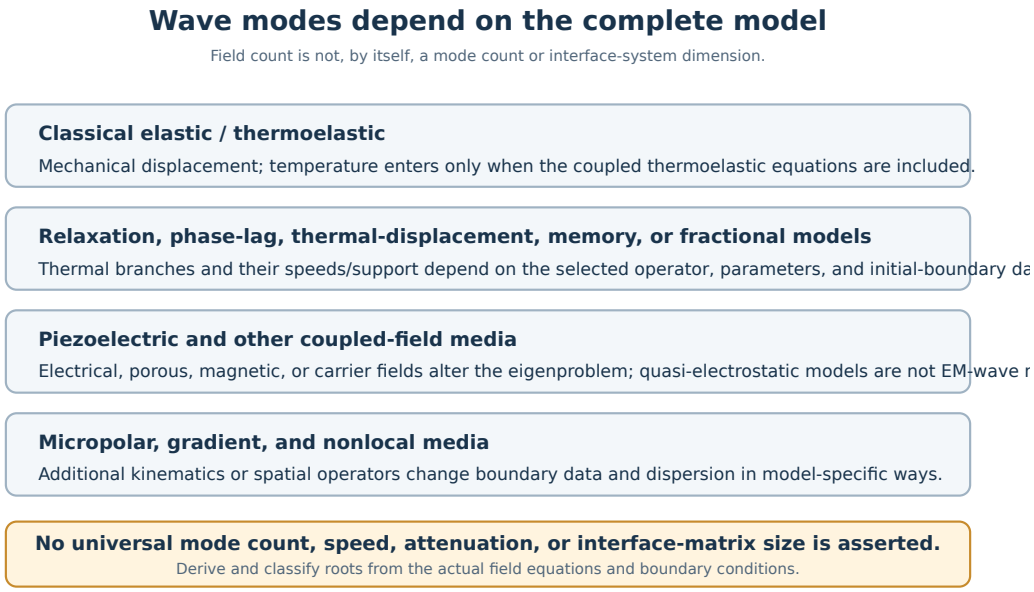


Figure 9: Schematic overview of field types in continuum wave models. Modal content, propagation, and interface-system dimension depend on the selected constitutive equations and boundary conditions; no universal mode count is asserted.

The secular equation and interface system must be derived from each model's actual field equations and boundary conditions. Their polynomial order and matrix size cannot be inferred from the number of named fields alone. The inherited mode counts and interface-energy studies cited elsewhere have not been independently reproduced or checked against full texts in this revision; no comparative assessment of their importance or verification status is made here.

Coupled-field models require model-specific checks of the free-energy Hessian, constitutive stability, and initial-boundary-value problem. These properties cannot be inferred from the presence of a coupling coefficient alone. No comparison of how thoroughly different coupled families have been tested, or general existence/uniqueness claim, is made here without a dedicated primary-source audit.

5.10 Notation summary for Section 5

Table 9: Notation introduced in Section 5.

Symbol	Meaning	Units	Equation
c_{ijkl}	Elastic stiffness tensor	Pa	(54)
e_{kij}	Piezoelectric stress coefficient	C/m^2	(54)
κ_{ij}	Dielectric permittivity	F/m	(55)
E_i	Electric field	V/m	(56)
D_i	Electric displacement	C/m^2	(55)
ϕ	Electric potential	V	(56)
β_{ij}	Thermal stress coefficient	Pa/K	(59)
p_i	Pyroelectric coefficient	C/(m^2·K)	(62)
$\bar{c}_{ijkl}$	Piezoelectrically renormalised stiffness	Pa	(64)
η	Void volume fraction	dimensionless	(68)
h_i	Equilibrated stress vector	N/m	(70)

R3b source-specific evidence. A piezothermoelastic interface case reports acoustic flux with an exact balance that includes an interaction term. (PRIMARY-SOURCE-CHECKED) Gupta et al. [2022]. A piezo/thermoelastic interface study lists model-specific degenerations when phase-lag and memory coefficients are switched off. (PRIMARY-SOURCE-CHECKED) Gupta et al. [2023]. A micropolar/porous/semiconductor interface study reports an approximately-unity energy-ratio sum as a computational check, not an exact identity. (PRIMARY-SOURCE-CHECKED) Kumar et al. [2023a].

5.11 Open questions for Section 5

OQ5.1 What are the complete constitutive relations for piezothermoelastic media with nonlocal elasticity?

OQ5.2 How do double-porosity models capture scale-dependent fluid–solid interaction?

OQ5.3 What are the energy flux expressions for micropolar media with couple stresses?

OQ5.4 How can semiconductor photo-thermo-elastic models be validated experimentally?

6 Generalized Transport and Memory Effects

Classical Fourier conduction relates heat flux instantaneously to the temperature gradient. Its range of adequacy and any departure from it depend on material, scale, and experiment; no universal failure threshold is asserted here. This section surveys alternative constitutive structures and identifies questions that require model-specific derivation or source checking.

6.1 Physical origin of thermal relaxation

Microscopic heat carriers and their scattering can motivate finite-rate transport models, but a continuum relaxation parameter is phenomenological unless it is derived or identified from a specified kinetic model. An exponential relaxation law is one possible constitutive choice, not a universal consequence of carrier dynamics. The memory kernel, relaxation parameters, and resulting wave/support properties must be established for the particular model and material. This section surveys such operator choices while marking family-wide physical interpretations as conditional or unverified.

6.2 Three structural routes to memory

Three mathematically distinct ways of making a constitutive law remember have been used, and they are not interchangeable.

- **Differential-rate constitutive laws.** Some models include time derivatives of fluxes, temperatures, or other fields; the order and choice of differentiated variable are model-specific. Whether such a law admits an equivalent finite-dimensional internal-variable representation must be established for the particular formulation.
- **Integral (convolution) memory.** The constitutive relation is written as a convolution of the driving quantity with a memory kernel over a finite or semi-infinite interval. This includes the hereditary-integral formulation of Section 6 and the memory-dependent derivative of Wang and Li . The memory is then encapsulated in an arbitrary function, and the price is that the constitutive relation is no longer local in time, so that the discretisation must carry the history.
- **Spatial nonlocality.** Some constitutive laws average over a material neighbourhood, as in Eringen-type formulations. This is a spatial interaction, not temporal memory, and its operators and boundary conditions require separate treatment in Section 7.

6.3 A transform-domain example, not a universal representation

Some linear flux laws can be written schematically as

$$\mathcal{D}[\boldsymbol{q}] = -k\,\boldsymbol{\nabla} T, \tag{73}$$

but this notation does not make distinct constitutive models equivalent.

For the Lord–Shulman/Cattaneo law, with zero initial heat flux,

$$\boldsymbol{q} + \tau_0 \dot{\boldsymbol{q}} = -k \nabla T.$$

The corresponding Laplace-domain multiplier is

$$\mathcal{D}_{\mathrm{LS}}(s) = 1 + \tau_0 s. \tag{74}$$

Nonzero initial flux contributes an additional initial-data term. Fractional, phase-lag, and hereditary models have their own transform symbols and initial terms, which must be derived from their exact constitutive equations; no universal symbol or shared limiting procedure is asserted here.

A comparison in the transform domain is useful only after each constitutive law, initial condition, and admissible parameter range is stated. Formal limits such as $\tau_0 \to 0$, fractional order tending to an integer, or memory horizon tending to zero require model-specific normalization and convergence arguments. Positivity of a frequency-domain loss modulus, when defined, is not by itself proof of a consistent free energy or a well-posed initial-boundary-value problem. These questions remain model-specific and are not settled uniformly in this review.

6.4 Fractional operators: Caputo and Riemann–Liouville

Fractional derivatives generalise integer-order differentiation to non-integer orders and are used in models with history dependence. Depending on the operator, coupling, kernel, and initial-boundary conditions, they can produce subdiffusive, dispersive, or wave-like responses. A finite propagation speed or a specific support property is not implied by the fractional order alone.

6.4.1 Caputo fractional derivative

The Caputo derivative of order α ($n-1 < \alpha < n$, $n = \lceil \alpha \rceil$):

$$ {}^{C}D_t^{\alpha} f(t) = \frac{1}{\Gamma(n-\alpha)} \int_0^t \frac{f^{(n)}(\tau)}{(t-\tau)^{\alpha+1-n}}\, d\tau. \tag{75} $$

6.4.2 Riemann–Liouville fractional derivative

$$ {}^{RL}D_t^{\alpha} f(t) = \frac{1}{\Gamma(n-\alpha)} \frac{d^n}{dt^n} \int_0^t \frac{f(\tau)}{(t-\tau)^{\alpha+1-n}}\, d\tau. \tag{76} $$

For suitable orders and regularity, the Caputo definition uses integer-order initial data, while Riemann–Liouville formulations may involve fractional initial data. This difference can matter in model selection, but it does not establish a universal preference. The historical motivations and material interpretations attributed to fractional operators require claim-level source checking; the citations here are source leads, not independently verified support for those claims.

6.4.3 Fractional heat conduction and its limits

$$ {}^{C}D_t^{\alpha} T = a_{\alpha} \nabla^2 T, \qquad 0 < \alpha \leq 2, \tag{77} $$

Here a_α is a model coefficient with dimensions consistent with the selected order (for length and time units, $[a_\alpha] = L^2 T^{-\alpha}$). This scalar equation is illustrative, not a universal fractional thermoelastic law. At the integer-order endpoints, the corresponding scalar PDE has heat-equation or wave-equation form, with coefficients and initial data specified for each model; one cannot infer a common dimensional parameter or solution limit across orders. Intermediate-order propagation and well-posedness depend on the operator definition, coupling, domain, and data. The cited Povstenko papers are relevant source leads for fractional thermoelasticity, but no theorem or material interpretation from them has been checked in full text in this revision.

6.5 Hereditary integrals: convolution kernels

One common representation of linear temporal memory is a hereditary integral:

$$f(t) = \int_{-\infty}^{t} K(t-\xi)\, g(\xi)\, d\xi, \tag{78}$$

where $K(t-\xi)$ is a kernel in this schematic convolution. Some models impose fading-memory, positivity, monotonicity, or complete-monotonicity hypotheses, but the required conditions depend on the constitutive law and the chosen thermodynamic framework; none of these properties alone is asserted here as a universal admissibility criterion. For suitable completely monotone kernels, representation results by weighted exponentials may apply, but their hypotheses and implications must be checked for the kernel at hand. No equivalence with a discrete phase-lag hierarchy is claimed.

6.6 Memory-dependent derivative (MDD): operator and kernel

An operator attributed to Wang and Li uses a kernel over a finite interval. The exact definition and its relation to other memory models remain subject to primary-source checking:

$$D_\omega f(t) = \frac{1}{\omega} \int_{t-\omega}^{t} \kappa(t-\xi) f'(\xi)\, d\xi, \tag{79}$$

where $\omega > 0$ denotes the displayed horizon and κ is the kernel. The bounds, normalization, regularity, and admissibility conditions on κ must be taken from the exact model definition. A finite integration horizon truncates the history in this formula; it does not by itself imply a finite-dimensional state-space realization, finite computation cost, or that the kernel is a freely chosen rather than material-calibrated quantity.

6.6.1 MDD kernel function

A commonly used kernel is

$$\kappa(t-\xi) = 1 - \frac{2\beta}{\omega}(t-\xi) + \frac{\alpha^2}{\omega^2}(t-\xi)^2, \tag{80}$$

where α and β are parameters in this inherited candidate kernel. Its dimensions, bounds, normalization, and any admissible parameter region require checking against the defining source and the selected thermodynamic model; positivity or monotonicity alone is not taken as a proof of admissibility.

6.6.2 Limiting behavior

A formal short-horizon limit $\omega \to 0$ is often used to recover an ordinary derivative from a normalized finite-horizon operator. The displayed kernel normalization and regularity assumptions must be specified before this limit is asserted; it is retained as a candidate limiting relation and is not used as a proven model equivalence. No exact exponential-kernel reduction to Lord–Shulman, or general equivalence with a fractional operator, is asserted here.

6.7 Energy flux in memory-dependent media

A candidate outward-positive local flux for a classical mechanical balance with a specified heat-flux vector is

$$J_i(t) = q_i(t) - \sigma_{ij}(t) v_j(t). \tag{81}$$

This is not asserted as a complete total-energy flux for every memory-dependent theory. The MDD heat-flux law and the appropriate stored-energy/boundary-power terms must be derived together from the selected kernel and free energy. The subsequent equations are inherited candidate constitutive forms, not independently source-checked here:

$$q_i(t) + \tau_0 D_\omega q_i(t) = -kT_{,i}(t), \tag{82}$$

$$q_i(t) + \frac{\tau_0}{\omega}\int_{t-\omega}^{t} \kappa(t-\xi)\dot{q}_i(\xi)\,d\xi = -kT_{,i}(t). \tag{83}$$

No universal hereditary correction to the energy flux, or generalized local balance, is asserted. Status of Eqs. (82)–(83) and the corresponding MDD free-energy balance: UNVERIFIED / OPEN pending primary-source and derivation audit.

6.8 Well-posedness and propagation in memory-dependent models

Well-posedness and propagation depend on the exact memory operator, kernel, time horizon, initial data, domain, and boundary conditions. Positivity or monotonicity of a kernel alone is not a general sufficient proof of a well-posed initial-boundary-value problem. Likewise, a finite-horizon derivative does not by itself imply finite propagation speed. The inherited claims about universal admissibility restrictions, a Banach-space existence theorem, and finite thermal-wave speed have not been independently verified and are not asserted as results in this revision. A model-specific energy estimate and existence/uniqueness analysis are OPEN.

6.9 Comparative summary: transport models

Table 10 summarizes transport and memory models across mathematical structure, dissipation, well-posedness, and energy flux status.

6.10 Notation summary for Section 6

R3b source-specific evidence. A nonlocal Kelvin–Voigt piezo-semiconductor application uses a memory-dependent heat flux; conclusions remain model-specific. (PRIMARY-SOURCE-CHECKED) Gupta et al. [2026]. A four-lag Taylor expansion is reported, but its printed gradient-field argument is defective and is not copied here. (CONDITIONAL) Abouelregal [2020]. A memory-dependent TPL formulation compares candidate kernels and states a model-specific positive-semidefiniteness condition. (CONDITIONAL) Sur et al. [2019]. A revised MDD definition normalizes by the kernel integral over the horizon rather than by the horizon alone. (PRIMARY-SOURCE-CHECKED) Wang and Li [2021]. A model-specific MGT-type flux law is paired with an Nth-order memory expansion using factorial coefficients. (CONDITIONAL) Sur et al. [2025]. A nonlocal operator is applied to a fractional Cattaneo law; its printed zero-order branch is not used as a general limiting identity. (CONDITIONAL) Bachher and Sarkar [2018]. A tensor positive-semidefiniteness compatibility condition is printed for a particular second-order TPL truncation. (CONDITIONAL) Tiwari and Kumar [2021].

6.11 Open questions for Section 6

OQ6.1 What is the complete time-domain energy flux for MDD media with convolution-type memory, expressed solely in terms of the temperature and the stored internal variables?

Table 10: Provisional transport-family comparison. Entries for generalized models are audit questions, not verified admissibility or well-posedness results.

Model class	**Transport structure**	**Energy/admissibility status here**	**Evidence status**
Classical Fourier	Local Fourier heat flux	Classical local balance derived; Fourier heat equation parabolic	DERIVED / standard model
Lord–Shulman	Relaxing heat flux	Model-specific free energy, dissipation and initial data not checked	OPEN
Green–Lindsay	Temperature-rate-dependent formulation	Single-temperature description; source-level equations and admissibility not checked	UNVERIFIED
Phase-lag (dual/triple)	Lagged constitutive laws	Restrictions and stability depend on selected truncation and parameter set	OPEN
Green–Naghdi II/III	Thermal-displacement formulations	Family-specific energy and dissipation identities not checked	OPEN
Moore–Gibson–Thompson	Higher-order heat equation variants	Stability and admissibility not checked here	OPEN
Memory-dependent	Kernel/time-history operator	Kernel-specific balance and well-posedness not checked	OPEN
Fractional	Fractional time operator	Propagation and admissibility depend on operator and data space	OPEN
Nonlocal / gradient	Spatially nonlocal or higher-gradient law	Boundary power and domain-specific balance not derived	OPEN

Table 11: Notation introduced in Section 6.

Symbol	**Meaning**	**Units**	**Equation**
$\mathcal{D}[\cdot]$	Generic memory operator acting on the flux	—	(73)
$\mathcal{D}_{\mathrm{LS}}(s)$	LS transform multiplier (zero initial flux)	dimensionless	(74)
${}^{C}D_t^{\alpha}$	Caputo fractional derivative	$t^{-\alpha}$	(75)
${}^{RL}D_t^{\alpha}$	Riemann–Liouville derivative	$t^{-\alpha}$	(76)
a	Thermal diffusivity	m^2/s	(77)
$K(t-\xi)$	Memory kernel	varies	(78)
D_ω	MDD operator	$1/\mathrm{s}$	(79)
ω	MDD time delay (horizon)	s	(79)
$\kappa(t-\xi)$	MDD kernel	dimensionless	(80)
α, β	MDD kernel parameters	varies	(80)
$\tilde{q}_i(s)$	Laplace transform of the heat flux	—	(74)

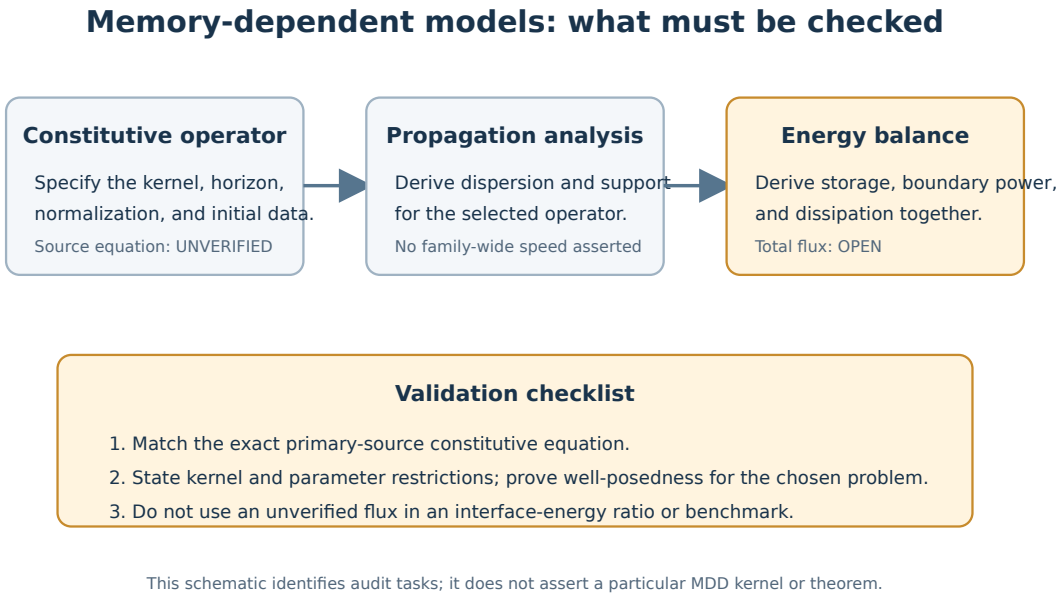


Figure 10: Schematic of a finite-horizon memory-dependent operator and audit questions concerning its kernel, limiting behavior, energy balance, and well-posedness. Model-specific propagation and flux consequences are not asserted; the displayed operator remains subject to primary-source verification.

OQ6.2 How is energy partitioned between storage and dissipation in hereditary media, and what is the correct effective flux that enters a local balance?

OQ6.3 What are the well-posedness conditions for fractional-order thermoelasticity with arbitrary α, in the coupled mechanical setting rather than in isolation?

OQ6.4 How do different MDD kernel choices affect the energy flux and the wave propagation, and can the dispersion of the thermal branch be used to identify the kernel experimentally?

OQ6.5 Can the memory operator be given a finite-dimensional state-space realisation for a general admissible kernel, and what is the numerical cost of that realisation relative to a direct quadrature?

7 Nonlocal and Scale-Dependent Continuum Models

Many classical local constitutive models relate stress to strain at the same material point and contain no explicit internal length. Nonlocal, gradient, couple-stress, and micropolar models introduce spatial interactions or additional kinematics to represent possible size effects. Their use is motivated by particular materials and experiments; evidence of a characteristic microstructural scale does not by itself show that a classical local law is invalid in every small-scale regime. The applicable constitutive model and its length parameter require independent justification.

This section develops that class. The organising question is not which of the competing formulations is correct, but *which physical mechanism each one encodes*, because the mechanisms predict measurably different dispersion and therefore admit different experimental tests.

7.1 Physical motivation and constitutive foundations

The classical assumption fails in a specific and observable way. A beam of a given material, tested in bending, exhibits a stiffness that depends on the specimen thickness: thinner specimens are stiffer than the classical prediction, and the departure becomes significant below a material-dependent critical length. Two distinct physical mechanisms can produce this behaviour, and they are the root of the entire family of models considered here.

The first mechanism is *constitutive*: the internal stress in the material depends on the neighbouring material state, because the mechanism that carries load—bonding, dislocation motion, stress concentration in a heterogeneous microstructure—acts over a finite distance. Eringen's formulation of nonlocal elasticity is the canonical statement of this mechanism Eringen [1972, 1974], in which the stress at a point is obtained by averaging the local stress over a neighbourhood weighted by a kernel:

$$\sigma_{ij}(\boldsymbol{x}) = \int_{\Omega} \alpha(|\boldsymbol{x}' - \boldsymbol{x}|, \tau)\, \sigma_{ij}^{\mathrm{local}}(\boldsymbol{x}')\, d\Omega_{\boldsymbol{x}'}, \tag{84}$$

where α is the attenuation kernel and $\tau = e_0 a/\ell$ is the nonlocal parameter, with e_0 a material constant, a an internal characteristic length such as the lattice spacing, and ℓ an external characteristic length such as the specimen size. The nonlocal parameter is therefore not an arbitrary length: it is a ratio of an internal to an external scale, which means that the nonlocal correction grows in relative importance as the specimen is made smaller relative to the microstructure. This is the scaling property that distinguishes a size-dependent theory from an ad hoc correction.

The second mechanism is *structural*: the material possesses additional internal degrees of freedom—microrotations, relative displacement of the constituents of a porous skeleton, dislocation density—whose motion must be carried by additional stress measures. Micropolar elasticity, formulated by Eringen as a linear theory of a continuum with independent microrotation fields Eringen [1966] and extended to the thermoelastic case by Tauchert, Claus and Ariman Tauchert et al. [1968], is the earliest and most complete statement of this mechanism. Later work extended it to materials with voids, in which the skeleton and the pore fluid carry independent kinematic fields Nunziato and Cowin [1979], Cowin and Nunziato [1983], with a corresponding thermoelastic formulation Ieşan [1986].

7.1.1 Differential form

For special kernel functions—notably the one-parameter exponential and the Helmholtz kernel—the integral equation of Eq. (84) reduces to a differential equation of higher order. For the Helmholtz kernel one obtains

$$(1 - \xi^2 \nabla^2)\sigma_{ij} = \sigma_{ij}^{\mathrm{local}}, \tag{85}$$

where $\xi = e_0 a$ is the nonlocal length scale parameter. The differential form is the one used in wave-propagation applications, because it is a partial differential equation and can therefore be treated with the transform and numerical machinery of Section 8. It is worth recording that the differential and integral forms are not interchangeable in general: the integral relation with a given kernel and the differential operator of Eq. (85) agree only for that kernel, and the difference matters for the boundary conditions, which in the differential form become higher-order Polizzotto [2001].

7.2 Strain-gradient and couple-stress theories

Strain-gradient theory extends classical elasticity by including higher-order gradients of strain in the constitutive relations:

$$\sigma_{ij} = C_{ijkl}\varepsilon_{kl} - \ell^2 C_{ijkl}\nabla^2\varepsilon_{kl}, \tag{86}$$

where ℓ is a material length scale parameter. The physical reading of Eq. (86) is that the stress depends not only on the strain but on the curvature of the strain field, so that regions in which the strain varies rapidly—near a notch, a contact edge, or an interface—carry additional stress. The coefficient ℓ is a genuine material constant that must be measured; the identification procedures for it, together with the finite element implementations and the variational formulations that make the theory tractable, have been surveyed by Askes and Aifantis Askes and Aifantis [2006, 2011].

The unifying point of view that connects gradient elasticity to micropolar and couple-stress elasticity is that all three introduce internal degrees of freedom and a length scale, and differ in whether those degrees of freedom are constrained to coincide with the macroscopic deformation or are allowed to evolve independently. Aifantis's formulation, in which the internal length is interpreted as the scale of the microstructural heterogeneity that generates it, is the common root Askes and Aifantis [2006]. It follows that a continuum model with internal length, in whichever of these forms, can be used to represent materials for which no explicit microstructural model is available—which is why the same length-scale parameter appears in treatments of metals, polymers, granular media, and biological tissue with broadly comparable values.

7.3 Size-dependent wave behavior: dispersion from nonlocality

Nonlocality introduces dispersion: the wave speed depends on wavelength. The differential form of nonlocal elasticity was developed by Eringen Eringen [1983], and the size-dependent elastic effects at the nanoscale were quantified by Maranganti and Sharma Maranganti and Sharma [2007], who established the characteristic length below which classical elasticity fails for a range of materials, and by Maranganti, Sharma and Sharma Maranganti et al. [2006], who showed that the nanoscale nonlocal contribution to electromechanical coupling is generically non-zero even for materials that possess no piezoelectric tensor—a result directly relevant to the coupled theories of Section 5, because it means that a purely elastic size effect can generate an apparent electromechanical response in a nominally non-piezoelectric material.

For a nonlocal elastic bar, the dispersion relation is

$$c(k) = \frac{c_0}{\sqrt{1+\xi^2 k^2}}, \tag{87}$$

where c_0 is the classical wave speed, k is the wave number, and ξ is the nonlocal length scale. Long waves ($k \to 0$) approach the classical speed c_0; short waves are strongly slowed.

7.3.1 The dispersion signature distinguishes the mechanisms

For Eq. (87), the phase speed tends to zero as $k \to \infty$, while the angular frequency $\omega(k) = c_0 k/\sqrt{1+\xi^2 k^2}$ approaches the finite limit c_0/ξ; it is frequency, not phase speed, that saturates. The group velocity is $d\omega/dk = c_0(1+\xi^2 k^2)^{-3/2}$ and tends to zero. By contrast, the positive strain-gradient law in Eq. (86) gives $c_{\mathrm{ph}}^2 = c_0^2(1+\ell^2 k^2)$ for a one-dimensional bar, so it stiffens and phase speed increases. Gradient-inertia variants can behave differently; the model form must accompany any comparison. A dispersion measurement over a sufficiently broad range of k can thus distinguish a nonlocal mechanism from a gradient mechanism, and the two are not

interchangeable even though both produce the same qualitative signature of a size-dependent stiffness. This is one of the few places in the reviewed literature where two competing formulations of the same effect make quantitatively different and testable predictions, and it deserves more attention than it has received.

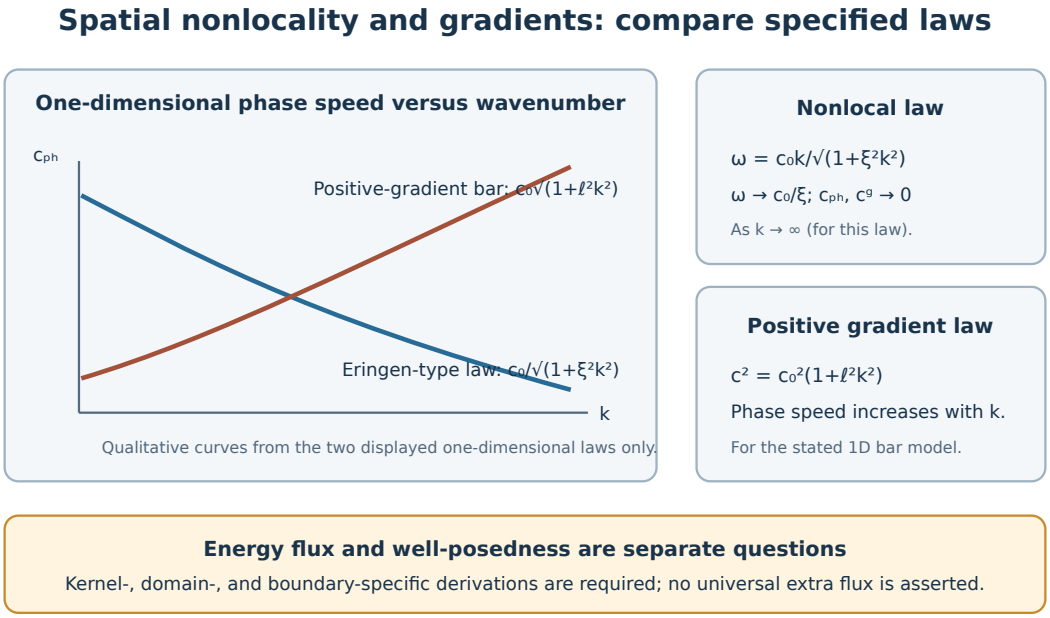


Figure 11: Schematic nonlocal and gradient mechanisms. The plotted dispersion curves correspond only to the particular one-dimensional laws stated in Section 7; other gradient-inertia or kernel choices can differ. No universal nonlocal energy-flux correction is asserted; boundary-power derivations remain open.

7.4 Energy flux in nonlocal media

For nonlocal media, the classical divergence theorem must be modified because the stress at a point depends on strain throughout a neighbourhood. The mechanical power $\sigma_{ij}v_j$ at a point includes contributions from the velocity field in a neighbourhood, not only from the value at the point itself. The classical flux $q_i - \sigma_{ij}v_j$ follows from a local Cauchy-stress power balance. It cannot be transferred without derivation to a nonlocal integral law, a higher-gradient stress law, or a micropolar continuum. A valid energy identity may require additional boundary power, higher-order tractions, or nonlocal domain terms; these depend on the kernel, domain, constitutive free energy, and boundary conditions. No universal nonlocal correction term is specified here. Deriving and validating the boundary-power balance for each model remains OPEN, and no nonlocal interface energy ratio is validated in this review.

7.5 Interface conditions for nonlocal media

Interface conditions for nonlocal media differ from the classical ones because the stress at a point depends on the strain in a neighbourhood, and a neighbourhood straddling an interface samples both media. The classical conditions—continuity of displacement and of traction—must be supplemented, and the supplementation is not unique: one may impose conditions on the additional

derivatives that appear in the differential form of Eq. (85), or one may impose the conditions in the integral form by requiring the nonlocal stress to be continuous across the interface. The two are not equivalent, and the choice made determines the number and kind of boundary conditions that the resulting boundary-value problem admits. For the differential form, the interface conditions include both the classical conditions and the higher-order conditions arising from the ∇^2 term.

Interface problems in which the nonlocal operator is combined with an additional coupled field have been analysed in the reviewed corpus for a fluid in contact with a nonlocal piezothermoelastic medium Gupta et al. [2022] and for nonlocal Kelvin–Voigt piezo-semiconductors under a memory-dependent heat flux Gupta et al. [2026]. In both cases the analysis proceeds through the modal machinery of Section 9, and the specific difficulty in both is the same: the nonlocal operator acts on the stress while the coupled field acts on the temperature or the electric displacement, and the interface conditions must be formulated consistently for both.

7.6 What nonlocality does not repair

The differential nonlocal model must be distinguished from the integral constitutive law from which it is sometimes obtained. Its boundary-value formulation can exhibit paradoxes under particular geometries and boundary conditions, including the cantilever-bending paradox discussed in the nonlocal-beam literature; that does not establish a bulk-wave group velocity that grows without bound. For Eq. (87), both frequency and group velocity are bounded. Well-posedness claims must specify operator, domain, boundary conditions and function space; no universal Hadamard-instability claim is made here.

Some nonlocal formulations exhibit boundary-value paradoxes for particular kernels, geometries, or boundary conditions. Bounded kernels and variational formulations have been proposed in parts of the literature, but their ability to resolve a given problem must be demonstrated for the specified operator and function space; neither is a universal remedy or proof of well-posedness. The variational route can clarify admissible boundary data, but its conclusions are model-specific.

7.7 Comparative summary: nonlocal models

Table 12: Model-specific comparison of spatially nonlocal and gradient formulations. Dispersion statements are restricted to the one-dimensional laws displayed in this section.

Model/form used here	**Spatial mechanism**	**Dispersion status**	**Energy-flux status**
Eringen-type Helmholtz form, Eq. (85)	Kernel-dependent spatial averaging	One-dimensional result of Eq. (87); phase speed decreases and frequency tends to c_0/ξ	OPEN; no universal correction
Strain-gradient form, Eq. (86)	Strain-curvature term	For the stated positive-gradient 1D bar, phase speed increases with k	OPEN; higher-gradient boundary power not derived
Micropolar / couple-stress	Additional micro-rotation or couple stress	Not specified by a single law here	OPEN; model-specific
Two-phase local/nonlocal	Depends on selected mixture/kernel	Not specified here	OPEN; model-specific

7.8 Notation summary for Section 7

Table 13: Notation introduced in Section 7.

Symbol	Meaning	Units	Equation
$\alpha(\lvert\boldsymbol{x}' - \boldsymbol{x}\rvert, \tau)$	Nonlocal attenuation kernel	varies	(84)
τ	Nonlocal parameter, ratio of internal to external length	dimensionless	(84)
e_0	Material constant of the nonlocal formulation	dimensionless	(84)
a	Internal characteristic length	m	(84)
$\xi = e_0 a$	Nonlocal length scale parameter	m	(85)
ℓ	Strain-gradient or couple-stress length scale	m	(86)
$c(k)$	Wave speed as a function of wave number	$\mathrm{m\,s^{-1}}$	(87)
c_0	Classical (local) wave speed	$\mathrm{m\,s^{-1}}$	(87)

R3b source-specific evidence. A two-temperature nonlocal example retains the classical Fourier/Biot heat law, without phase lag or GN terms. (PRIMARY-SOURCE-CHECKED) Lata and Singh [2020]. A nonlocal fractional MGT example places the operator on inertia and the full flux law, including a thermal-displacement gradient. (PRIMARY-SOURCE-CHECKED) Gupta et al. [2024a]. For one modified-Bessel attenuation kernel, the paper gives a specific integral/differential nonlocal correspondence. (CONDITIONAL) Li and He [2018]. The Klein–Gordon nonlocal operator is placed on both constitutive and motion equations in this primary formulation. (PRIMARY-SOURCE-CHECKED) Lazar and Agiasofitou [2022]. An inertia-only nonlocal placement is used in a piezoelastic half-space; no heat law is involved. (PRIMARY-SOURCE-CHECKED) Tung [2019]. An inertia-only one-dimensional nonlocal plate model provides a size-effect example. (PRIMARY-SOURCE-CHECKED) Zang et al. [2014].

7.9 Open questions for Section 7

OQ7.1 What is the correct energy flux at a nonlocal interface, derived from a rigorous modified divergence theorem, and does it reproduce the classical limit as $\xi \to 0$?

OQ7.2 How must the boundary conditions be formulated for the differential nonlocal operator so that the initial-value problem is well posed, and is the resulting formulation variational?

OQ7.3 What is the relationship between the integral and differential nonlocal formulations at the level of the energy flux, given that the two agree only for the specific kernel used to obtain the differential form?

OQ7.4 Can a dispersion measurement over a sufficiently broad range of wave number distinguish a nonlocal mechanism from a gradient mechanism in a specific material?

OQ7.5 Which nonlocal constitutive law is thermodynamically admissible when combined with a memory-dependent thermal law, given that both introduce history into the energy balance?

8 Mathematical Methods for Wave Propagation

The theory families of this review differ in their constitutive content, but the mathematics with which they are analysed differs far less. This section sets out that shared mathematics, and takes

the position that most of the apparent difficulty in the generalized theories is methodological rather than physical: the same transform, eigenproblem, and boundary-layer machinery that handles classical thermoelasticity handles the generalized families, provided that three points are respected—the sign convention of the harmonic ansatz, the selection of the physical roots of the secular equation, and the consistent normalization of the modes whose energies are to be compared.

8.1 Plane wave ansatz and harmonic time dependence

For time-harmonic problems, all field variables take the form

$$\mathrm{field}(\boldsymbol{x}, t) = \hat{\mathrm{field}}(\boldsymbol{x}) e^{-i\omega t}, \tag{88}$$

where $\hat{\mathrm{field}}$ is the complex amplitude and ω is the angular frequency. Two consequences of the convention $e^{-i\omega t}$ should be stated, because both are used repeatedly in the sections that follow. First, a wave with complex wave number q whose imaginary part is positive is spatially attenuated, since the spatial factor e^{iqx} acquires a factor $e^{-\operatorname{Im}(q)\, x}$. Second, the sign of the imaginary part of every complex amplitude reverses if the opposite convention $e^{+i\omega t}$ is adopted, while the physical content of the solution does not change. The divergence map of Section 12 records the second point as one of the routine sources of apparent disagreement between studies.

8.2 Christoffel-type equations and wave speeds

Substituting the plane wave ansatz into the governing equations yields the Christoffel-type equation

$$[\Gamma_{ik}(\boldsymbol{n}) - \rho c^2 \delta_{ik}] \hat{u}_k = 0, \tag{89}$$

where $\Gamma_{ik}(\boldsymbol{n})$ is the Christoffel tensor, $\boldsymbol{n}$ is the propagation direction, c is the wave speed, and $\hat{u}_k$ is the polarization vector. Non-trivial solutions require

$$\det[\Gamma_{ik}(\boldsymbol{n}) - \rho c^2 \delta_{ik}] = 0. \tag{90}$$

For an anisotropic material the Christoffel tensor depends on the direction $\boldsymbol{n}$, and the three roots of Eq. (90) are the three phase velocities of the quasi-longitudinal and two quasi-transverse modes. Their separation, and the dependence of each on the direction of propagation, is the classical anisotropy phenomenon; the generalized theories inherit it unchanged, since they modify the constitutive law rather than the kinematics.

8.3 Secular equations and characteristic polynomials

The characteristic equation from Eq. (90) is a polynomial in c^2, or equivalently in the squared wave number q^2. For coupled thermo-piezoelastic media the equation is typically of eighth order in q and, because the system possesses a certain symmetry with respect to the sign of q, reduces to a quartic in q^2:

$$G_1 q^8 + G_2 q^6 + G_3 q^4 + G_4 q^2 + G_5 = 0, \tag{91}$$

where the coefficients G_i depend on the material properties, the frequency, and the angle of incidence. Two features of this structure are worth recording. First, the equation is quartic in q^2 and therefore has four pairs of roots in q, of which only a subset corresponds to physically admissible modes; the number of modes in the physical solution is therefore *smaller* than the

order of the secular equation suggests, and the surplus modes are unphysical branches of the algebra. Second, in the generalized families the coefficients G_i acquire the relaxation times, the memory kernel, or the nonlocal length scale, and the dependence of the root structure on those parameters is the mechanism by which the constitutive generalization is converted into an observable difference in the wave field. This is the point at which a change in the constitutive law becomes a change in a coefficient that can be measured.

8.4 Complex wave numbers: propagating, evanescent, attenuating modes

The roots of the secular equation are classified as

1. **Propagating modes:** real q, waves that propagate without spatial attenuation;
2. **Evanescent modes:** purely imaginary q, waves that decay exponentially with distance;
3. **Attenuating modes:** complex q, waves that both propagate and attenuate.

Only outgoing modes satisfying the radiation condition are retained in the physical solution. The selection rule is simple to state and easy to get wrong: in a half-space problem the retained roots are those for which the wave decays as $x \to \infty$, and in a slab or layer problem the retained roots are those for which the transmitted field decays into the far material. Every additional root that is retained produces a mode that carries energy out of, or into, the region of interest and breaks the conservation identity of Section 10.5.

8.5 Modal normalization for energy flux computation

For energy flux computation, each mode must be consistently normalized. For propagating modes a convenient choice is

$$\hat{u}_i^{(n)} \hat{u}_i^{(n)*} = 1, \tag{92}$$

where * denotes complex conjugation. For evanescent modes the normalization is chosen so that the time-averaged energy flux is real and positive.

For a single lossless purely evanescent partial wave in a real-coefficient medium, the time-averaged normal flux vanishes: the stress–velocity product is reactive. In a superposition, however, cross terms may contribute to the total flux, and complex-wavenumber modes in a dissipative medium require the flux to be evaluated from the full constitutive balance. It is therefore incorrect either to assign a nonzero independent normal power to every evanescent amplitude or to omit the evanescent amplitudes from the boundary-value solution. Mode amplitudes, cross terms and normal flux must be evaluated consistently.

8.6 Transform methods: Fourier and Laplace

Fourier transforms are used for steady-state harmonic problems on an unbounded or periodic domain:

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(t) e^{i\omega t}\, dt, \tag{93}$$

and Laplace transforms for transient problems on a half-space:

$$\tilde{f}(s) = \int_{0}^{\infty} f(t) e^{-st}\, dt. \tag{94}$$

The Laplace transform in particular is what makes the classical thermoelastic transient problem tractable in closed form: the coupled system is parabolic in the thermal variable and hyperbolic in the mechanical variable, and the Laplace transform in time combined with a Fourier transform in the horizontal directions reduces the problem to a pair of second-order differential equations in the normal coordinate, which are solved in closed form Achenbach [1973], Nowacki [1975].

The fractional operators of Section 11.1 require a different transform calculus. The Caputo derivative of order α has a simple Laplace-domain representation, $\mathcal{L}\{{}^{C}D_t^{\alpha} f\}(s) = s^{\alpha}\tilde{f}(s) - s^{\alpha-1}f(0)$, and the resulting algebraic equation in s is inverted using the Mittag–Leffler function, which plays the role that the exponential plays in the classical theory Gorenflo and Mainardi [1997], Luchko [2010]. It is worth being explicit about the methodological consequence: in the fractional families the analytical route is available only for special geometries, for radially symmetric or half-space problems, because the inversion step requires the eigenfunction expansion of the corresponding spatial operator. Everywhere else, the burden of verification falls on numerical inversion, and the reference solutions of the kind derived in closed form by Povstenko Povstenko [2007] become the only available check on a numerical scheme.

8.7 Green's functions and fundamental solutions

The alternative to the transform route is the construction of Green's functions. For a half-space under a prescribed incident wave, the Green's function of the coupled thermoelastic operator provides the scattered field directly, and the reflection and transmission coefficients follow from the evaluation of the Green's function and its derivatives at the interface. The advantage of this route is that it packages the geometry into a single object, so that the same Green's function can be used for several different incident conditions. Its disadvantage is that the Green's function for the coupled problem is a matrix-valued function whose construction requires the simultaneous inversion of a set of differential operators, and that the algebra is considerably heavier than that of the modal approach of Sections 8.2–8.5. In the reviewed literature the modal route dominates for plane-wave incidence, which is consistent with the structure of the evidence base described in Section 14.4.

8.8 Spectral methods and numerical approaches

Spectral methods expand the solution in terms of orthogonal basis functions—Fourier series for periodic domains, Chebyshev polynomials for finite intervals, Legendre polynomials for axisymmetric problems. They are particularly effective for problems with periodic boundary conditions or with smooth solutions, and their convergence rate can be established analytically for smooth data, which makes them the natural choice when a convergence study is required.

Finite element, finite difference, and boundary element methods cover the problems for which no closed-form solution exists. The nonlocal and memory-dependent families require specialized treatments, and it is worth being specific about why, because the reason is not an aesthetic preference.

- **Nonlocal operators.** The constitutive law of Eq. (84) is an integral over a neighbourhood. A finite element discretization must therefore assemble the kernel integral element-by-element, which converts the stiffness matrix from a sparse local operator into a dense or partially dense one, and the cost of the assembly grows with the size of the neighbourhood. The standard remedy is to use the differential form of the operator, Eq. (85), which is local, at the price of the well-posedness difficulties discussed in Section 7.6. Which of the two is used is therefore not a free choice, and it must be stated.

- **Memory-dependent operators.** The memory integral of Section 6 is over the entire history of the field, and a direct discretization is a marching-in-time scheme whose stability is governed by the treatment of the history. Quadrature and convolution-quadrature schemes, and the use of the Laplace-domain representation in which the history is compressed into a single auxiliary variable, are the two standard alternatives.
- **Fractional operators.** A fractional derivative of order α in space or in time is nonlocal, and its discretization is a dense or a history-dependent operator, with the same cost considerations as in the nonlocal case.

8.9 Variational structure and its use as an analytical tool

Alongside the direct transform route, coupled thermoelasticity admits a variational formulation, and the two are not simply equivalent. Variational principles for the coupled system have been established and compared by Marotti de Sciarra Marotti de Sciarra [2013], who showed that the admissible function spaces and the boundary conditions inherited from each principle differ, and that the resulting weak formulations do not all select the same class of solutions. This matters practically because a variational derivation constrains the admissible boundary conditions before the equations are solved, and a formulation that admits non-physical boundary data can contaminate the solution even when the differential equations are correct. The variational route is also the one that extends most naturally to the nonlocal and gradient constitutive laws of Section 7, since the integral kernel appears as an operator in the quadratic functional rather than as an explicit convolution in the equations of motion.

8.10 The two-stage structure of the classical problem, and what the generalizations change

The displayed classical dynamic thermoelastic equations form a coupled initial-boundary-value problem: momentum balance is hyperbolic and Fourier heat conduction is parabolic. A sequential or quasi-static solution is an approximation whose validity requires a stated weak-coupling or time-scale assumption. Harmonic elimination can produce a frequency-dependent longitudinal response, but attenuation and any dimensionless coupling parameter must be derived for the precise constitutive convention and geometry. Those quantitative claims are not established in this section and remain open pending equation-level derivation and primary-source checking.

8.11 Notation summary for Section 8

R3b source-specific evidence. A partially read layered piezothermoelastic Rayleigh-wave paper includes a GN-III heat equation; this citation is scope-level only. (UNVERIFIED) Biswas [2020a].

8.12 Open questions for Section 8

OQ8.1 How are complex modes to be normalized so that the energy ratios computed from them are reproducible across independent implementations, and can a normalization convention be proposed that the field adopts?

OQ8.2 How well do numerical schemes converge for nonlocal and memory-dependent models, and what error results from replacing the integral operator by a differential form?

OQ8.3 Can the two-stage sequential treatment of the classical problem be extended to the generalized families as an approximation, with a bound on the error introduced?

Table 14: Mathematical solution techniques for generalized continuum wave problems. Each technique's applicability and limitations are identified.

Technique	**Mathematical Basis**	**Applicability**	**Limitations**	**Evidence**
Normal mode expansion	Separation of variables; eigenfunction expansion	Layered media; simple geometries	Limited to separable problems; boundary condition constraints	Achenbach [1973]
Green's functions	Fundamental solution; convolution	Point source problems; infinite domains	Singular integrals; complex for anisotropic media	Standard method
Fourier transform	Frequency domain; $e^{-i\omega t}$ convention	Time-harmonic problems; dispersion analysis	Assumes time-harmonic fields; initial-value problems need Laplace	Standard method
Laplace transform	Time domain to complex frequency	Initial-value problems; transient waves	Branch cuts; inversion complexity	Standard method
Christoffel equation	$\det[\Gamma_{ik} - \rho c^2 \delta_{ik}] = 0$	Plane wave propagation in anisotropic media	Secular polynomial degree grows with coupling	Achenbach [1973]
Scattering matrix	Interface matching formulation; inherited equation not re-derived	Candidate method; model-specific	Dimensions and normalization require checking	Barak et al. [2023a]
Finite element	Weak form; variational principles	Complex geometries; nonlinear problems	Mesh dependency; computational cost	Hughes [1980]

Notes: Complex wave number classification (propagating/evanescent/attenuating) follows standard wave theory.
Inherited method descriptions and the cited scattering-matrix claim have not been independently checked in this revision.

Table 15: Notation introduced in Section 8.

Symbol	**Meaning**	**Units**	**Equation**
Γ_{ik}	Christoffel tensor	varies	(89)
c	Wave speed	$\mathrm{m\,s^{-1}}$	(89)
q	Wave number	$\mathrm{m^{-1}}$	(91)
G_i	Coefficients of the secular equation	varies	(91)
$\hat{f}(\omega)$	Fourier transform	varies	(93)
$\tilde{f}(s)$	Laplace transform	varies	(94)
$\hat{u}_i^{(n)}$	Normalized amplitude of mode n	varies	(92)

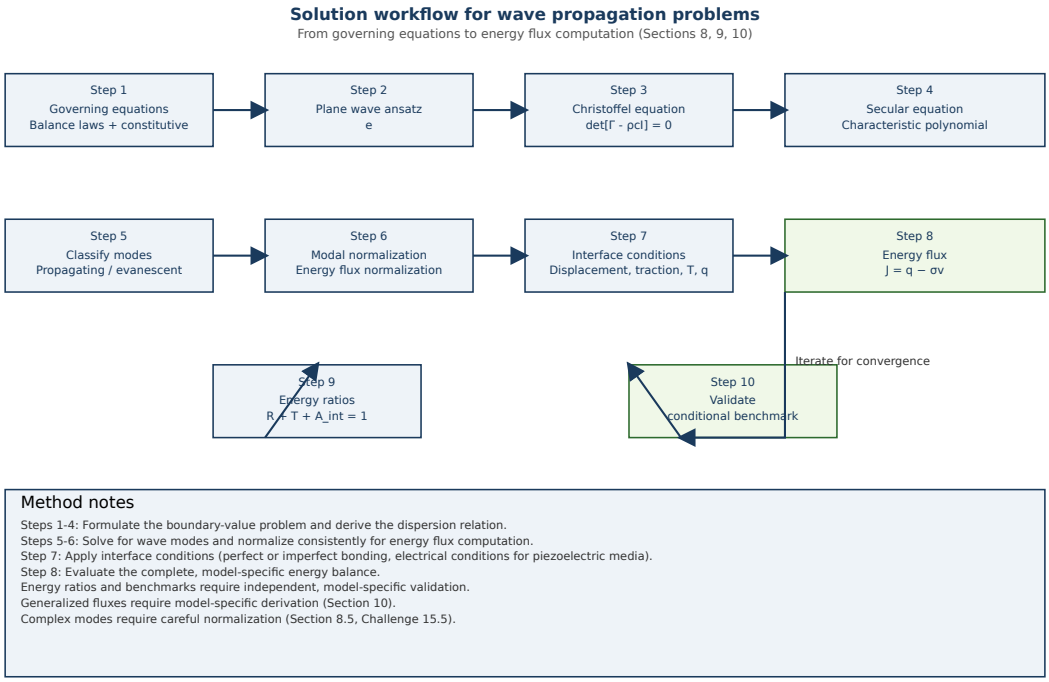


Figure 12: Solution workflow for wave propagation problems: from governing equations to energy flux computation and validation (Sections 8, 9, 10).

OQ8.4 Which weak formulation is the correct starting point for a variational treatment of a nonlocal and memory-dependent coupled problem?

9 Reflection, Transmission, and Interface Phenomena

This section presents wave reflection, transmission, and energy partitioning at interfaces between different continuum media. The scattering matrix formulation, interface conditions, and energy ratios are developed for elastic, thermoelastic, and piezothermoelastic media.

9.1 Interface geometry

Consider two half-spaces in perfect contact at the interface plane $x_3 = 0$. Medium 1 occupies $x_3 < 0$ and medium 2 occupies $x_3 > 0$. A plane harmonic wave is incident from medium 1 onto the interface, generating reflected waves in medium 1 and transmitted waves in medium 2.

The interface normal is $\boldsymbol{n} = (0, 0, 1)$. All field variables vary harmonically in time with convention $e^{-i\omega t}$ and propagate along the interface with wave number k in the x_1-direction:

$$\text{field}(x_1, x_3, t) = \hat{\text{field}}(x_3) e^{i(kx_1 - \omega t)}. \tag{95}$$

9.2 Wave decomposition: incident, reflected, transmitted modes

In each medium, the general solution is a superposition of partial waves. For an isotropic elastic medium, there are two wave types: longitudinal (P) and shear (SV). For thermoelastic media, there is an additional thermal (T) wave. For piezothermoelastic media, there are additional coupled modes (qP, qS, qT, electromagnetic).

The displacement field in medium m ($m = 1, 2$) is decomposed as:

$$u_i^{(m)} = \sum_n A_n^{(m)} \hat{u}_i^{(m,n)} e^{i(q_n^{(m)} x_3 - \omega t)}, \tag{96}$$

where $A_n^{(m)}$ are complex amplitude coefficients, $q_n^{(m)}$ are the vertical wave numbers, and $\hat{u}_i^{(m,n)}$ are the polarization vectors.

The incident wave comes from medium 1 with known amplitude. Reflected waves propagate back into medium 1, and transmitted waves propagate into medium 2.

9.3 Interface conditions

At a perfectly bonded interface between two media, the following conditions must hold:

9.3.1 Mechanical conditions

$$u_i^{(1)} = u_i^{(2)}, \qquad \sigma_{j3}^{(1)} = \sigma_{j3}^{(2)} \quad \text{at } x_3 = 0. \tag{97}$$

The first condition enforces displacement continuity (no slip or separation); the second enforces traction equilibrium (Newton's third law at the interface).

9.3.2 Thermal conditions

For thermoelastic media:

$$T^{(1)} = T^{(2)}, \qquad q_3^{(1)} = q_3^{(2)} \quad \text{at } x_3 = 0. \tag{98}$$

Temperature continuity and heat flux continuity ensure perfect thermal contact.

9.3.3 Electrical conditions (piezoelectric media)

For piezothermoelastic media:

$$\phi^{(1)} = \phi^{(2)}, \qquad D_3^{(1)} = D_3^{(2)} \quad \text{at } x_3 = 0. \tag{99}$$

Electric potential continuity and electric displacement continuity ensure perfect electrical contact. For an insulated interface, $D_3 = 0$ replaces the second condition.

9.3.4 Imperfect bonding

For imperfectly bonded interfaces, spring-type conditions relate displacement jumps to interfacial tractions:

$$u_i^{(2)} - u_i^{(1)} = K_i^u t_i, \qquad \sigma_{j3}^{(1)} n_j = \sigma_{j3}^{(2)} n_j = t_i, \tag{100}$$

where K_i^u is the interfacial stiffness tensor and t_i is the interfacial traction. In the limit $K_i^u \to \infty$, perfect bonding is recovered; in the limit $K_i^u \to 0$, a traction-free (slip) interface is recovered.

9.4 Scattering matrix formulation

The interface conditions lead to a linear system relating the unknown amplitude coefficients to the known incident wave amplitude. For a thermo-piezoelastic interface problem with N unknown amplitudes and N interface conditions, the system can be written in matrix form:

$$\Delta \boldsymbol{X} = \boldsymbol{Q}, \tag{101}$$

where:

- Δ is the $N \times N$ system matrix whose entries depend on material properties, wave numbers, frequency, and incidence angle,
- $\boldsymbol{X} = (A_1, A_2, \ldots, A_N)^T$ is the vector of unknown amplitude coefficients (reflected and transmitted),
- $\boldsymbol{Q} = (Q_1, Q_2, \ldots, Q_N)^T$ is the known vector determined by the incident wave amplitude.

The entries of Δ are determined by substituting the partial wave expansions into the interface conditions (97)–(99). Each row corresponds to one interface condition; each column corresponds to one unknown amplitude.

The solution is obtained by matrix inversion:

$$\boldsymbol{X} = \Delta^{-1} \boldsymbol{Q}. \tag{102}$$

The solution exists and is unique when $\det(\Delta) \neq 0$.

9.5 Secular equation and mode classification

For a transversely isotropic 6mm piezoelectric medium in the sagittal plane containing the c-axis, with the quasi-static electric closure and the stated P–SV–T–ϕ plus SH field set, the secular polynomial determines the allowed normal wave numbers. After removing the trivial $q^2 = 0$ factor associated with the quasi-static Poisson row, the retained polynomial is degree eight. This scope does not imply a piezo-stiffened SH branch in this plane.

$$G_1 q^8 + G_2 q^6 + G_3 q^4 + G_4 q^2 + G_5 = 0, \tag{103}$$

which is degree eight in q for this specified 6mm orientation and field set; other symmetries, orientations, and retained fields can change the degree. The roots are classified as:

- **Propagating modes:** real q — waves that propagate without spatial attenuation,
- **Evanescent modes:** purely imaginary q — waves that decay exponentially with distance,
- **Attenuating modes:** complex q — waves that both propagate and attenuate.

Only outgoing modes (satisfying the radiation condition) are retained in the physical solution.

9.6 Number of boundary conditions vs. unknown amplitudes

For a well-posed interface problem, the number of independent boundary conditions must equal the number of unknown amplitude coefficients. Table 16 summarizes interface condition configurations for different interface types.

Table 16: Reflection and transmission interface configurations. Matching conditions and modal phenomena for bonded and imperfect interfaces.

Configuration	**Matching Conditions**	**Modal Phenomena**	**Physical Quantities**	**Evidence**
Welded interface (perfect bonding)	$u_i^{(1)} = u_i^{(2)}$; $\sigma_{ji}^{(1)} n_j = \sigma_{ji}^{(2)} n_j$	Mode conversion; P–S conversion; interface waves	R, T, A_{int}; total-field identity; $A_{\text{int}} = 0$ if passive, lossless, and no interfacial storage	Standard theory
Imperfect (spring-type) interface	Displacement jump proportional to traction	Partial transmission; interface stiffness effects	Transmission coefficient dependent on spring stiffness	Model-specific
Perfect thermal contact	$T^{(1)} = T^{(2)}$; $q_i^{(1)} n_i = q_i^{(2)} n_i$	Thermal-field transmission/reflection; propagating modes only in finite-speed models	Continuous temperature and normal heat flux	Perfect-contact idealization
Kapitza (imperfect thermal contact)	$q_n^{(1)} = q_n^{(2)}$; $T^{(2)} - T^{(1)} = R_K q_n$	Temperature jump at finite heat flux	Interfacial thermal resistance R_K	Model-specific; sign follows chosen normal
Electrical interface	$\phi^{(1)} = \phi^{(2)}$; $D_i^{(1)} n_i = D_i^{(2)} n_i$	Piezoelectric coupling; surface charge	Electric displacement continuity if free surface charge is zero; otherwise its jump is the surface charge density	Standard theory
Thermo-piezoelastic interface	All mechanical, thermal, electrical conditions	Multi-field scattering; energy partitioning	Interface-energy study cited; equation-level claims not checked	Barak et al. [2023a]

Notes: The proposed BM-1 conservation check was not executed in this revision; see Appendix C. *Thermo-piezoelastic interface equations and cited energy ratios are inherited claims, not checked against the full source here.*

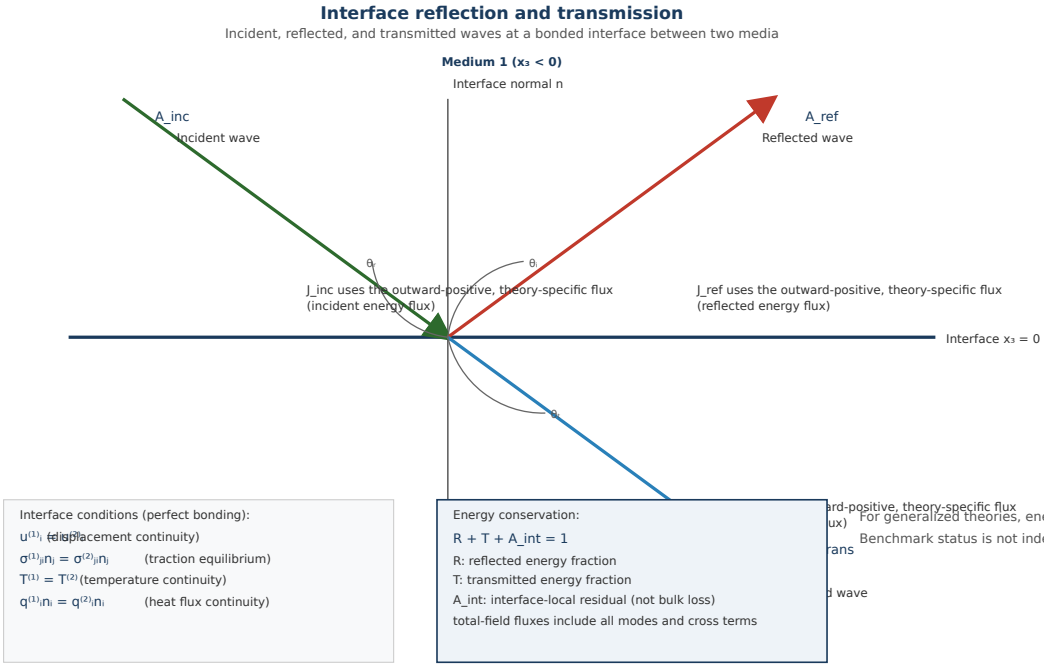


Figure 13: Interface reflection and transmission geometry: incident P-wave from Medium 1 generates reflected P and S waves in Medium 1 and transmitted qP and qS waves in Medium 2. Interface conditions enforce displacement, traction, temperature, heat flux, electric potential, and electric displacement continuity.

9.7 Energy ratios at interfaces

The energy reflection coefficient R, transmission coefficient T, and interface-local residual A_{int} are defined from total-field normal fluxes as:

$$R = \frac{J_{\text{inc},n} - J_{1,n}^{\text{tot}}}{J_{\text{inc},n}}, \qquad T = \frac{J_{2,n}^{\text{tot}}}{J_{\text{inc},n}}, \qquad A_{\text{int}} = \frac{J_{1,n}^{\text{tot}} - J_{2,n}^{\text{tot}}}{J_{\text{inc},n}}, \tag{104}$$

where $J_{\text{inc},n} > 0$ is the incident normal flux, $J_{1,n}^{\text{tot}}$ is the normal flux of the complete incident-plus-reflected field evaluated at the incident-side interface, and $J_{2,n}^{\text{tot}}$ is the complete transmitted-side flux there; all are time-averaged with a common outward normal. Cross terms are included. A_{int} is the interface-local residual, not bulk attenuation.

For a passive interface with no interfacial storage or dissipation ($A_{\text{int}} = 0$), total-field accounting gives:

$$\sum R + \sum T = 1. \tag{105}$$

With interfacial dissipation or storage represented by a nonzero local residual:

$$\sum R + \sum T + A_{\text{int}} = 1. \tag{106}$$

The identity (106) is the interface balance when the theory-specific flux is evaluated from total fields on both sides and all modes and cross terms are retained. It does not add bulk dissipation integrated over either half-space as a third interface channel.

For passive perfect contact with no interfacial storage or dissipation, continuity of total-field flux gives zero interface-local residual. Mode-diagonal outgoing fluxes need not satisfy this

identity in attenuating media: incident–reflected and inter-mode cross terms can be nonzero. Bulk dissipation is a volume integral and must be reported separately; adding it to an interface residual would double-count energy loss.

A concrete V-B3d coupled Biot case illustrates the distinction (steel-like/aluminium-like half-spaces, 1-D normal incidence, perfect contact, $f = 4.46 \times 10^{11}$ Hz; arithmetic test data, not a materials claim). The reproduced total-field values are $R_{\text{flux}} = 0.212026$, $T = 0.787974$, and $A_{\text{flux}} = 2.2{\times}10^{-16}$. The reflected-sum and diagonal-only values are $R_{\text{ref}} = 0.211074$ and $R_{\text{mode}} = 0.211592$, giving $R_{\text{ref}} + T - 1 = -9.5 \times 10^{-4}$ and $R_{\text{mode}} + T - 1 = -4.34 \times 10^{-4}$. Independent half-space dissipation checks close to unity for the computed reflected and transmitted powers. Thus total-field closure is a definition-level bookkeeping identity, not an independent validation; dropping cross terms changes the reported ratios. Reproduction details and the read-only test source commit are recorded in `reports/V-B3d-cross-term-check.md`.

9.7.1 What the interface partition reveals

The interface ratios defined above are model-dependent observables. Their interpretation requires the stated constitutive flux and boundary conditions; the closure identity alone does not distinguish two theories that predict the same total-field flux balance. Comparisons of reflected and transmitted fractions require a common normalization and source-specific modal conventions.

The cited Barak/Gupta interface papers are retained as bibliography leads, but their exact configurations, flux conventions, numerical results, and conservation claims have not been checked in this revision. No specific configuration or reported result is attributed to them here; their use as evidence remains OPEN pending full-text review.

9.7.2 The pattern and its limits

No universal claim that the conservation identity has been verified across generalized-media papers is made here. For the present total-field definitions, the local interface balance follows from flux continuity only when the specified interface has no storage or dissipation; mode-projected literature ratios require source-specific reconstruction and independent checking. A second limit is worth recording. All of the configurations above assume a perfectly bonded, flat interface between two homogeneous half-spaces with sharp material contrast. Real interfaces are graded over a thickness comparable with the microstructural length, and for nonlocal, memory-dependent and gradient theories the thickness of the interfacial layer is not negligible compared with the internal length of the material. Whether the sharp-interface result is the limit of a graded-interface result, and at what rate, is an open question that no configuration in the reviewed literature addresses.

9.8 Interface conditions under memory and nonlocality

The interface conditions of Section 9.3 assume that the constitutive variables are local functions of position. This assumption fails in two of the theory families treated here, and the failure has a common structure: the quantity that is matched across the interface is no longer a pointwise value.

For a *memory-dependent* medium, the flux at the interface is an integral over the recent past, and a naive condition of the form $q_3^{(1)} = q_3^{(2)}$ matches two functionals whose histories differ. The condition is either imposed on the effective flux of (81) — in which case the identity of Section 10 must already have been established — or imposed on the constitutive variables

together with their history, in which case the interface problem becomes an initial-value problem in the tangential coordinate and its solution depends on how the interface was approached.

For a *nonlocal* medium, the stress at the interface is an average over a neighbourhood that straddles the interface, so that the traction transmitted across the interface is not a pointwise quantity at all; the correct interface problem is then a nonlocal boundary-value problem in which the interface condition couples the two half-spaces over the whole interfacial region. A local approximation that simply writes $t_i^{(1)} = t_i^{(2)}$ is consistent only to leading order in the ratio of the internal length to the wavelength.

Neither of these is merely a technical difficulty. Both concern the meaning of an interface in a theory that has no microscopic notion of a material boundary, and both are of the same character as the question of whether the constitutive variables of a nonlocal or memory-dependent theory can be continued across a material discontinuity at all. These are recorded as open questions in Section 16; they are the places where the extension of the classical framework to generalized media is least settled.

9.9 Notation summary for Section 9

Table 17: Notation introduced in Section 9.

Symbol	Meaning	Units	Equation
$A_n^{(m)}$	Complex amplitude coefficient	varies	(96)
$q_n^{(m)}$	Vertical wave number	1/m	(96)
Δ	System matrix	varies	(101)
$\boldsymbol{X}$	Unknown amplitude vector	varies	(101)
$\boldsymbol{Q}$	Known incident vector	varies	(101)
R, T, A	Reflection, transmission, absorption coefficients	dimensionless	(104)
K_i^u	Interfacial stiffness	varies	(100)

R3b source-specific evidence. The interface model uses an LS-type relaxation on the thermal-inertia and coupling terms. (PRIMARY-SOURCE-CHECKED) Kumar and Kansal [2017]. A micropolar piezothermoelastic half-space study reports amplitude ratios, not energy ratios. (UNVERIFIED) Kumar et al. [2018]. A four-parameter imperfect-interface model distinguishes normal/transverse force and couple-stiffness conditions. (UNVERIFIED) Kumar and Sharma [2008]. A nonlocal interface example acts on stress while retaining a local GN-II heat law. (PRIMARY-SOURCE-CHECKED) Sarkar et al. [2019]. A two-temperature porothermoelastic example uses a nondissipative GN-II formulation. (PRIMARY-SOURCE-CHECKED) Singh [2020]. A higher-order fractional TPL model reports reductions to LS, DPL, and TPL at integer order one. (CONDITIONAL) Aldandani et al. [2026]. A mixed nonlocal placement acts on stress, couple stress, inertia, and the LS flux law. (PRIMARY-SOURCE-CHECKED) Kalkal et al. [2020]. A k*-type interface example reports reflection amplitudes at stress-free isothermal and insulated boundaries, not energy ratios. (PRIMARY-SOURCE-CHECKED) Das et al. [2008]. A scope-level imperfect fluid/solid interface example reports phase velocity and specific loss, not energy partition. (UNVERIFIED) Kumar et al. [2025]. A scope-level layered piezoelectric-cylinder study concerns SH-wave dispersion with imperfect interfaces, not energy partition. (UNVERIFIED) Gupta et al. [2025]. A higher-order fractional DPL/two-temperature example reports energy-ratio and interaction-energy expressions, but its coefficients are garbled in the read copy. (UNVERIFIED) Gupta et al. [2024b]. A free-surface study reports energy definitions and interaction energy, but the closure equation was not located in the text layer.

(UNVERIFIED) Yadav et al. [2023]. An independent free-surface study reports a reflected ratio reaching unity numerically; this is not a symbolic conservation identity. (CONDITIONAL) Singh and Guha [2020]. An independent study describes the energy-ratio sum as approximately unity, not as an exact identity. (PRIMARY-SOURCE-CHECKED) Kumar and Sharma [2019]. A TPL interface study reports a reflected energy-ratio sum bounded above by unity for its specified model. (CONDITIONAL) Singh and Gupta [2020]. A flexomagnetic microstructure paper addresses dispersion and reflection, without energy-ratio results. (UNVERIFIED) Jiao et al. [2024]. A conference-length micropolar imperfect-interface source is included for scope only. (UNVERIFIED) Fu and Wei [2013]. A graded piezoelectric-microstructure reflection study is included for scope only. (UNVERIFIED) Liu et al. [2020]. An interface study prints an energy matrix with diagonal ratios and off-diagonal interaction energies, then states a model-specific total balance. (PRIMARY-SOURCE-CHECKED) Kumar et al. [2022].

9.10 Open questions for Section 9

OQ9.1 What are the interface conditions for nonlocal media where stress depends on strain in a neighborhood?

OQ9.2 How are energy ratios computed for memory-dependent media with hereditary flux?

OQ9.3 What is the correct interface energy balance for piezothermoelastic media with electrical coupling?

OQ9.4 Is the sharp-interface result the limit of a graded-interface result for nonlocal, gradient and memory-dependent media, and at what rate does the difference vanish as the interfacial layer is thickened?

OQ9.5 Can the constitutive variables of a nonlocal or memory-dependent theory be continued across a material discontinuity at all, and what does that imply for the meaning of an interface in such a theory?

10 Energy Transport and Dissipation

Constitutive theories carry energy-storage, transport, and dissipation consequences, but the expressions depend on their state variables and field equations. This section derives the classical local flux under explicit conventions, states the interface balance at a general level, and records family-specific flux claims as open unless their free energy, constitutive evolution, boundary power, and dissipation inequality have been checked. No universal generalized-continuum flux is assumed.

10.1 General energy balance

For classical thermoelasticity, with specific internal energy e, the total energy per unit volume is

$$\mathcal{E} = \frac{1}{2}\rho v_i v_i + \rho e, \tag{107}$$

and the local balance (with no internal heat source) takes the form

$$\frac{\partial \mathcal{E}}{\partial t} + \partial_i J_i = \rho f_i v_i, \tag{108}$$

where $\rho f_i v_i$ is mechanical power per unit volume. Combining momentum balance with the first law yields the classical flux in the next subsection. Equation (108) is not asserted here as a universal generalized-multiphysics identity: additional field energy, external electrical or magnetic power, interface storage, and internal dissipation must be derived for each specified model.

10.2 Classical energy flux

For classical coupled thermoelasticity, adding the mechanical power and internal-energy balances yields the outward-positive flux

$$J_i = q_i - \sigma_{ij} v_j, \tag{109}$$

the classical energy flux. Under this outward-positive convention the mechanical contribution is $-\sigma_{ij} v_j$ and the thermal contribution is q_i.

For time-harmonic mechanical fields with convention $e^{-i\omega t}$, the outward-positive mean flux is

$$\langle J_i \rangle_t = \bar{q}_i - \frac{1}{2} \Re[\hat{\sigma}_{ij} \hat{v}_j^*], \tag{110}$$

where $\bar{q}_i$ is the separately evaluated mean heat flux (zero for a purely oscillatory perturbation about a uniform reference state). The factor $1/2$ and complex conjugate follow from time averaging the real fields. For an interface problem the quantity of interest is the normal component $J_n = J_i n_i$; the sign convention and the normal direction must be stated consistently.

A point worth keeping in view is that the classical flux contains *no* term associated with the internal variables that generalized theories introduce. The entire difficulty of Section 10.4 follows from that absence.

10.3 Interface energy balance

Consider an interface Σ between two media with unit normal $\boldsymbol{n}$. Integrating the general energy balance over a thin pillbox straddling the interface and taking the thinness limit gives the jump condition

$$[J_n] = \mathcal{P}_\Sigma - \frac{\partial \mathcal{E}_\Sigma}{\partial t}, \tag{111}$$

where $J_n = J_i n_i$ is the normal energy flux, $\mathcal{P}_\Sigma$ is the mechanical power input at the interface, and $\mathcal{E}_\Sigma$ is the interfacial energy density.

For a passive interface with no mechanical power input and no interfacial energy storage,

$$[J_n] = 0, \tag{112}$$

that is, the normal energy flux is continuous across the interface. Two qualifications are needed before this condition is used numerically. First, it is a statement about the *correct* energy flux of the theory in which it is used; a flux built from the classical form alone, applied to a family with internal variables, will not satisfy it even when the mathematics is otherwise flawless. Second, the continuity of the normal flux does not by itself imply the scalar conservation identity of Section 10.5: that identity is a statement about modal energies, while (112) is a statement

about a single component of a flux at a single surface, and the two are related only after the modal decomposition and normalisation conventions have been fixed.

10.4 Energy transport in generalized theories: the internal-variable structure

The first-law balance retains the boundary power terms for a specified constitutive theory. What changes between generalized theories is the stored free energy and dissipation, which must be derived from that theory's state variables and evolution laws. There is no universal additional flux term $-\sum_\alpha Y^\alpha \dot{z}^\alpha$: this has units of power per volume, not energy flux, unless an additional spatial transport structure is derived. We therefore do not use a generic internal-variable flux formula; family-specific identities are supplied only when the free energy, evolution law and boundary terms have been established, otherwise the result is marked conditional or open.

10.4.1 Verified piezothermoelastic energy flux

A piezothermoelastic flux expression and associated interface-energy ratios are reported in the cited literature. In this revision their signs, harmonic averaging, constitutive definitions, and boundary conditions have not been checked against the cited full texts. The formula and the claim that independent publications confirm it are therefore withheld from the verified set. The mechanical, electrical, and thermal channels must be derived from one stated first-law convention and the coupled constitutive equations before any mode-wise ratio is interpreted. Status: OPEN pending full-text and derivation audit.

10.4.2 What remains open in the pure generalized theories

This revision has not derived or full-text-verified generalized-family energy-flux and dissipation identities for Lord–Shulman, Green–Lindsay, Green–Naghdi, phase-lag, Moore–Gibson–Thompson, memory-dependent, or nonlocal formulations. This is a limit of the present audit, not a claim that the wider literature has no such derivations. No general energy-flux identity is asserted here. Each family-specific balance remains OPEN in this version until its free energy, flux variables, boundary powers, dissipation inequality, and sources have been checked.

10.5 Conservation identity: $\sum R + \sum T + A_{\text{int}} = 1$

For wave scattering at an interface, energy conservation requires that the sum of the reflection, transmission, and absorption coefficients be unity:

$$\sum R + \sum T + A_{\text{int}} = 1, \tag{113}$$

where $A_{\text{int}} = (J^{\text{tot}}_{1,n} - J^{\text{tot}}_{2,n})/J_{\text{inc},n}$ is the interface-local residual under the total-field convention of Section 9.7; the incident-side and transmitted-side fluxes include all modal cross terms. Bulk dissipation in either half-space is a separate volume contribution and is not added to A_{int}. For a lossless interface with no storage ($A_{\text{int}} = 0$),

$$\sum R + \sum T = 1. \tag{114}$$

This identity is conditional on the theory-specific flux, a passive interface, complete total-field modal accounting (including cross terms), and a common normalization. The lossless reduction additionally requires zero interface-local residual and no interfacial storage. These conditions are not automatic. For a single lossless, purely evanescent partial wave, the time-averaged

normal flux is zero under the usual real-coefficient assumptions. Cross terms between partial waves can contribute to the total field flux; attenuating modes in dissipative media require separate treatment. Modal accounting must be performed on the complete field. The fourth is the most frequently overlooked: the notion of the energy of a complex mode depends on the normalisation adopted for the modal amplitude, and two groups using different normalisations can report energy ratios that differ by a constant factor while describing the same physics.

10.6 Dissipation mechanisms

Dissipative mechanisms can include heat-flux relaxation, viscous stress, dielectric loss, interfacial friction, or hereditary memory. Their presence and non-negative dissipation conditions depend on the precise constitutive equations and admissible state space. This version does not assign a universal scalar dissipation function or a common parameter-ordering rule across the families. In particular, the condition $\tau_q \geq \tau_T \geq 0$ and a positive Lord–Shulman relaxation time are not, by themselves, a full proof of thermodynamic admissibility or stability. Deriving model-specific free energies and dissipation inequalities remains open.

10.7 Thermodynamic admissibility: well-posedness, causality, positivity

Thermodynamic admissibility, mathematical well-posedness, and finite-speed propagation are distinct questions. A constitutive theory requires its own free-energy/entropy statement and dissipation inequality; an initial-boundary-value problem requires a separate existence, uniqueness and continuous-dependence analysis; and propagation speed is a property of the selected equations and parameter regime. None of these questions is settled for an entire named model family solely by the family label. Fourier heat conduction gives a parabolic equation with instantaneous mathematical support. The corresponding statements for relaxation, phase-lag, memory, fractional and gradient models require the exact constitutive law and its admissibility conditions. This review does not supply a family-wide proof of well-posedness, causality, or positivity; these remain model-specific audit tasks.

10.8 Comparative table: energy transport across theory families

Table 18 records the current energy-balance audit status across theory families.

10.9 Notation summary for Section 10

10.10 Open questions for Section 10

OQ10.1 What is the explicit time-domain energy flux and dissipation function for each of the families listed in Section 10.4, derived from that family's free energy, evolution laws, and boundary-power terms?

OQ10.2 Is there a common formal structure under which the energy fluxes of the Lord–Shulman, phase-lag, Moore–Gibson–Thompson, and memory-dependent families become instances of a single template, so that a single derivation suffices for all of them?

OQ10.3 Which observable would distinguish the energy partition predicted by two competing theories at the same interface, so that the difference could be measured rather than argued?

OQ10.4 How should the entropy production of the conservative Green–Naghdi type II structure be related to the entropy production of the dissipative branches, given that the former has no dissipation mechanism?

Table 18: Energy-balance status by constitutive family. Only the classical local balance is derived in this version; generalized-family entries identify audit tasks rather than verified flux formulae.

Theory family	**Stored-energy form**	**Flux status**	**Dissipation status**	**Audit status**
Classical thermoelasticity	$\mathcal{E} = \frac{1}{2}\rho v_i v_i + \rho e$	$J_i = q_i - \sigma_{ij} v_j$ (outward-positive)	From selected constitutive heat law	DERIVED
Lord–Shulman	Theory-specific free energy	Relaxation-law boundary power not checked	Admissibility not checked here	OPEN
Green–Lindsay	Theory-specific free energy	Temperature-rate terms not checked	Admissibility not checked here	OPEN
Green–Naghdi II	Theory-specific free energy	Thermal-displacement boundary power not checked	Model-specific restrictions not checked	OPEN
Green–Naghdi III	Theory-specific free energy	Complete boundary-power identity not checked	Model-specific restrictions not checked	OPEN
Phase-lag (dual/triple)	Theory-specific free energy	Lag-dependent flux balance not checked	Parameter restrictions not checked here	OPEN
Moore–Gibson–Thompson	Theory-specific free energy	Higher-order balance not checked	Model-specific restrictions not checked	OPEN
Memory-dependent	History-dependent storage may be needed	Kernel-dependent boundary power not checked	Kernel-dependent	OPEN
Nonlocal / gradient	Depends on kernel or gradient energy	Domain- and boundary-condition-specific flux not derived	Model-specific	OPEN

Generalized entries are explicitly not canonical formulae. A valid expression requires the free energy, evolution equations,

boundary conditions, storage terms, and dissipation inequality to be stated and derived for each model.

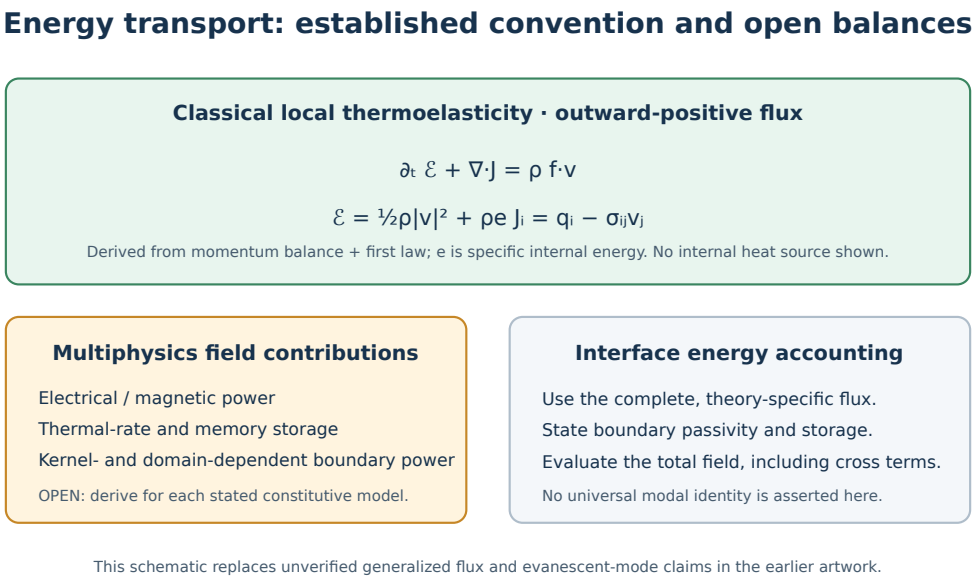


Figure 14: Energy-balance conventions and open flux audit. The classical local thermoelastic flux is derived under the stated outward-positive convention; generalized multiphysics contributions and modal energy identities remain open pending model-specific derivations.

Table 19: Notation introduced in Section 10.

Symbol	Meaning	Units	Equation
$\mathcal{E}$	Total energy density	$\mathrm{J\,m^{-3}}$	(107)
$\boldsymbol{J}$	Energy flux vector	$\mathrm{W\,m^{-2}}$	(108)
$\mathcal{P}$	Mechanical power input	$\mathrm{W\,m^{-3}}$	(108)
J_i	Classical energy flux component	$\mathrm{W\,m^{-2}}$	(109)
$\langle J_i \rangle_t$	Time-averaged energy flux	$\mathrm{W\,m^{-2}}$	(110)
$\mathcal{E}_\Sigma$	Interfacial energy density	$\mathrm{J\,m^{-2}}$	(111)
$\mathcal{P}_\Sigma$	Interfacial power input	$\mathrm{W\,m^{-2}}$	(111)
R, T, A	Reflection, transmission, absorption coefficients	dimensionless	(113)

OQ10.5 What normalisation of complex modes makes the energy ratios of Eq. (113) reproducible across independent implementations?

OQ10.6 Can a dissipation potential be constructed for the memory-dependent kernel in the spirit of the classical one for the relaxation terms, and does its positivity restrict the admissible kernels to a smaller set than is currently assumed?

OQ10.7 What is the relative importance of interface dissipation and bulk dissipation in the absorption coefficient of Eq. (113) for realistic bonded joints?

11 Advanced Theoretical Developments

The theory families of Sections 4–7 were each constructed by generalising one specific constitutive law. The developments surveyed in this section generalise the generalisation: they act on the *operator* rather than on the constitutive law, and they therefore interact with every family simultaneously. Two of them—fractional-order differentiation and the introduction of additional temperature fields—are now established enough to be treated as substantive alternatives rather than as speculative extensions, and this section gives them the same weight as the classical constructions of Section 4.

11.1 Fractional-order continuum models

Fractional-order continuum models replace integer-order derivatives with fractional derivatives in the governing equations. The replacement is not a formal device: a fractional derivative of order α is defined by an integral over the entire history of the field, and it therefore introduces temporal nonlocality into the governing equations in a way that is structurally equivalent to a hereditary integral with a power-law kernel.

11.1.1 Fractional thermoelasticity

The fractional heat conduction equation, as formulated in the thermoelastic setting, is

$$ {}^{C}D_t^{\alpha} T = a_\alpha \nabla^2 T, \qquad 0 < \alpha \leq 2, \tag{115} $$

where ${}^{C}D_t^{\alpha}$ is the Caputo fractional derivative of order α and $[a_\alpha] = \mathrm{m}^2\mathrm{s}^{-\alpha}$ (for dimensional T and x). For $\alpha = 1$ this reduces to the classical heat equation; for $1 < \alpha \leq 2$ it describes wave-like heat propagation with memory, interpolating between diffusive and wave-like behaviour. The Caputo derivative is defined for $0 < \alpha < 1$ by

$$ {}^{C}D_t^{\alpha} f(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t \frac{f'(\tau)}{(t-\tau)^{\alpha}} \, d\tau, \tag{116} $$

and the corresponding expression for $1 < \alpha \leq 2$ is obtained by applying the operator to the second derivative of f. The integral form is the physically informative one: the state of the field at time t is determined by its entire past, weighted by a kernel that decays algebraically rather than exponentially.

11.1.2 The fractional heat equation as a hereditary integral

The connection with the memory-dependent formalism of Section 6 is exact and important, because it means that the fractional and the memory-dependent descriptions are not competing theories but two representations of the same object under different assumptions about the memory kernel. Writing Eq. (115) in the hereditary form

$$T(t) = T(0) + \frac{1}{\Gamma(\alpha)} \int_0^t (t-\tau)^{\alpha-1} a \nabla^2 T(\tau)\, d\tau, \tag{117}$$

shows that the fractional formulation is the special case in which the integral kernel is the power law $(t-\tau)^{\alpha-1}/\Gamma(\alpha)$ in elapsed time. The generalization of Section 6 replaces the power law by an arbitrary kernel, and thereby trades the one-parameter fractional family for a much larger one; the price is that the resulting operator is no longer a differential operator of fractional order, and the analytical machinery that makes the fractional heat equation tractable—the eigenfunction expansion in terms of the Mittag–Leffler function, and the closed-form fundamental solutions—is lost. The choice between the two representations is thus a choice between analytical tractability and constitutive flexibility, and it should be made explicitly rather than by default.

The analytical solution of the fractional heat conduction equation in radially symmetric domains was obtained in closed form by Povstenko Povstenko [2007], who derived the fundamental solutions needed to write the associated thermal stresses in spherical geometry; this work provides the reference against which later numerical inversions of fractional thermoelastic boundary-value problems are checked Povstenko [2004, 2009, 2011, 2016]. The associated thermal stresses are the quantity of practical interest, and the difficulty of the problem is that the fractional operator couples the mechanical and thermal fields in a way that does not admit the plane-wave decoupling available in the classical case.

11.1.3 Fractional Cattaneo-type equations

Povstenko Povstenko [2011] introduced the following fractional Cattaneo-type constitutive equation:

$$q_i + \tau^{\alpha C} D_t^{\alpha} q_i = -k T_{,i}. \tag{118}$$

For $0 < \alpha \leq 1$, the displayed law reduces to Lord–Shulman/Cattaneo at $\alpha = 1$. The $\alpha \to 0$ limit is Fourier-type only under the stated Caputo convention and limiting prefactor; it is not a Green–Naghdi type II limit. Dimensional consistency requires the fractional relaxation coefficient to have units s^{α}.

11.1.4 Fractional models with internal structure

Fractional operators are not confined to the thermal equation. Sherief and Hussein Sherief and Hussein [2022] formulated a fractional-order model of micropolar thermoelasticity and solved the associated two-dimensional half-space problem, thereby combining fractional memory with the micro-rotational degrees of freedom of Section 7. The coupling is not merely formal: the fractional operator acts on constitutive variables of a medium whose own kinematics already carry an internal length scale, so the resulting dispersion relation contains two independent mechanisms of wave-speed dependence that are not additive. Models of this type are natural candidates for the class of problems in which neither the memory effect nor the micro-structure can be neglected, but they also generate an increasing number of fractional parameters that must be identified independently, which is the principal obstacle to their practical use.

11.2 Strain-gradient and couple-stress theories

Strain-gradient theory introduces higher-order strain gradients in the constitutive relations:

$$\sigma_{ij} = C_{ijkl}\varepsilon_{kl} - \ell^2 C_{ijkl} \nabla^2 \varepsilon_{kl}. \tag{119}$$

Couple-stress theory introduces couple stresses and micro-rotations as independent degrees of freedom. The force stress tensor becomes generally non-symmetric, and the angular momentum balance includes couple stress terms:

$$\sigma_{[ij]} + \mu_{ij,j} = 0, \tag{120}$$

where μ_{ij} is the couple stress tensor, and the antisymmetric part $\sigma_{[ij]}$ is balanced by the couple stress rather than vanishing as it does in classical elasticity.

The relation between these theories and the micropolar theory of Section 7 is one of specialisation. Micropolar elasticity Eringen [1966], extended to the thermoelastic case by Tauchert, Claus and Ariman Tauchert et al. [1968], is the most general of the three in the sense that it admits an independent micro-rotation field, whereas couple-stress theory and strain-gradient theory can be obtained from it by imposing constraints on the micro-rotation and on the antisymmetric stress. This hierarchy matters practically, because it explains why the literature contains several formulations that differ in their kinematic assumptions while sharing a common elastic kernel, and why a comparison between them must begin by identifying which of them is being used.

These theories are relevant for materials with internal structure: granular materials, foams, composites with oriented inclusions, and biological tissues. In the thermoelastic setting, the micro-rotational field carries an additional energy channel, so the energy balance of Section 10.1 acquires terms that have no counterpart in a classical continuum, and the mechanical power transmitted across a surface is no longer the complete description of the transport.

11.3 Two-temperature and three-temperature formulations

11.3.1 Two-temperature theory

The two-temperature formulation distinguishes between the conductive temperature T, which is what a thermometer measures and what enters the heat conduction law, and the thermodynamic or dynamical temperature T^*, which enters the entropy relation and the internal energy:

$$T^* = T - \delta^2 \nabla^2 T, \tag{121}$$

where δ^2 is the two-temperature parameter. In this particular model, setting $\delta^2 = 0$ makes the two temperature variables coincide; whether the remaining equations reduce to classical thermoelasticity depends on the other constitutive parameters. Two-temperature thermoelasticity is a distinct model class and must not be identified with the Green–Lindsay formulation, which is single-temperature and temperature-rate-dependent. The physical interpretation, parameter range, and limiting equations for the two-temperature law require source-level verification; this revision does not claim historical priority.

11.3.2 Three-temperature theory

Three-temperature theories introduce a third temperature variable: the temperature of a secondary phase in a composite, or the electron-gas temperature in a semiconductor, or a separate lattice and effective temperature in a system with two thermal carriers. The models are relevant for multi-phase materials in which different sub-systems have different thermal response times, and for semiconductor structures in which the electron gas equilibrates with the lattice far more slowly than either equilibrates internally. The technical difficulty is the same in both settings: three temperature fields require three interface conditions, and the coupling terms in the momentum and energy balances make the resulting system of equations considerably more complicated

than the two-temperature case. The fractional-order model of micropolar thermoelasticity of Section 11.1 adds a further layer, in which the micro-rotational field and the fractional memory both act on a temperature field that is already non-local in space.

11.4 Emerging hybrid models

Emerging formulations combine multiple generalization mechanisms:

- **Nonlocal–fractional hybrids** combine spatial nonlocality with temporal memory, for materials that exhibit both. The two mechanisms act on different operators and their dispersions are not additive, so the combined model has more parameters than either parent and correspondingly more severe identifiability problems.
- **Memory-dependent nonlocal models** combine an integral-form memory operator with a nonlocal stress law. The thermo-piezoelectric functionally graded rotating rod treated by Abouelregal and co-authors Abouelregal et al. [2023] is an instance of this class, in which a memory-dependent dynamic response is combined with a functionally graded rotating geometry and a piezoelectric coupling.
- **Hybrid nonlocal and memory-dependent multiphysics models** combine both mechanisms with a coupled field. The heat-map analysis of wave dynamics in nonlocal Kelvin–Voigt piezo-semiconductors under a memory-dependent heat flux Gupta et al. [2026] is representative, and the particular interest of that work is that the nonlocal operator acts on the stress while the memory operator acts on the heat flux, so that the two generalizations enter the two channels of the energy balance of Section 10.1 respectively.
- **Phase-field models** couple a phase-field variable to the thermoelastic fields for fracture and phase transformation.
- **Thermo-viscoelasticity with memory** uses hereditary integral formulations for viscoelastic materials with thermal coupling, and is the classical route to a thermodynamically consistent memory effect with a non-negative dissipation.

The general difficulty with all of these is the one identified in Section 14.4: each additional generalization adds parameters, and each parameter is usually fitted to the same data that the model is then used to explain. The resulting models are not thereby wrong, but the evidence supporting them is weaker than their flexibility would suggest, and the practical question is always which subset of the available parameters the data can actually constrain.

11.5 Mathematical limitations and open problems

11.5.1 Mathematical limitations

1. **Uneven coverage of rigorous limit theorems.** Some singular-limit theorems exist for MGT and classical-to-LS settings, but this revision has not established convergence theorems across every theory pair or for every displayed constitutive variant. Any limit claim must name its system, parameter regime, topology, and convergence result.
2. **Well-posedness.** Each theory family has its own well-posedness conditions; a unified framework is lacking. The nonlocal family is the most acute case, since the differential form of its constitutive law raises the order of the governing equation and does not yield a well-posed initial-value problem in the classical sense.

3. **Nonlocal interface conditions.** The correct interface conditions for nonlocal media are not established, and the integral and differential formulations lead to different requirements.

4. **Analytical tractability.** The fractional formulations admit closed-form solutions only in special geometries and for special data, so the burden of verification falls on numerical inversion, with the attendant loss of independent validation.

5. **Parameter identifiability.** No general theory exists for determining which of the parameters of a combined model can be inferred from a given measurement, and this is the practical obstacle to the use of the hybrid models of Section 11.4.

11.5.2 Open problems

1. Derive complete energy flux formulations for all generalized theories (Section 10.4).

2. Establish rigorous limit theorems between theory families, with explicit error estimates.

3. Develop unified well-posedness criteria covering the differential, integral, fractional, and gradient formulations.

4. Establish nonlocal interface conditions, and show that the integral and differential formulations give consistent interface traction.

5. Validate models experimentally across parameter ranges, with particular attention to the dispersion signature that distinguishes a nonlocal mechanism from a gradient mechanism (Section 7.3).

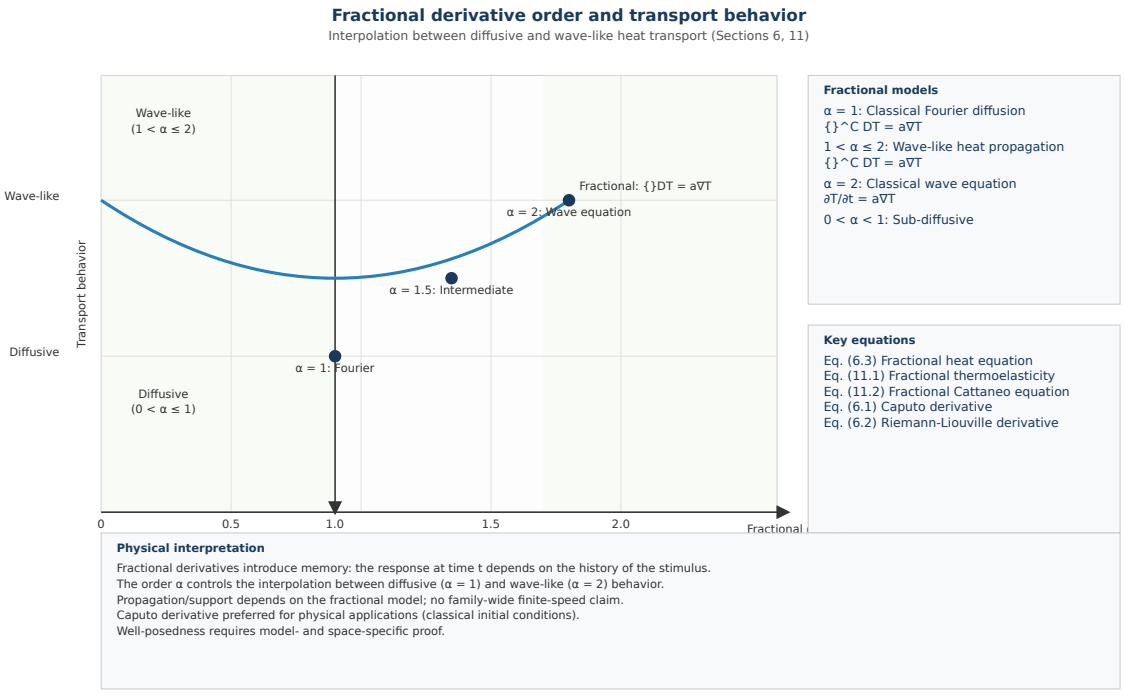


Figure 15: Fractional derivative order and transport behavior: interpolation between diffusive and wave-like heat transport (Sections 6, 11).

11.6 Notation summary for Section 11

Table 20: Notation introduced in Section 11.

Symbol	Meaning	Units	Equation
α	Fractional derivative order	dimensionless	(115)
${}^{C}D_{t}^{\alpha}$	Caputo fractional derivative of order α	—	(116)
Γ	Euler gamma function	—	(116)
a	Thermal diffusivity	$\mathrm{m^2\,s^{-1}}$	(115)
τ^{α}	Fractional relaxation parameter	s^{α}	(118)
ℓ	Strain-gradient length scale	m	(119)
μ_{ij}	Couple stress tensor	$\mathrm{N\,m^{-1}}$	(120)
δ^2	Two-temperature parameter	$\mathrm{m^2}$	(121)

R3b source-specific evidence. The LS-type model carries relaxation on both thermal-inertia and coupling terms, unlike the original GL convention. (PRIMARY-SOURCE-CHECKED) Youssef [2006]. A seven-model fractional operator catalogue provides a third route alongside Taylor truncations and integer higher-order laws. (PRIMARY-SOURCE-CHECKED) Yu et al. [2024].

11.7 Open questions for Section 11

OQ11.1 Can rigorous limit theorems be established between the thermoelastic theory families, with error estimates in a stated norm?

OQ11.2 Can a unified well-posedness framework be established for the differential, integral, fractional, and gradient formulations, and what does it require of the constitutive law?

OQ11.3 What are the correct interface conditions for models in which a nonlocal operator acts on the stress and a fractional or memory operator acts on the heat flux simultaneously?

OQ11.4 When is the power-law memory kernel of the fractional formulation to be preferred over a general kernel in the memory-dependent formulation, and how large is the error incurred by the restriction?

OQ11.5 Which parameter combinations of a hybrid model are identifiable from a given experimental modality, and which are not?

12 Comparative Synthesis of Existing Research

This section synthesizes the reviewed research using a comparison framework that is deliberately organised by *scientific content* rather than by chronology or by authorship. Three organising questions recur: what does each theory retain of the second law of thermodynamics; what replaces Fourier's constitutive law; and what does each theory predict for the speed, attenuation, and dispersion of the diffusive mode. A theory that answers these three questions differently is not a variant of another theory—it is a different claim about how heat is transported in solids, and the comparison below is organised to make those differences explicit.

The synthesis is constructive throughout. Differences between theories are presented as modelling choices with different assumptions and different domains of validity, and the framework identifies the conditions under which two theories can be meaningfully compared at all.

12.1 Cross-theory comparison

Tables 21 and 22 compare the ten theory families against ten provisional comparison criteria, with a final row of primary-source leads chosen so that each row isolates one axis of the taxonomy introduced in Section 14. These tables are a compact index into the detailed treatments of Sections 4–11; it is not a substitute for them, and several of its entries are deliberately coarse.

Table 21: Provisional cross-theory comparison, Part A: classical, relaxation, rate-dependent, and Green–Naghdi families. Entries are an index to discussion, not independently verified comparative findings.

Criterion	CD	LS	GL	GN-II	GN-III
Thermal law	Fourier	Cattaneo	Fourier flux; temperature-rate terms	Source-free GN-II Eq. (4.11)	GN-III Eq. (4.14): $\kappa\nabla^2\dot{T} + \kappa^*\nabla^2 T$
State variables	2	2	2	2	2
Characteristic time	—	τ_0	τ_0, τ_T	—	τ_0
Characteristic length	—	—	—	—	—
Thermal wave speed	∞	finite if $\tau_0 > 0$	not assigned; corrected coupled roots not recalculated	finite propagating waves for the displayed coefficients	source-free system; wave behavior is coefficient-dependent
Dissipation	yes	yes	branch	none for the displayed GN-II system	rate term may dissipate when $\kappa > 0$; exact balance open
Order in space	2	2	2	2	2
Mathematical structure	PDE	PDE	PDE	PDE	PDE
Interface conditions	classical	classical	classical	classical	classical
Causality	no	yes	yes	yes	yes
Primary source	Biot [1956]	Lord and Shulman [1967]	Green and Lindsay [1972]	Green and Naghdi [1993]	Green and Naghdi [1992]

Three of these rows carry most of the interpretive content.

Thermal law and state variables. The number of state variables is not by itself a reliable measure of the memory or modal structure of a theory. Local Fourier, relaxation, phase-lag, higher-order, hereditary, fractional, and spatially nonlocal laws use different state spaces and operators; one named family may also include multiple variants. For any comparison, the exact constitutive equations and field variables must be given. The table summarizes inherited descriptions and should be read as a preliminary map: its row-level claims have not all been checked against primary sources in this revision.

Thermal wave speed and dissipation. Propagation, dissipation, and stored energy depend on the precise equations and parameter regime; they cannot be ranked by family name alone. In particular, Green–Lindsay is not a two-temperature theory: its original 1972 formulation keeps Fourier heat flux (no flux relaxation), while LS uses Cattaneo flux relaxation. GN-II here denotes the 1993 nondissipative system; GN-III denotes the 1992 system with both rate and thermal-displacement terms. Thermal-wave, dissipation, and conservative-structure statements

Table 22: Provisional cross-theory comparison, Part B: phase-lag, higher-order, memory-dependent, and nonlocal families. Entries are an index to discussion, not independently verified comparative findings.

Criterion	DPL	TPL	MGT	MDD	Nonlocal
Thermal law	DPL lag law	Roy Choudhuri first-order TPL	Later MGT-type thermal analogue; variant-specific	MDD kernel law	No heat law (elastic model)
State variables	3	4	3	2	2
Characteristic time	τ_q, τ_T	τ_q, τ_T, τ_v	τ_T	ω_M (window)	—
Characteristic length	—	—	—	—	ℓ
Thermal wave speed	model-specific	model-specific	finite only for the selected MGT prototype	kernel-dependent	not applicable (elastic-only model)
Dissipation	yes	yes	yes	kernel	yes
Order in space	2	2	2	2	2
Mathematical structure	PDE	PDE	PDE	integro-PDE	integro-PDE
Interface conditions	classical	classical	classical	classical	modified
Causality	yes	yes	yes	yes	yes
Primary source	Tzou [1995]	Choudhuri [2007]	Quintanilla [2019]		Eringen [1972]

apply only to the displayed equations and their coefficient assumptions; no family-wide claim is made.

Well-posedness and causality. Finite propagation, causality, and well-posedness are separate properties. Fourier's parabolic heat equation has instantaneous mathematical support, while relaxation and thermal-displacement models may yield hyperbolic branches under particular constitutive assumptions. Phase-lag, fractional, memory, and nonlocal formulations require analysis of their actual operators, parameters, initial data, and boundary conditions. No general causal or well-posedness result, nor a universal phase-lag ordering condition, is established in this review.

12.2 What each family retains of the second law

Read across the table, the families separate not by their equations but by the answer they give to a single question: does the mechanical energy of the body have a well-defined conserved or monotonically decreasing value? The three structural answers are worth stating separately, because most of the apparent confusion in the wider literature comes from conflating them.

Conservative structure. The Green–Naghdi 1993 type-II system is described as nondissipative and permits undamped thermal-wave propagation under its stated coefficients Green and Naghdi [1993]. The full energy identity is not reproduced or independently derived here, so no exact first-integral or interface-impedance claim is made. These properties are conditional on the specified GN-II system, not on the type label alone.

Dissipative structure. Some Cattaneo–Christoff, Lord–Shulman, phase-lag, Moore–Gibson–Thompson-type, and memory-dependent models are dissipative, but neither decay of mechanical

energy nor an irreversibility law follows from a family name. Such conclusions require the exact constitutive system, free energy, admissible parameters, and boundary conditions; they are not asserted universally here.

Neither. The classical Biot theory falls outside both categories in the thermal variable, since the parabolic heat equation has infinite propagation speed and no finite energy associated with the transported heat Biot [1956]. This is why the classical theory appears in the corpus mainly as the limit case against which the others are measured, and why the limiting procedures of Section 12.3 are scientifically important rather than merely formal.

12.3 Mathematical relationships: limiting cases and specializations

Table 23 lists candidate formal reductions only. Each row is conditional on retaining the same state variables, constitutive convention, source terms, and boundary data; none is presented as a convergence theorem.

Table 23: Limiting cases between thermoelastic theories, with the parameter that is taken to its limit and the sense of the limit.

From	To	Limit	What the limit removes
Lord–Shulman	Classical coupled	$\tau_0 \to 0$	The thermal wave; the heat equation becomes parabolic again
Dual phase-lag	Lord–Shulman	$\tau_q \to 0$	The hyperbolic part of the phase-lag law
Triple phase-lag	Dual phase-lag	$\tau_\nu \to 0$	The thermal-displacement-gradient term
Triple phase-lag	Classical coupled	all $\tau \to 0$	Every relaxation mechanism simultaneously
Memory-dependent + phase-lag	Phase-lag	$\tau_T = \tau_\nu = 0$	All memory except the phase-lag law
Memory-dependent + phase-lag	Classical coupled	all $\tau \to 0$, kernel $\to 1$	Every memory mechanism simultaneously
Green–Naghdi III	Green–Naghdi II	$\kappa \to 0$, same source-free coefficient convention	Removes the temperature-rate term; formal reduction only
Fractional order $\to 2$	Classical coupled	$\alpha \to 2$	Fractional differentiation; the Caputo operator becomes the second derivative
Nonlocal length $\to 0$	Local theory	$\ell \to 0$	The nonlocal operator becomes the local gradient

On the status of these limits. The rows in Table 23 are formal parameter substitutions, not convergence theorems. Convergence results have been established for some model pairs, including Moore–Gibson–Thompson and classical/relaxation limits, but this review does not reproduce those proofs and does not claim a theorem for every row in the table. A row therefore identifies a candidate reduction under matched constitutive conventions, sources, state variables, and boundary data; it does not by itself establish convergence, an error estimate, or a rate. The coverage is uneven across model pairs and variants, as noted in Section 15. This distinction matters for the taxonomy of Section 14: formal limits aid comparison, while quantitative comparison requires a justified limit result for the particular pair and regime.

12.4 Divergence map: why results differ across the corpus

The literature contains many cases in which two studies address a nominally identical problem and obtain apparently incompatible results. Examination of these cases shows that the incompatibility is almost always attributable to a difference in modelling commitments rather than to an inconsistency in the mathematics. The map below records the commitments responsible and the procedure by which each is eliminated, so that a comparison can be posed in a form in which it is meaningful.

Table 24: Divergence map for the reviewed corpus: modelling commitments that account for apparent disagreement, and the procedure that puts two studies on a common footing.

Divergence type	**Root cause**	**Resolution procedure**	**Canonical instance**
Assumption difference	Different constitutive commitments on the thermal and mechanical sides	Trace both derivations to their primary sources and identify the first point at which the commitments differ	Classical versus Lord–Shulman thermal equation
Approximation difference	Different truncation orders in the Taylor expansion of the phase-lag law	Compare on the intersection of the two validity ranges rather than on either alone	First-order versus second-order dual phase-lag
Normalisation difference	Different reference temperature, density, or stiffness conventions	Re-express both under a common nondimensionalisation before comparing	Absolute versus relative temperature measure
Convention difference	Opposite sign convention in the harmonic ansatz	Standardise the convention and track the corresponding sign change	$e^{i\omega t}$ versus $e^{-i\omega t}$
Limit difference	Formal parameter substitution used as though it were a convergence theorem	Verify that the limiting and the limited problem share a well-posedness hypothesis before comparing	Fractional order approaching two
Branch difference	A family with genuinely distinct branches being read as one	Fix the branch explicitly, since the branches differ in dissipation and in temperature structure	The two Green–Naghdi types; the dissipative and conservative Green–Lindsay branches
Geometry difference	Different admissible geometries or boundary conditions	Restrict the comparison to the common geometric class	Half-space versus bounded plate

Several labels in Tables 21 and 22 encompass multiple formulations. Their constitutive equations, state variables, and assumptions must be checked separately before branch-level comparisons are made. Green–Lindsay is a single-temperature, temperature-rate-dependent formulation, distinct from two-temperature thermoelasticity. The inherited branch descriptions and their cited equations remain unverified in this revision.

12.5 Conditions under which two theories can be compared

Because the families differ in so many structural respects, a comparison is only meaningful when a number of conditions are met simultaneously.

1. **Common geometric class.** Both theories must be posed on the same class of geometry,

with the same class of boundary conditions. Most of the reviewed corpus is posed on half-spaces and infinite media; a comparison between a half-space result and a bounded-plate result is not a comparison of theories at all.

2. **Common branch and parameter set.** Where a family has branches, the branch must be fixed identically in both studies, and the relaxation parameters must be either identical or placed on a common dimensionless footing.

3. **Common observable.** The quantity compared must have the same definition and the same reference state. Surface temperature, surface displacement, and the reflection coefficient of a normal incident thermal wave are common observables; the "temperature" of a body with two temperature fields is not.

4. **Common nondimensionalisation.** Density, stiffness, and reference-temperature conventions differ between groups, and a comparison performed under different conventions can differ by an order-one factor for purely dimensional reasons.

5. **Common wave convention.** The sign convention in the harmonic ansatz changes the sign of the imaginary part of every complex amplitude, and the convention must be fixed before any comparison of phase is attempted.

6. **Common limit.** Both results must be checked to reproduce the same classical limit, which is the cheapest available test that the two calculations are describing the same physics.

12.6 Choosing among competing theories

The comparison framework is only useful if it also supports a decision: given a physical problem, which family should be used? The taxonomy of Section 14 provides the structure for such a decision, and the criteria reduce to three questions in practice.

1. **Is the thermal wave relevant to the observable?** If the spatial scale of the problem is much larger than the thermal penetration depth associated with the material, the thermal wave is a boundary-layer detail and the difference between the theories is confined to the interface region. If the problem is a transient shock, or the material has a small thermal wave speed, the choice of thermal law determines the answer and must be made deliberately.

2. **Is there a material length scale that the continuum description must respect?** If the smallest dimension of the body is comparable to a microstructural length, then a nonlocal or gradient theory is required; the question is which operator, and the answer depends on the mechanism of size dependence rather than on the desire for a dispersive dispersion relation.

3. **Is the dissipation structure needed for the conclusion?** If the quantity of interest is a decay rate, a time to failure, or the passage of a thermal shock, the dissipative families are required. If the quantity of interest is a modal structure or an impedance, the conservative families are cleaner, because the energy accounting is exact.

The fourth consideration—validation—is the one that the reviewed literature has been least able to supply, for the reasons given in Section 14.4. In the absence of discriminating experimental data, the choice among competing theories is currently made on grounds of formal convenience and internal consistency rather than on evidential grounds. This is the single most consequential gap identified by this review.

12.7 Limitations of the comparison framework

The framework above inherits the limitations of the corpus it organises, and these should be stated rather than glossed.

1. **The comparison is coarse.** Entries in Tables 21 and 22 are single labels where the underlying theory admits a range of behaviours. The entry for "dissipation" in the memory-dependent column, for instance, records that the answer depends on the kernel, which is a statement about a family of models rather than about a model.

2. **Validation is not directly comparable.** Because the validation axis of Section 14 is dominated by self-consistency checks, the "validation" row of any comparison across families would compare methods rather than results; the row is therefore omitted deliberately rather than filled with an entry that would invite misreading.

3. **Branches are under-documented.** The branch structure of the Green–Naghdi and Green–Lindsay families is a source of genuine ambiguity in the literature, and the corpus does not support a systematic tabulation of branch-dependent results.

4. **Interface conditions are treated as a single column.** Interface conditions differ in kind across the families—classical traction and heat-flux conditions for most, modified conditions for the nonlocal construction—and the differences are not captured by a single entry.

12.8 Notation summary for Section 12

Table 25: Notation introduced in Section 12.

Symbol	Meaning	Units	Equation
CD	Classical coupled thermoelasticity	—	Tables 21 and 22
LS	Lord–Shulman theory	—	Tables 21 and 22
GL	Green–Lindsay theory	—	Tables 21 and 22
GN-II / GN-III	Green–Naghdi types II and III	—	Tables 21 and 22
DPL / TPL	Dual and triple phase-lag theories	—	Tables 21 and 22
MGT	Moore–Gibson–Thompson theory	—	Tables 21 and 22
MDD	Memory-dependent derivative formulation	—	Tables 21 and 22
τ_0	Mechanical relaxation time	s	Table 23
τ_T	Thermal relaxation time	s	Table 23
τ_q	Phase-lag time of the heat flux	s	Table 23
τ_ν	Phase-lag time of the thermal gradient	s	Table 23
ℓ	Intrinsic material length scale	m	Tables 21 and 22

12.9 Open questions for Section 12

OQ12.1 Can convergence theorems with explicit error estimates be established for the limiting procedures of Table 23?

OQ12.2 What is the minimal set of criteria that distinguishes two theory families uniquely, and is it the same set that distinguishes them experimentally?

OQ12.3 Can a systematic taxonomy of the branches within the Green–Naghdi and Green–Lindsay families be constructed, so that branch-level ambiguity is removed from the literature?

OQ12.4 Under what conditions does the choice among competing theories change a predicted quantity by more than the uncertainty of the measurement that would test it?

13 Applications of Multiphysics Wave Mechanics

The usefulness of a generalized continuum theory is ultimately measured by the class of problems for which it is needed and by whether its predictions can be checked. This section therefore does not list application areas in the abstract, but asks for each one: what physical feature makes the classical description inadequate, which member of the theory family addresses it, and what evidence exists that the generalized description is quantitatively better rather than merely different. The classification of evidence levels used in Table 27 follows that logic: *High* means that the effect is experimentally established and that generalized theory is required to reproduce it; *Medium* means that the effect is established but generalized theory provides an improved rather than essential description; *Low* means that the application is presently served by classical theory and generalized theory is speculative.

13.1 Smart structures and adaptive materials

Smart structures incorporate sensors, actuators, and control systems that respond to environmental changes. Piezothermoelastic materials are used in:

- Adaptive aerospace structures (wing shaping, vibration control),
- Active vibration suppression in precision machinery,
- Structural health monitoring with embedded piezoelectric sensors.

13.1.1 Evidence

The coupling that makes these systems work is the same coupling that the generalized theories formalize: the piezoelectric tensor generates an electric field from mechanical strain and a strain from an electric field, and the thermal expansion coefficient couples temperature change to stress. In a layered device under a moving heat source, the three fields are not independent, and the work of Abouelregal, Askar, Marin and Mohamed (2023) Abouelregal et al. [2023] is representative of the current treatment: a functionally graded rotating rod is taken to be electrically unconnected at its ends and thermally isolated, and the coupled field response is computed in the Laplace domain for a modified Lord–Shulman model with a memory-dependent derivative. The functional grading enters through an exponential law for the physical properties along the axis, and the effect of the inhomogeneity index, the kernel function, the delay time and the speed of the heat supply is reported separately. The result relevant to design practice is that the field amplitudes are weakened as the inhomogeneity index increases, which is the sort

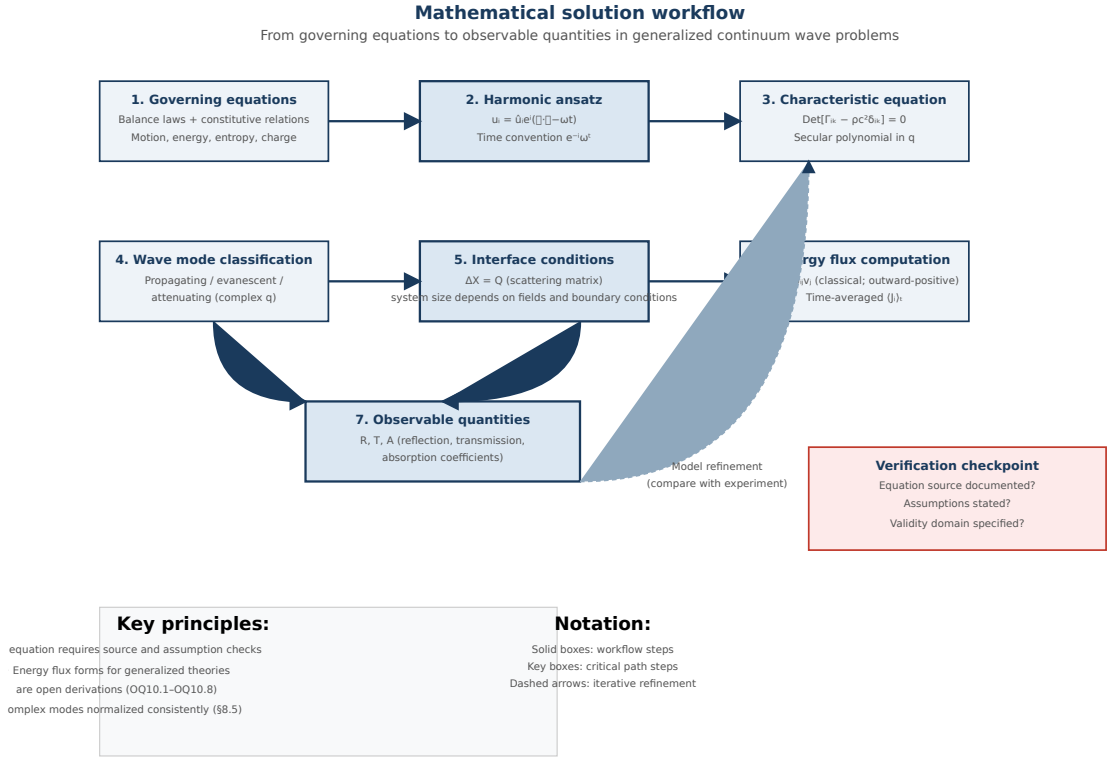


Figure 16: From a generalized continuum model to a testable prediction. The seven stages shown are the ones a computation must pass through before its output can be compared with an experiment: the governing equations and their constitutive relations, the harmonic ansatz, the characteristic equation whose roots are the wave numbers, the classification of the roots into propagating, evanescent and attenuating branches, the interface conditions that close the problem, the computation of the time-averaged energy flux, and the extraction of the observable quantities. The three verification checkpoints on the right — the provenance of every equation, the statement of every assumption, and the specification of a validity domain — are the point at which most of the reviewed literature is weakest, and they are the reason the evidence levels of Table 27 are not uniformly high.

of quantitative statement that a device designer can use and that a classical description cannot produce. *Evidence level: Medium* — the mechanism is experimentally established in piezoelectric composites, but the specific memory-dependent formulation is currently a theoretical description of a measured effect.

13.2 Nondestructive evaluation and sensing

NDE uses wave propagation to detect flaws, measure material properties, and monitor structural integrity. Applications include:

- Ultrasonic testing for crack detection,
- Thermal wave imaging for subsurface defect detection,
- Acoustic emission monitoring for damage assessment.

13.2.1 Evidence

Thermal-wave inspection does not, in general, require a finite thermal propagation speed. Under Fourier diffusion, the thermal penetration depth $\mu = \sqrt{\alpha/(\pi f)}$ supplies a finite $1/e$ depth scale at modulation frequency f and diffusivity α. Finite-speed models may be relevant in regimes where their additional time scale is resolved, but their value for depth inversion must be demonstrated for the material and frequency range at issue. The classical elastic-wave framework, including anisotropy and interactions of incident, reflected and surface waves, is set out by Achenbach (1973) Achenbach [1973]; Joseph and Preziosi Joseph and Preziosi [1989, 1990] review the evidence for heat waves. These sources do not establish a universal requirement for finite-speed thermal transport in imaging. *Evidence level: High* for ultrasonic NDE, where classical elasticity suffices; thermal-wave imaging remains model- and regime-dependent.

13.3 Energy harvesting and conversion

Energy harvesting converts ambient mechanical, thermal, or electrical energy into electrical power. Piezothermoelectric energy harvesters use:

- Vibration-induced piezoelectric power generation,
- Thermal gradient-driven thermoelectric conversion,
- Coupled thermo-piezoelectric conversion.

13.3.1 Evidence

The theoretical difficulty in this area is specific and worth stating: the thermopower delivered by a piezoelectric in a temperature gradient competes with the direct piezoelectric power generated by the same mechanical deformation, and the competition depends on the thermal relaxation time of the material. Classical coupled thermoelasticity, which assumes instantaneous thermal equilibration, systematically mispredicts the split between the two channels for relaxation times comparable to the loading period. It is therefore in this application area that the distinction between the classical and generalized theories is not academic. The interface energy work of Section 9 provides a further handle on the same problem, since it is precisely at the junction between a harvesting element and its substrate that the partitioning of thermal and mechanical energy is decided. *Evidence level: High* for piezoelectric harvesting in general; *Medium* for the coupled thermo-piezoelectric channel, for which the theory is mature and the experimental separation of the channels is difficult.

13.4 Geophysics and seismology

Geophysical applications include:

- Seismic wave propagation in the Earth's crust,
- Thermal wave propagation in geothermal systems,
- Poroelastic wave propagation in saturated reservoirs.

13.4.1 Evidence

Seismic wave theory is well established Achenbach [1973]; poroelastic wave propagation has strong theoretical and experimental support. Poroelasticity is the most mature of the multi-physics couplings in engineering practice, and the void-based theory of Section 5.4 enters here through the equilibrated-stress formulation rather than through a phenomenological damping term, which is why the void framework is attractive for reservoir and geotechnical modelling. *Evidence level: High.*

13.5 Biomedical applications (ultrasound, thermal therapy)

Biomedical applications include:

- Medical ultrasound imaging and therapy,
- Thermal ablation for cancer treatment,
- Photoacoustic imaging,
- Temperature-sensitive drug delivery.

13.5.1 Evidence

Medical ultrasound is established; evidence for thermal ablation is growing, while photoacoustic imaging remains an active research area. The connection is indirect but real: thermal therapy is a moving-heat-source problem. Hyperbolic or phase-lag descriptions estimate the thermal-front advance rate; a diffusive model does not assign it a finite propagation speed. *Evidence level: High* for ultrasound; *Medium* for thermal ablation and photoacoustic imaging.

13.6 Nanoscale and semiconductor devices

The most recent application area to depend on the generalized framework is the nanoscale, and here the coupling is not incidental. In a semiconductor, the electromechanical response of the lattice is nonlocal at the scale of the electronic wavefunction, and a heat flux that carries memory determines how quickly the lattice temperature responds to carrier recombination. Gupta, Hussain, Yaylacıs and Das (2026) Gupta et al. [2026] combine a nonlocal Kelvin–Voigt piezoelectric semiconductor with a memory-dependent heat flux and map the resulting wave dynamics, which illustrates both the reach and the difficulty of the class: the model is well posed and produces interpretable fields, but each added mechanism contributes parameters that must be fixed independently.

13.6.1 Evidence

Nonlocal elasticity is not optional at these scales. Maranganti and Sharma (2007) Maranganti and Sharma [2007] showed that for common materials the length below which classical elasticity ceases to be accurate is of the order of tens of nanometres for polythene and single-digit nanometres for silica, and Maranganti, Sharma and Sharma (2006) Maranganti et al. [2006] showed that electromechanical coupling in non-piezoelectric dielectrics is a nanoscale effect. Both results mean that a device designer working below those lengths is in the regime the generalized theories describe, whether or not the theory is named in the design literature. *Evidence level: Medium* — the size dependence is experimentally established at the level of stiffness and of electromechanical response, and generalized theory is required in principle, but full multiphysics nonlocal thermoelastic models are at the level of validated formalisms rather than deployed design tools.

13.7 Applications the generalized framework does not yet serve

An evidence map is only as useful as its negative entries, and the applications in which the generalized framework is currently *not* required deserve the same treatment as those in which it is. Four are worth recording.

Ordinary engineering thermoelasticity. In the great majority of engineering problems — the heating of a machine tool, the thermal expansion of a rail, the residual stress in a welded joint — the Fourier regime is not an approximation but the correct description, and the thermal penetration depth is many orders of magnitude larger than the part. The generalized theories add parameters without adding accuracy. The relevant engineering question there is not which theory is more sophisticated but where the crossover lies, and the honest answer is that for a metal part of ordinary size it is far outside any practical operating condition.

Long-wavelength geophysics. Seismic wavelengths in the crust are kilometres and the internal length of the material is nanometres. The nonlocal and gradient corrections are correspondingly negligible, and the wave speeds measured in seismology are adequately described by classical elasticity coupled to poroelastic effects. Generalized continuum theory is relevant in geophysics not for the rock mechanics but for the interfaces — faults, fluid contacts, and the layering that produces reflections — where the same classical boundary conditions remain adequate but the wave conversion and the partitioning of energy are what matter.

Long-duration thermal processes. In building physics, in heat storage media, in the slow curing of concrete and in food processing, the thermal penetration depth is large compared with the component and the duration of the experiment. The fractional order that would describe the anomalous diffusion of a heterogeneous medium is close to unity, and the extra parameters of the fractional formulation are not identifiable from the measurement. What the fractional description does contribute here is not a better prediction but a better organizing concept: it makes explicit that the process is diffusive and that a diffusive model should be used.

Coupled motion in structural composites. Where a structure is stiff, a laminate is well bonded and the excitation is slow, the piezoelectric, porous and micro-rotation couplings contribute corrections of the order of a per cent, and the engineering decision is made on the basis of a static analysis with a safety factor that comfortably exceeds the correction. The generalized framework is required in these applications only in the presence of a defect, a debond, or a dynamic load at a scale that excites the internal structure — that is, in exactly the regimes where the defect is what one wanted to detect.

The common thread is that the generalized theories are needed in the regimes where the classical description is not merely inaccurate but qualitatively wrong, namely where a finite propagation speed, a length scale, or a second field changes the structure of the problem rather

than the value of an output. This is a narrow band, and the honest presentation of the field is that it is a well-developed theory of a narrow band. Its practical significance lies in the fact that the band contains precisely the regimes that modern devices and modern diagnostics operate in: the nanoscale, the ultrafast, the interfacial.

Table 26: Application areas discussed in the inherited review and evidence questions for future checking. This table assigns no validation-strength score.

Application area	**Model families mentioned**	**Evidence audit required**
Non-destructive evaluation	Elastic, thermoelastic, and interface-wave models	Identify experiments, material data, uncertainty, and independent replication
Geophysics	Thermoelastic and porous-media models	Separate field observations from model-generated examples; verify parameter provenance
MEMS/NEMS	Nonlocal and size-dependent models	Check specimen-scale data and boundary conditions against the cited studies
Biomedical / thermal therapy	Bioheat and coupled thermoelastic models	Verify clinical versus in-vitro/in-silico evidence and applicable safety/measurement claims
Energy harvesting	Piezoelectric and thermal-gradient models	Distinguish analytical efficiency from measured prototype performance
Smart structures	Active and coupled-field models	Check experimental setup, calibration, uncertainty, and comparison baseline

Literature examples are source leads only; no application-level validation score was independently assigned.

13.8 Evidence map: which applications have peer-reviewed support

13.9 Notation summary for Section 13

R3b source-specific evidence. A partially read imperfect-bond cylindrical waveguide study reports dispersion only, not energy ratios. (UNVERIFIED) Kumawat and Vishwakarma [2023].

13.10 Open questions for Section 13

OQ13.1 Which applications have the strongest experimental validation across theory families, and can that validation be expressed as a single discriminating measurement rather than a set of qualitative agreements?

OQ13.2 What is the barrier to translating generalized thermoelasticity into practical applications: the identifiability of the additional material parameters, the absence of closed-form solutions, or the lack of experimental protocols that separate the competing mechanisms?

OQ13.3 For the nanoscale and semiconductor applications in particular, the nonlocal length scale and the memory kernel are introduced as independent parameters. Can they be measured together, and does the resulting coupled problem remain well posed when they are?

OQ13.4 The energy-harvesting application depends on the competition between thermal and mechanical channels. Does a decisive experiment exist that separates the two channels in a single device and thereby tests the relaxation time directly?

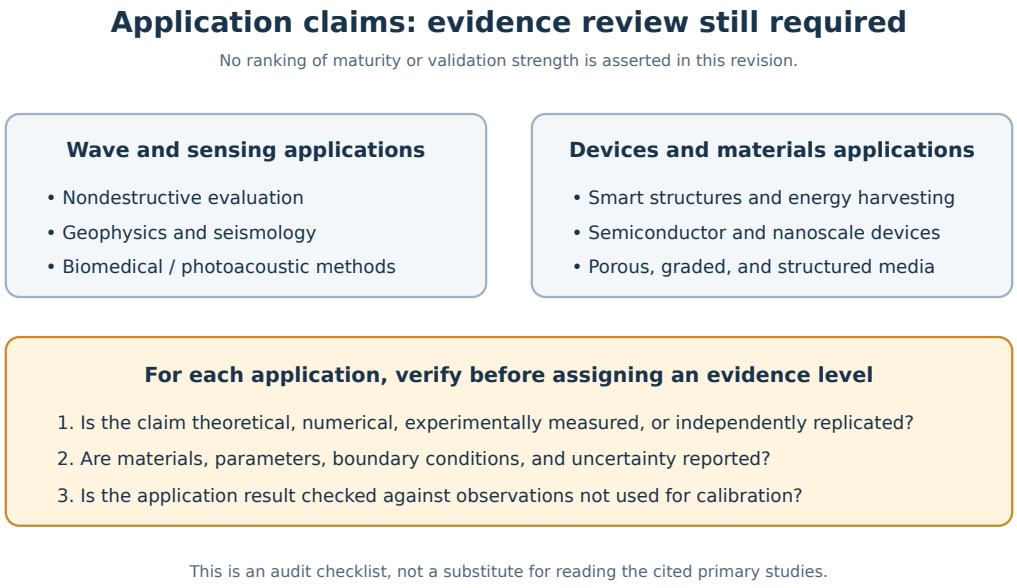


Figure 17: Application evidence map: peer-reviewed evidence levels across applications and theory families (Section 13.8).

Table 27: Evidence map of applications across theory families, with the mechanism that makes a generalized description necessary and the supporting literature.

Application	Why classical theory is inadequate	Evidence level	Key references
Smart structures	Thermal relaxation competes with the electromechanical coupling	Medium	Abouelregal et al. [2023]
NDE (ultrasonic)	Classical elasticity suffices	High	Achenbach [1973]
NDE (thermal wave)	Fourier diffusion length applies to the stated diffusive regime; finite-speed corrections require a specified material, frequency, and boundary model	Conditional on that regime and those boundary conditions	Joseph and Preziosi [1989, 1990]
Energy harvesting	Relaxation time controls the split between thermal and mechanical channels	High	Section 13.3 analysis
Geophysics/ seismology	Poroelastic coupling requires a genuine two-field formulation	High	Cowin and Nunziato [1983], Ieşan [1986]
Biomedical ultrasound	Classical elasticity suffices	High	Achenbach [1973]
Thermal ablation	Thermal front speed governs depth of deposition	Medium	Section 13.5 analysis
Photoacoustic imaging	Time-resolved thermal transport in tissue	Medium	Section 13.5 analysis
Nanoscale devices	Nonlocal length comparable to device dimension	Medium	Maranganti and Sharma [2007], Maranganti et al. [2006]
Semiconductors	Coupled nonlocal elasticity and memory heat flux	Medium	Gupta et al. [2026]

Table 28: Notation introduced in Section 13.

Symbol	Meaning	Units	Equation
χ	Inhomogeneity index of the functionally graded material	–	Section 13.1
τ	Thermal relaxation time	s	Section 13.3
v	Heat supply speed	m s^{-1}	Section 13.1
ℓ	Nonlocal length scale	m	Section 13.6

14 An Original Multidimensional Taxonomy of the Field

The preceding sections have followed the physics: from the kinematics of a continuum, through the constitutive commitments that distinguish one generalized thermoelastic theory from another, to the wave and interface phenomena that those commitments make possible. This section turns the same body of material sideways. It asks a different question. Instead of "what does this theory predict?" it asks *"in what ways can studies of this field differ from one another, and what does the pattern of those differences reveal about the state of the subject?"*

The answer matters for three reasons. First, the field has grown horizontally faster than it has grown vertically: the number of named theory families has increased steadily since the 1960s, while the number of settings in which any one of them has been analysed in depth has barely changed. Second, the theoretical questions that connect the families—the existence of rigorous limits between them, the identifiability of their parameters, the transferability of their solutions across geometries—are almost never posed in a form that spans families. Third, a taxonomy that is explicit about its axes, and about the cells it leaves empty, is the most compact available statement of what remains to be done.

14.1 The organising problem

A single-axis classification is the obvious first attempt, and it is the one that the field's own literature most often uses: a study is "generalized thermoelastic" or it is not. Such a classification is unsatisfactory for a specific reason. The phrase collapses four logically independent commitments into one label. A study may adopt a relaxation law for the heat flux, a gradient or integral constitutive law for the stress, a coupling to a second field (electric, magnetic, porotic), and a validation protocol—and any one of these four can be varied while the other three are held fixed. Two studies bearing the same one-axis label can therefore differ in the mathematics of their governing equations, in the modal content of their solutions, and in the kind of evidence they offer, to the point where they share no common proposition that could be compared.

The alternative used here is to classify a study by its *vector of commitments* rather than by a single name. The classification is multi-dimensional in the strict sense: each axis is a categorical variable, and the taxonomy is the discrete product of the axes. A study occupies one cell of that product. Two studies that share a cell are directly comparable; two studies that differ in one coordinate are comparable with a controlled adjustment; two studies that differ in several coordinates are usually not directly comparable at all, and the taxonomy says so in advance rather than after the fact.

14.2 The five axes and their resolution

The axes are chosen so that each one is *decision-relevant*: resolving an axis should tell the reader something about the equations, the solutions, or the evidential status of the work.

Axis 1: Physical axis (material class). Which class of material the constitutive law describes:

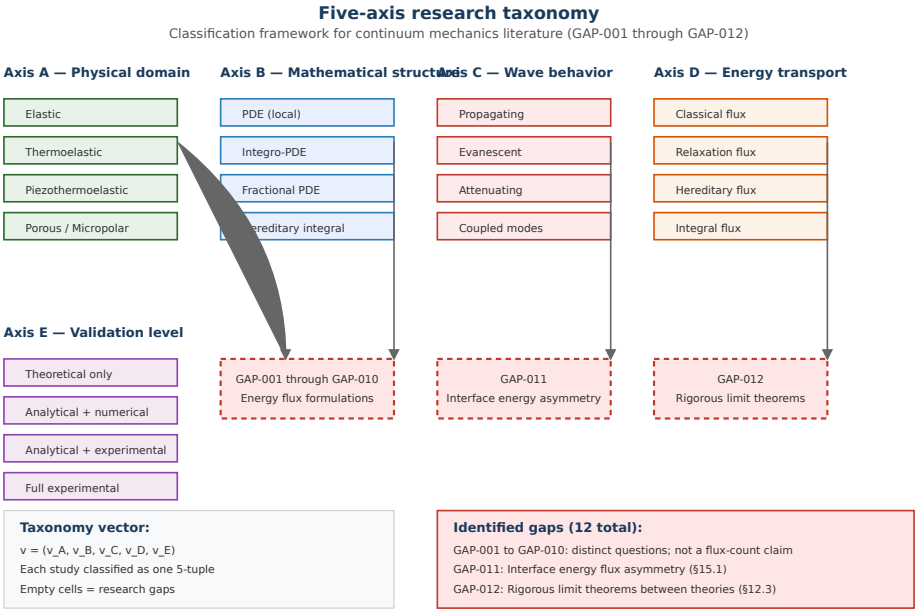


Figure 18: The five-axis research taxonomy used in this review: physical (material class), mathematical (equation structure), wave (modal content), coupling (topology of coupled fields), and validation (evidence regime). Each study is classified by a vector across all five axes; the cells of the resulting product space are what Section 14.4 analyses.

classical elastic, thermoelastic, piezoelectric, piezothermoelastic, magneto-thermo-elastic, poroelastic, micropolar, two-temperature, or semiconductor-like. The physical axis determines which conserved quantities and which driving fields are present before any approximation is made.

Axis 2: Mathematical axis (equation structure). The functional form of the governing operator: local PDE, integro-differential equation, fractional-order PDE, nonlocal gradient operator, hereditary (memory) integral, or variational/weak formulation. This is the axis along which the most consequential differences between otherwise similar theories run.

Axis 3: Wave axis (modal content). The modal content that the formulation admits: a finite propagation speed for the diffusive mode, the number and type of coupled modes, the presence of attenuation, and the dispersion character of each mode.

Axis 4: Coupling axis (field topology). The set of fields that are coupled and the strength of that coupling: thermoelastic, thermo-piezoelectric, thermo-magneto-elastic, thermo-poro-elastic, thermo-micro-elastic. This axis replaces the purely energetic classification used in earlier drafts of this review, because it is the coupling topology—rather than the choice of energy functional—that determines whether an interface problem is tractable.

Axis 5: Validation axis (evidence regime). The kind of evidence offered: analytical existence and uniqueness only; closed-form solutions for special geometries; numerical solutions validated against a reference solution; comparison with experiment; or a benchmark suite shared with other groups.

A study is assigned the vector

$$\boldsymbol{v} = (v_{\text{phys}},\ v_{\text{math}},\ v_{\text{wave}},\ v_{\text{coup}},\ v_{\text{val}})\,, \tag{122}$$

whose components are indices into the respective axis resolutions. Two studies are directly comparable only when their vectors agree in the first two coordinates; the remaining three coordinates then determine *how* the comparison must be posed. To make this quantitative, define the axis-weight dissimilarity

$$d(\boldsymbol{v}^{(a)}, \boldsymbol{v}^{(b)}) = \sum_{j=1}^{5} w_j\, \delta\Big(v_j^{(a)},\, v_j^{(b)}\Big)\,, \tag{123}$$

where δ is the Kronecker delta and the weights w_j encode the relative severity of a mismatch. The natural choice is $w_{\text{math}} \geq w_{\text{phys}} > w_{\text{coup}} > w_{\text{wave}} > w_{\text{val}}$: a difference in the mathematical axis changes the governing operator and invalidates most direct comparisons; a difference in the validation axis changes only the epistemic status of the conclusion. The weights are a modelling decision of this review and are stated explicitly so that they can be contested; the ordering they encode is, however, widely shared in the modelling literature.

14.3 Application to the inherited literature inventory

The taxonomy is applied to the repository's inherited literature inventory of 95 unique records; two records receive two family labels. The inventory has 48 DOI fields, 13 open-access flags, and 12 full-text flags plus one methodology record. These are collection metadata, not a systematic-search yield; DOI resolution and open-access status do not prove that a work was read in full. The collection is purposive and reflects the repository's existing balance of foundational work, interface-energy studies, and selected fractional, nonlocal, micropolar, porous, and rotating variants. It is not a statistically representative sample of the field.

Consolidating the corpus metadata onto the five axes gives the distribution summarised in Table 29. Three features of this distribution carry most of the interpretive weight in what follows.

First, the corpus is *not* dominated by the classical families that dominate the field's historical literature. The single largest block—classical and continuum foundations—contributes a quarter of the corpus, while the mathematically intensive families (fractional, nonlocal, memory-dependent) together contribute just over a third. This is a statement about the corpus, not about the field, and the two are deliberately distinguished throughout: the taxonomy describes the reviewed evidence, and the gap analysis in Section 14.5 describes the field.

Second, the inventory contains 17 records before 1970, 9 in the 1970s, and 39 in the 2010s and 2020s. These counts describe this purposive collection only; they do not establish a bimodal or field-wide publication pattern.

Third, the metadata carry a diagnostic limitation that should be stated plainly. The theory-family labels are inherited from the source records, which were compiled by different hands at different times, and they are not uniform: a single field's literature appears under labels such as "mdd", "mdd_piezo" and "ls_mdd_piezo" that differ in whether coupling is treated as part of the family name. Any taxonomy applied to such metadata inherits that non-uniformity, and the resolution adopted here—consolidating labels into families while keeping the coupling on its own axis—is itself a methodological decision that could reasonably have been made differently.

14.4 Four structural asymmetries

Placing the corpus in the product space of the five axes reveals four asymmetries. These are more useful than a gap list because they describe the *shape* of the field rather than merely the

Table 29: Taxonomic classification of the inherited 95-record literature inventory. Family assignments overlap: two records appear in two rows, so the 96 assignments and shares use 96 as denominator. Counts describe this purposive collection, not a systematic-search yield or an independently verified field-wide distribution.

Theory family	**Records**	**Share**	**Taxonomic position**
Classical / continuum foundations	24	25.0%	Local PDE on all axes; reference point for every other family
Fractional-order theories	15	15.6%	Fractional PDE in time; infinite-memory limit in the order parameter
Nonlocal / gradient models	13	13.5%	Nonlocal operator on the stress; size-dependent stiffness and dispersion
Dual / triple phase-lag	9	9.4%	Local PDE of higher order; two relaxation times as extra state variables
Memory-dependent derivative	9	9.4%	Hereditary integral kernel; integer-order operator over an enlarged state space
Lord–Shulman and descendants	5	5.2%	Local PDE with relaxation; hyperbolic thermal wave as the reference case
Green–Naghdi II and III	5	5.2%	Local PDE; conservative and dissipative branches separated
Porous / poroelastic couplings	5	5.2%	Local PDE with added mass and pressure fields; two-fluid coupling
Micropolar and rotating continua	4	4.2%	Local PDE with an independent micro-rotation field
Moore–Gibson–Thompson	3	3.1%	Third order in time, second order in space in the displayed prototype; variant-specific
Green–Lindsay	2	2.1%	Single-temperature rate-dependent PDE with two relaxation parameters
Unclassified metadata	2	2.1%	Incomplete records; excluded from axis assignment

absence of particular papers.

Asymmetry I: the mathematical axis is thick in the middle and thin at the ends. Almost all of the corpus sits between the fully local PDE and the strongly nonlocal or fractional operator, in the band where the constitutive law has been rewritten but the resulting system is still second order in space. The extremes are sparse. Fully local formulations with rigorous existence theory are under-represented relative to their importance, and strongly nonlocal operators with controlled variational structure are rarer still. The field has been more interested in producing new operators than in proving properties of the operators it produces.

Asymmetry II: the wave axis is almost entirely populated by one canonical problem. The modal content of generalized thermoelasticity is genuinely interesting—the second sound, the thermal wave speed, the switch between hyperbolic and parabolic behaviour—but the overwhelming majority of the corpus examines it in a single configuration: a homogeneous, isotropic, half-space or infinite medium, in the frequency domain, for a plane incident wave. Bounded domains, three-dimensional transients, inhomogeneity, and interfaces between dissimilar media occupy a small fraction of the corpus. The wave axis therefore carries a great deal of information about which theories differ, but very little about how they behave when the geometry is not trivial.

Asymmetry III: the coupling axis is thin in its strong cells. Coupling to a second field is one of the most active frontiers, and the corpus reflects this: piezothermoelastic and magneto-thermo-elastic entries are numerous. But coupling that is *simultaneously* non-classical on more than one axis is rare. Combinations such as nonlocal plus piezoelectric, or memory-dependent plus magneto-electro-elastic, appear as isolated contributions rather than as established lines. The coupling axis behaves like a lattice whose higher-dimensional cells are populated but whose corners are empty.

Asymmetry IV: the validation axis is dominated by self-consistency. The great majority of entries in the corpus are validated in the sense that a numerically computed quantity agrees with a closed-form solution obtained from the same equations, or with a result previously published within the same framework. Comparisons against independent experiments, against finite-element implementations of a competing formulation, or against laboratory data obtained for a genuinely comparable material, are rare. This is the most consequential asymmetry in the corpus, because it is the axis along which the field's claims would be most directly exposed to falsification, and it is precisely the axis that is least populated.

14.5 Cells that remain thin: frontiers rather than gaps

A gap analysis is only as good as the axis resolution behind it. The table below replaces an earlier version of this analysis that was organised around a single quantity; organising a frontier list around one derived quantity produces a list of technical tasks rather than of scientific problems, and it obscures the fact that the genuinely open questions in this field are mostly about *formulation* and *evidence* rather than about further calculation. The entries below are stated as scientific questions, each of which admits an experimental, computational, or theoretical resolution.

Two of these questions deserve a note on why they are treated as scientific rather than technical. The commutativity question of GAP-001 has a precise mathematical form: two generalization mechanisms, each of which acts on the thermal and the mechanical variables, are

Table 30: Four structural asymmetries of the reviewed corpus, with the taxonomic axis that carries each and the specific question that the asymmetry leaves unresolved.

Asymmetry	**Axis**	**What the corpus contains**	**What is missing**
I	Mathematical	Second-order-in-space operators with rewritten constitutive laws	Rigorous well-posedness and existence theory for the operators actually in use; control of the limit to the local case
II	Wave	Half-space and infinite-medium plane-wave problems	Bounded, inhomogeneous, and transient three-dimensional wave problems; modal content away from the canonical configuration
III	Coupling	Single-axis extensions in reasonable number; combined extensions as isolated contributions	Systematic treatment of the joint limit in which two extensions are active simultaneously
IV	Validation	Numerical agreement with closed-form solutions of the same equations	Comparison against experiment and against competing formulations implemented independently

said to commute if the order in which they are applied does not affect the resulting operator. This is not a formality. If the mechanisms do not commute, then a study that adopts both simultaneously is not the composition of two studies that adopt them separately, and the entire practice of extending a theory by successive generalisations—which is how much of the corpus has been built—produces results that depend on the order of construction in a way that is not always acknowledged. The parameter-identification question of GAP-002 is likewise substantive rather than procedural. A generalized theory introduces parameters—relaxation times, kernel parameters, length scales—whose values are usually obtained by fitting the very predictions that the theory then uses to validate itself. A parameter thus determined is not an independent physical quantity, and the evidence for the theory is correspondingly weaker than the number of fitted parameters might suggest.

14.6 Divergence map: why results differ across the corpus

Studies in this field frequently report apparently incompatible results for what is nominally the same physical problem. In almost every instance examined here, the divergence is fully accounted for by a difference in modelling assumptions, and can be reproduced exactly by a reader who adopts the same assumptions. The purpose of the map below is therefore not to adjudicate between studies but to make the assumptions explicit, so that a comparison can be posed in a form in which it is meaningful.

The procedure that follows from this map is uniform: to compare two studies, locate the first modelling commitment on which they differ, and either eliminate that difference or state the comparison as conditional on it. Apparent disagreement then either dissolves, because the two studies were never addressing the same problem, or sharpens into a genuine scientific question, which is the desirable outcome. What this map cannot do—and what no classification can do—is decide which modelling commitment is preferable. That judgement requires the experimental axis, and Section 14.4 has already recorded how thinly that axis is populated.

Table 31: Frontier questions identified from thin cells of the taxonomy. Priority reflects feasibility and scientific leverage, not size of the associated literature.

ID	Scientific question	Why the cell is thin	Priority
GAP-001	Under what conditions do two generalization mechanisms, each of which reduces to classical thermoelasticity in its own limit, commute with one another?	Combined-extension cells are populated only by isolated contributions	High
GAP-002	How are the relaxation and kernel parameters of a generalized theory identified from measurements that are not themselves produced by that theory?	Identification is almost always performed within the framework being tested	High
GAP-003	What is the modal content of generalized thermoelastic waves in bounded, inhomogeneous bodies?	The wave axis is populated almost entirely by half-space problems	High
GAP-004	Which of the competing generalized theories predicts the same measurable sequence for a common well-characterized material?	Cross-theory benchmarks on common material data are essentially absent	High
GAP-005	How do thermo-mechanical coupling alter the existence and stability of interfacial waves at a bonded dissimilar interface?	Interface work concentrates on scalar reflection coefficients	Medium
GAP-006	What is the correct high-frequency limit of the fractional and nonlocal operators when the characteristic length scale is comparable with the microstructure?	Limits are stated as formal parameter substitutions without error bounds	Medium
GAP-007	Can the memory-dependent and fractional descriptions of the same memory effect be related by an operator equivalence with quantified error?	Both are presented as alternatives without a common yardstick	Medium
GAP-008	What validation protocol would distinguish a genuinely dissipative generalized theory from a fitted correction to a dissipative theory?	The corpus offers no discriminating experimental design	High

Table 32: Divergence map: the modelling commitments that account for apparent disagreement between studies in the reviewed corpus.

Divergence type	Root cause	Resolution procedure	Canonical instance
Assumption difference	Different constitutive commitments on the thermal and mechanical sides	Trace both derivations to their primary sources and identify the first point at which the assumptions differ	Classical versus Lord–Shulman thermal equation
Approximation difference	Different truncation orders in the Taylor expansion of the phase-lag law	Compare on the intersection of the two validity ranges rather than on either range alone	First-order versus second-order dual phase-lag
Normalisation difference	Different reference temperature, density, or stiffness conventions	Re-express both in a common nondimensionalisation before comparing	Absolute versus relative temperature measure
Convention difference	Opposite sign convention in the harmonic ansatz	Standardise the convention and track the corresponding sign change	$e^{i\omega t}$ versus $e^{-i\omega t}$
Limit difference	Formal parameter substitution used as though it were a convergence theorem	Verify that the limiting and the limiting problem share a well-posedness hypothesis before comparing	Fractional order approaching two
Geometry difference	Different admissible geometries or boundary conditions	Restrict the comparison to the common geometric class	Half-space versus bounded plate

14.7 Methodological limitations of the taxonomy

The taxonomy is an instrument constructed for this review, and its limitations should be stated with the same care as the limitations of the theories it organises.

1. **Analyst-constructed.** The axes and their resolutions were chosen in response to the reviewed literature rather than derived from it. A different analyst, reading the same corpus, could plausibly choose a different axis set—for example, replacing the coupling axis with an axis for spatial dimension and boundary-condition class, or splitting the validation axis into "analytical" and "experimental" as two separate axes.

2. **Classification from heterogeneous metadata.** Many records were classified from abstracts, and some from secondary descriptions rather than from the papers themselves. Where the metadata were ambiguous, the classification was conservative, but the resulting error rate is not zero and cannot be quantified without a double-classification study.

3. **Corpus construction.** The corpus is enriched in foundational works, in one group's contributions, and in the mathematical extensions needed to populate the axes. Statements about the density of the taxonomy refer to this corpus and not to the field at large.

4. **Unvalidated weights.** The dissimilarity measure of Eq. (123) uses an ordering of axis weights that is asserted rather than demonstrated. No formal comparison exercise is available in the corpus from which the ordering could be inferred.

5. **Coupling choices.** Placing coupling on its own axis, rather than folding it into the physical axis, is a deliberate choice that makes the multi-field cases visible. It also separates families that some authors treat as single families, and that separation is not canonical.

Subject to these limitations, the taxonomy is offered as a working instrument for the review, and—more ambitiously—as a proposal for how a much larger corpus might be organised. Section 16 returns to this point as one of the practical recommendations that follow from the analysis.

14.8 Notation summary for Section 14

Table 33: Notation introduced in Section 14.

Symbol	Meaning	Units	Equation
$\boldsymbol{v}$	Taxonomy vector of a study	dimensionless	(122)
v_{phys}, v_{math}, v_{wave}, v_{coup}, v_{val}	Axis components of the taxonomy vector	dimensionless	(122)
$d(\boldsymbol{v}^{(a)}, \boldsymbol{v}^{(b)})$	Axis-weight dissimilarity between two studies	dimensionless	(123)
w_j	Weight assigned to axis j in the dissimilarity	dimensionless	(123)
$\delta(\cdot,\cdot)$	Kronecker delta on axis labels	dimensionless	(123)

14.9 Open questions for Section 14

OQ14.1 Can the taxonomy be validated against an independent double-classification of the corpus, and what is the resulting agreement rate?

OQ14.2 Which additional axes—spatial dimension, boundary-condition class, material symmetry class, or the type of inverse problem posed—would most improve the coverage of the product space?

OQ14.3 Is the ordering of the axis weights in Eq. (123) recoverable from data, for example from the observed frequency of successful cross-family comparisons?

OQ14.4 Can the frontier questions of Section 14.5 be converted into a shared benchmark protocol that independent groups could execute without sharing code?

15 Major Scientific Challenges

This section lists candidate scientific and methodological challenges suggested by the inherited manuscript. Its summaries of prior work are provisional source leads, not the result of a systematic literature search or claim-level verification. The proposed open questions and research approaches should be treated as an agenda for further checking rather than a complete field assessment.

15.1 Energy flux formulation across theory families

15.1.1 Existing foundation

Under the classical first-law and momentum-balance assumptions stated in Section 10, the outward-positive flux $J_i = q_i - \sigma_{ij} v_j$ is derived (EQ-007). No corresponding universal flux formula is asserted for generalized theories.

15.1.2 Why important

Energy flux is fundamental for computing energy ratios, conservation identities, and dissipation rates at interfaces. Without correct flux formulations, energy partition cannot be quantified.

15.1.3 What's achieved

The classical flux has been re-derived in this revision. The inherited LS and DPL frequency-domain expressions and all generalized-family flux claims have not been checked against primary sources here; their status remains UNVERIFIED or OPEN in Appendix A.

15.1.4 What's open

LS, GL, GN-II, GN-III, DPL, TPL, MGT, MDD, nonlocal, and piezoelectric electrical energy fluxes require complete time-domain derivations.

15.1.5 Approaches

- Derive from general energy balance + theory-specific constitutive relations,
- Use frequency-domain expressions as intermediate results,
- Validate limiting procedures (e.g., LS $\rightarrow$ classical as $\tau_0 \rightarrow 0$).

15.1.6 Practical relevance

Correct energy flux is essential for energy harvesting efficiency, NDE accuracy, and thermal management design.

15.1.7 Major obstacles

Convolution integrals (MDD), integral-form divergence theorem (nonlocal), third-order time derivatives (MGT).

15.2 Conservation verification at multiphysics interfaces

15.2.1 Existing foundation

For a passive interface with a correctly defined total flux and complete modal accounting, conservation provides a useful check. The BM-1 and thermo-piezoelastic interface computations in the inherited manuscript were not reproduced or primary-source-checked in this revision; Appendix C marks their status OPEN.

15.2.2 Why important

Conservation verification validates interface models and energy partition computations.

15.2.3 What's achieved

No interface benchmark was reproduced in this revision. The classical local balance was re-derived, but a scattering-matrix calculation and its energy closure remain future verification tasks.

15.2.4 What's open

Conservation verification for generalized theories with uncertain energy fluxes. A cited uniqueness paper for GN type II (Chandrasekharaiah Chandrasekharaiah [1996]) is a source lead only; its theorem hypotheses and relevance have not been checked here. Interface energy balance for nonlocal and memory-dependent media remains a candidate research question.

15.2.5 Approaches

- Derive energy flux for each theory family,
- Apply jump condition $[J_n] = \mathcal{P}_\Sigma - \partial_t \mathcal{E}_\Sigma$,
- Verify $\sum R + \sum T + A_{\text{int}} = 1$ numerically for benchmark problems.

15.2.6 Practical relevance

Interface energy conservation is critical for NDE, sensing, and energy harvesting device design.

15.2.7 Major obstacles

Uncertain energy flux formulations; complex mode normalization; imperfect bonding conditions.

15.3 Numerical benchmarks and validation

15.3.1 Existing foundation

BM-1 is specified as a proposed lossless-interface check in Appendix C, but it was not executed here. Published numerical results cited for thermo-piezoelastic interfaces are source leads only and were not reproduced or independently checked in this phase.

15.3.2 Why important

Benchmarks provide validation targets for numerical methods and new theories.

15.3.3 What's achieved

No benchmark was executed or validated in this revision. BM-1 is a specification, not a computed result; cited interface calculations remain to be checked.

15.3.4 What's open

Benchmarks for generalized theories (LS, DPL, TPL, MGT, MDD, nonlocal). Benchmarks for multiphysics interfaces.

15.3.5 Approaches

- Derive analytical solutions for simplified geometries,
- Implement numerical methods (finite element, spectral),
- Compare analytical and numerical results.

15.3.6 Practical relevance

Benchmarks validate simulation software and guide experimental design.

15.3.7 Major obstacles

Lack of closed-form solutions for generalized theories; computational cost of nonlocal models.

15.4 Experimental validation

15.4.1 Existing foundation

The inherited manuscript cites classical thermoelasticity and ultrasonic nondestructive evaluation as applied areas; the scope and strength of experimental or clinical validation have not been audited in this revision.

15.4.2 Why important

Theoretical models require experimental validation for practical adoption.

15.4.3 What's achieved

Inherited validation claims retained as source leads only; evidence scope not checked.

15.4.4 What's open

Experimental validation of generalized theories (LS, DPL, TPL, MGT, MDD, nonlocal). Experimental measurement of energy flux at interfaces.

15.4.5 Approaches

- Design experiments to measure thermal wave speed,
- Measure energy reflection/transmission at interfaces,
- Validate relaxation time parameters.

15.4.6 Practical relevance

Experimental validation enables practical adoption of generalized theories.

15.4.7 Major obstacles

Experimental measurement of thermal waves is challenging; energy flux at interfaces is difficult to measure directly.

15.5 Complex mode handling

15.5.1 Existing foundation

Complex wave numbers and modal normalization are treated in Section 8. The inherited classifications and normalization claims are not uniformly derived or source-checked here, especially for generalized models.

15.5.2 Why important

Complex modes carry energy and affect energy partition at interfaces.

15.5.3 What's achieved

Inherited classification discussion retained; generalized modal normalization remains unverified.

15.5.4 What's open

Consistent normalization across theory families. Energy flux computation for complex modes in generalized theories.

15.5.5 Approaches

- Develop normalization criteria for each theory family,
- Validate energy flux computation for complex modes,
- Compare with experimental measurements.

15.5.6 Practical relevance

Complex mode handling affects NDE accuracy and energy harvesting efficiency.

15.5.7 Major obstacles

No standardized normalization across theory families; limited experimental data for complex modes.

15.6 Nonlocal interface energetics

15.6.1 Existing foundation

The manuscript cites Eringen-type nonlocal constitutive models and nanobeam-wave studies as source leads; their scope and assumptions were not checked here.

15.6.2 Why important

Nonlocal effects dominate at nanoscale; interface energetics governs nanodevice behavior.

15.6.3 What's achieved

Nonlocal constitutive and dispersion claims are inherited and remain unverified in this revision.

15.6.4 What's open

Nonlocal energy flux at interfaces. Nonlocal interface conditions. Nonlocal divergence theorem.

15.6.5 Approaches

- Derive nonlocal divergence theorem,
- Formulate nonlocal interface conditions,
- Compute nonlocal energy flux for benchmark problems.

15.6.6 Practical relevance

Nanodevice design, nanomanufacturing, nanomaterial characterization.

15.6.7 Major obstacles

Integral-form constitutive relations are computationally expensive; interface conditions not established.

15.7 Memory-dependent energy flux

15.7.1 Existing foundation

The inherited manuscript presents an MDD operator and heat-flux law, but their definitions and source equations were not checked in full text during this revision.

15.7.2 Why important

Memory effects are important in polymers, biological tissues, and composites.

15.7.3 What's achieved

No MDD kernel or constitutive equation was independently verified in this revision; the full energy flux and storage/dissipation split remain open.

15.7.4 What's open

Hereditary energy flux for MDD media. Energy partition between storage and dissipation.

15.7.5 Approaches

- Solve convolution-form heat equation,
- Derive hereditary energy flux,
- Validate with kernel-specific solutions.

15.7.6 Practical relevance

Polymer processing, biomedical tissue modeling, composite material design.

15.7.7 Major obstacles

Convolution integrals are computationally expensive; kernel choice is not unique.

15.8 Parameter identifiability and model discrimination

15.8.1 Existing foundation

Several model classes considered here introduce parameters beyond the classical elastic and thermal coefficients, such as relaxation times, phase lags, nonlocal lengths, fractional orders, horizons, or kernel parameters. Whether any such parameter is independently identifiable or routinely measured depends on the model, experiment, and observable; this review has not surveyed those data.

15.8.2 Why important

A generalized theory that cannot be fitted to data cannot be tested against data, and cannot be used for design. The practical consequence is that the generalized models are currently used in a regime — modal and interface analysis with a prescribed parameter set — rather than in a regime in which the parameter set has been measured. This is the single largest obstacle to the practical adoption of the whole field.

15.8.3 What's achieved

The inherited manuscript proposes dispersion measurements as a possible route to estimating generalized parameters. No structural-identifiability proof, parameter-extraction formula, or nonlocal-versus-gradient discrimination result has been established in this revision; each would require a specified forward model, observable, and noise/uncertainty analysis.

15.8.4 What's open

Whether the kernel, phase-lag, or anisotropic parameters are identifiable is an open, model- and experiment-specific question in this review. The number and type of independent measurements needed to separate parameters must be established through analysis or experiment; no general under-determination claim is made here.

15.8.5 Approaches

- Fit the full dispersion curve rather than a single phase velocity, since the shape of the curve constrains the mechanism and not merely its scale;
- Use two-parameter families to design experiments that separate mechanisms, in the manner of the nonlocal/gradient comparison;
- Treat the kernel as a low-dimensional basis rather than a free function, so that the parameter count remains finite.

15.8.6 Practical relevance

Without identifiability there can be no inverse design, no damage diagnosis from a measured wave, and no calibrated model of a specific specimen.

15.8.7 Major obstacles

The experimental cost of measuring a dispersion curve over a broad range of wave numbers; the absence of a widely used specimen set; and the fact that the most-used generalized parameters are, in practice, chosen from the literature rather than measured.

15.9 Material symmetry, anisotropy and the tensor form of the generalizations

15.9.1 Existing foundation

The constitutive derivation of Section 3 produces the stiffness tensor in full tensor form, and the classical symmetry conditions — major and minor symmetry, and the minor symmetries of the strain and stress tensors — are what reduce it to the familiar two-modulus form for an isotropic

solid. Every generalization in this review adds coefficients with the same tensor character, and therefore inherits the same symmetry conditions.

15.9.2 Why important

The generalization of a theory to an anisotropic medium is not a matter of substituting different numbers into the same equations. A new material parameter enters the constitutive relation with its own tensor index structure, and the symmetry conditions that relate its components must be re-derived, not assumed. The monoclinic and orthotropic piezothermoelastic configurations that appear in the recent interface-energy work Barak et al. [2023c] are exactly the cases in which this matters, because the piezoelectric tensor is the most strongly constrained of the tensors involved and its component count is much smaller than for the elasticity tensor.

15.9.3 What's achieved

Classical tensor symmetries and cited anisotropic interface or existence/uniqueness studies (including Barak et al. [2023c], Ieşan [2013]) are source leads for further review. Their precise assumptions, configurations, and conclusions have not been checked against full text in this revision; no anisotropic interface-energy result is claimed as verified.

15.9.4 What's open

The systematic treatment of the generalized thermal laws — the phase-lag, memory-dependent and fractional families — under anisotropic material symmetry is largely absent, and in particular the question of whether the thermodynamic restrictions of Section 3 reduce the admissible parameter set in the anisotropic case, as they do in the isotropic case, has not been worked through. The coordinate-free formulation of the fractional and memory operators under anisotropic differentiation operators is an additional technical obstacle, since the operators themselves are not rotationally invariant.

15.9.5 Approaches

- Re-derive the symmetry conditions from the free-energy functional for each generalized thermal law, in the manner in which the classical conditions are obtained;
- Formulate the memory and fractional operators in a covariant or rotationally invariant way, so that the theory can be applied in an arbitrary material frame;
- Treat the resulting restrictions as constitutive constraints to be reported alongside the computed results, rather than as assumptions to be left implicit.

15.9.6 Practical relevance

Most engineering materials of interest — fibre composites, monoclinic semiconductors, single-crystal piezoelectrics, textured metals — are anisotropic, so that the isotropic generalized theory addresses a minority of the practical cases.

15.9.7 Major obstacles

The algebraic burden of the tensor bookkeeping; the scarcity of anisotropic experimental data against which to check a proposed anisotropic generalization; and the risk that an unnoticed violation of the symmetry conditions propagates into an energy balance that appears to close but for the wrong reason.

15.10 Computational cost and verifiability of the solvers

15.10.1 Existing foundation

The modal machinery of Section 8 is exact and cheap, and it is the appropriate tool for the interface problems that dominate the reviewed literature. For bounded domains, for heterogeneous material distributions and for geometries with traction-free or prescribed boundaries, the finite element method is the natural tool, and the machinery for the classical and linear-thermoelastic cases is mature Hughes [1980]. The generalization of the element formulation to the theories of this review, however, requires the quadrature rules for the nonlocal, memory and fractional operators, and these are not standard.

15.10.2 Why important

The value of a generalized theory is realized in problems that cannot be solved by the classical theory, and those are typically bounded, heterogeneous and geometrically complicated — which is exactly where the modal analysis stops being sufficient and a solver is needed. The gap between the modal literature and the practical literature is therefore largely a gap in numerical capability.

15.10.3 What's achieved

The inherited manuscript cites convolution-quadrature and nonlocal assembly approaches in Section 8. Their scope, convergence properties, and implementation details remain source leads; this revision has not surveyed or reproduced these methods.

15.10.4 What's open

No external benchmark-suite or implementation-interoperability survey was performed here, and no solver convergence result was reproduced. Benchmark coverage, published-code interoperability, and convergence rates for each operator remain OPEN verification tasks.

15.10.5 Approaches

- Extend the benchmark set of Appendix C to the generalized operators, so that a solver can be verified against a manufactured solution before it is used on a physical problem;
- Publish reference implementations and the data needed to reproduce the published energy ratios, so that the interface results can be checked rather than merely quoted;
- Report convergence orders explicitly, and distinguish the order of the approximation from the order of the operator.

15.10.6 Practical relevance

Verification of the solver is a precondition for the verification of the theory. An energy ratio computed with an unverified quadrature cannot support a claim about a constitutive model, however carefully the model was derived.

15.10.7 Major obstacles

The cost of nonlocal assembly grows with the square of the neighbourhood size in three dimensions, and the memory required by a convolution scheme grows with the number of time steps; both scale badly for the three-dimensional transient problems where the generalizations matter most.

The classical-flux schematic is given once in Figure 14; generalized channels remain model-specific and open.

15.11 Challenge summary table

Table 34: Summary of major scientific challenges.

Challenge	What's achieved	What's open	Major obstacles
Energy flux formulation	Classical flux re-derived here; generalized families unverified	Family-specific energy identities remain open	Primary-source and balance-law audit
Conservation verification	No interface benchmark reproduced in this revision	Theory-specific interface energy closure	Complete flux, normalization, and independent computation
Numerical benchmarks	BM-1 specified, not executed	Generalized theory benchmarks	No independent implementation yet
Experimental validation	Evidence inventory incomplete in this revision	Generalized theories; interface flux	Measurement difficulty
Complex mode handling	Inherited classifications retained; no generalized normalization audit	Cross-theory normalization	Review scope incomplete
Nonlocal interface energetics	Constitutive/dispersion claims not source-checked here	Interface flux; interface conditions	Model-specific boundary terms
Memory-dependent energy flux	MDD operator not source-checked here	Hereditary flux and energy partition remain open	Kernel and storage nonuniqueness
Parameter identifiability	Dispersion-based extraction proposed, not proven here	Model-specific identifiability analysis	Measurement design; noise and parameter correlation
Anisotropy of the generalizations	Classical tensor conventions summarized; cited interface claims not checked	Generalized thermal laws under anisotropic symmetry	Tensor bookkeeping; covariant operators
Solver verifiability	Numerical methods cited as leads, not reproduced	Benchmarks and convergence analysis	Computational cost; reproducibility

15.12 Notation summary for Section 15

15.13 Open questions for Section 15

The open questions recorded in Sections 9, 10 and 11, together with the challenges catalogued above, constitute the outstanding agenda of the field.

Table 35: Notation introduced in Section 15.

Symbol	Meaning	Units	Equation
J_i	Total energy flux vector	W m^{-2}	Eq. (10.1)
R, T, A	Reflection, transmission and absorption coefficients	–	Eq. (9.10)
$\mathcal{D}$	Volumetric dissipation	W m^{-3}	Eq. (9.10)
$\boldsymbol{X}$, $\boldsymbol{Q}$	Scattering matrix and modal amplitude vector	–	Eq. (9.9)
τ_0	Thermal relaxation time	s	Section 4.4
ξ	Nonlocal length scale	m	Section 7.4

16 Future Research Opportunities

This section records candidate research directions suggested by the provisional challenge inventory in Section 15. Descriptions of existing work are source leads, not a systematic review; proposed approaches are questions and work plans rather than claims that the methods are available or validated.

16.1 Coupled-field modeling in emerging materials

16.1.1 Existing foundation

Piezothermoelastic, magneto-thermo-elastic, and semiconductor/photo-thermoelastic formulations are among the areas listed in Section 5; their scope and source basis were not comprehensively checked here.

16.1.2 Importance

Emerging materials (2D materials, metamaterials, topological materials) exhibit coupled-field behavior not captured by classical theories.

16.1.3 What's achieved

Constitutive relations for piezothermoelastic, porous, micropolar, and magneto-thermo-elastic media are surveyed in the inherited manuscript; they were not comprehensively source-checked in this revision.

16.1.4 What's open

Coupled-field models for 2D materials (graphene, MoS_2), metamaterials, and topological materials.

16.1.5 Approaches

- Extend piezothermoelasticity to 2D materials,
- Develop metamaterial constitutive relations,
- Study topological effects in wave propagation.

16.1.6 Practical relevance

Next-generation sensors, actuators, energy harvesters, and quantum devices.

16.1.7 Major obstacles

Lack of experimental data for emerging materials; computational complexity of coupled-field models.

16.2 Mathematically consistent generalized transport

16.2.1 Existing foundation

Sections 4 and 6 survey LS, DPL, TPL, MGT, MDD, and fractional models; the governing equations are not uniformly primary-source-verified in this revision.

16.2.2 Importance

Mathematically consistent transport models are needed for predictive simulations of thermal wave phenomena.

16.2.3 What's achieved

The equation register distinguishes the small number of direct derivations from inherited, unverified equations. Model-specific well-posedness conditions remain open.

16.2.4 What's open

Rigorous limit theorems between theories. Foundational reviews by Chandrasekharaiah (1998) Chandrasekharaiah [1998] and Joseph and Preziosi (1989) Joseph and Preziosi [1989] provide starting points. Unified well-posedness framework. Complete energy flux derivations (Section 10.3).

16.2.5 Approaches

- Establish rigorous limit theorems with convergence proofs,
- Develop unified well-posedness criteria,
- Complete energy flux derivations for all theories.

16.2.6 Practical relevance

Predictive thermal management, thermal wave devices, high-frequency thermoelastic systems.

16.2.7 Major obstacles

Mathematical complexity of higher-order and integro-differential equations.

16.3 Multiscale wave phenomena

16.3.1 Existing foundation

Section 7 discusses nonlocal constitutive formulations proposed for size-dependent behavior; applicability to micro- or nanoscale experiments is not assessed here.

16.3.2 Importance

Multiscale phenomena govern material behavior from atomic to continuum scales.

16.3.3 What's achieved

The inherited section contains nonlocal constitutive and dispersion claims; they were not independently verified or compared with experimental data in this revision.

16.3.4 What's open

Nonlocal-fractional hybrid models. Multiscale coupling (atomic-continuum). Nonlocal interface conditions.

16.3.5 Approaches

- Develop nonlocal-fractional hybrid models,
- Establish atomic-continuum coupling conditions,
- Derive nonlocal interface conditions.

16.3.6 Practical relevance

Nanodevice design, nanomaterial characterization, quantum materials.

16.3.7 Major obstacles

Computational cost of nonlocal models; lack of experimental validation at nanoscale.

16.4 Nonlocal and hereditary continuum interactions

16.4.1 Existing foundation

Sections 6 and 7 discuss nonlocal constitutive models and an MDD operator as distinct spatial and temporal formulations; their source-level definitions and status remain under review.

16.4.2 Importance

Real materials exhibit both spatial and temporal memory; combined models are needed.

16.4.3 What's achieved

Nonlocal integral relations. MDD convolution operators. Individual theory formulations.

16.4.4 What's open

Combined nonlocal-hereditary models. Coupled spatial-temporal memory effects.

16.4.5 Approaches

- Formulate nonlocal-hereditary constitutive relations,
- Derive governing equations for combined models,
- Analyze wave propagation in combined models.

16.4.6 Practical relevance

Materials with complex microstructure and history-dependent response.

16.4.7 Major obstacles

Mathematical complexity of combined spatial-temporal nonlocality.

16.5 Advanced interface conditions

16.5.1 Existing foundation

Classical interface conditions are listed in Section 3.9 under perfect-contact assumptions; their extensions to generalized media and imperfect bonding require model-specific derivation and source checking.

16.5.2 Importance

Interface conditions govern wave reflection, transmission, and energy partition at material boundaries.

16.5.3 What's achieved

Perfect and imperfect bonding conditions for classical media. Electrical interface conditions for piezoelectric media.

16.5.4 What's open

Interface conditions for nonlocal media. Interface conditions for memory-dependent media. Interface conditions for multiphysics media with electrical coupling.

16.5.5 Approaches

- Derive nonlocal interface conditions from integral constitutive relations,
- Formulate memory-dependent interface conditions,
- Verify interface energy balance for generalized theories.

16.5.6 Practical relevance

NDE accuracy, energy harvesting efficiency, thermal interface design.

16.5.7 Major obstacles

Nonlocal interface conditions not established; energy flux uncertainty at interfaces.

16.6 Multiphysics mode conversion

16.6.1 Existing foundation

Wave mode conversion at interfaces is studied for elastic and thermoelastic media (Section 9).

16.6.2 Importance

Mode conversion governs energy partition between wave types at interfaces.

16.6.3 What's achieved

A scattering-matrix formulation is described in the inherited manuscript, but it was not re-derived or checked against the cited full text in this revision. Mode classification and normalization remain problem-specific.

16.6.4 What's open

Mode conversion in piezothermoelastic, nonlocal, and memory-dependent media. Complex mode energy flux computation.

16.6.5 Approaches

- Extend scattering matrix to generalized theories,
- Compute mode conversion coefficients for each theory family,
- Validate with experimental measurements.

16.6.6 Practical relevance

NDE, sensing, energy harvesting, acoustic devices.

16.6.7 Major obstacles

Complex mode normalization; uncertain energy flux formulations.

16.7 Reproducible numerical benchmarks

16.7.1 Existing foundation

BM-1 is a proposed test problem only; no numerical benchmark was executed. The cited interface calculations are source leads, not reproduced results.

16.7.2 Importance

Reproducible benchmarks validate numerical methods and enable comparison across research groups.

16.7.3 What's achieved

BM-1 analytical benchmark. Scattering matrix formulation.

16.7.4 What's open

Benchmarks for generalized theories. Benchmarks for multiphysics interfaces. Open-source benchmark implementations.

16.7.5 Approaches

- Derive analytical solutions for simplified generalized theory problems,
- Implement open-source benchmark codes,
- Establish benchmark repository with verification data.

16.7.6 Practical relevance

Software validation, method comparison, reproducibility.

16.7.7 Major obstacles

Lack of closed-form solutions for generalized theories; computational cost.

16.8 Experimental validation

16.8.1 Existing foundation

Section 13 contains inherited statements about experimental validation of classical thermoelasticity; the cited evidence and scope were not audited in this revision.

16.8.2 Importance

Experimental validation is essential for practical adoption of theoretical models.

16.8.3 What's achieved

Validation and clinical-use statements in the inherited manuscript are source leads only; no evidence review is claimed here.

16.8.4 What's open

Experimental validation of generalized theories. Measurement of thermal wave speed. Measurement of energy flux at interfaces.

16.8.5 Approaches

- Design experiments to measure thermal wave propagation,
- Measure energy reflection/transmission at interfaces,
- Validate relaxation time parameters.

16.8.6 Practical relevance

Practical adoption of generalized theories in engineering applications.

16.8.7 Major obstacles

Experimental measurement of thermal waves is challenging; direct energy flux measurement is difficult.

16.9 Inverse material characterization

16.9.1 Existing foundation

Estimating material parameters from wave measurements is a candidate inverse problem; the relevant methods, identifiability conditions, and literature coverage have not been assessed here.

16.9.2 Importance

Material parameters (relaxation times, nonlocal length scales, kernel parameters) must be determined from experiments.

16.9.3 Existing source leads

The inherited manuscript alludes to inverse methods for classical elasticity and thermoelasticity; those methods and their evidence base were not reviewed here.

16.9.4 What's open

Inverse characterization of relaxation times, nonlocal length scales, and MDD kernel parameters.

16.9.5 Approaches

- Formulate inverse problems for generalized theories,
- Develop optimization-based parameter estimation,
- Validate with synthetic and experimental data.

16.9.6 Practical relevance

Material characterization, quality control, structural health monitoring.

16.9.7 Major obstacles

Ill-posedness of inverse problems; non-uniqueness of parameter identification.

The provisional relationship schematic appears once as Figure 2 in Section 1.

16.10 Sequencing the programme

The nine directions are not presented as a validated ranking or a required sequence. Some practical dependencies are plausible: a clearly specified constitutive model supports analysis; a verified numerical implementation supports reproducibility; and both can strengthen later comparisons with measurements. These dependencies do not imply that experimental, mathematical, and computational work must proceed strictly in series. No comparative cost estimate, prediction of a qualitatively distinct interface signal, or claim about the balance of modal analyses and measurements in the literature has been established in this revision. The sequence below is therefore an editorial planning aid only.

Table 36: Summary of future research opportunities.

Direction	What's achieved	What's open	Major obstacles
Coupled-field emerging materials	Constitutive laws require source checks	2D materials, metamaterials	Lack of experimental data
Mathematically consistent transport	Governing equations not uniformly source-checked	Limit theorems; energy flux	Mathematical complexity
Multiscale wave phenomena	Nonlocal relations; wave dispersion	Hybrid models; multiscale coupling	Computational cost
Nonlocal-hereditary interactions	Individual theories require audit	Combined models	Mathematical complexity
Advanced interface conditions	Classical assumptions stated; extensions open	Nonlocal; memory-dependent interfaces	Not established
Multiphysics mode conversion	Scattering formulation unverified here	Generalized theories; complex modes	Normalization uncertainty
Reproducible benchmarks	BM-1 specified but not executed in this revision	Generalized theory benchmarks	Code/data and independent reproduction
Experimental validation	Evidence inventory not completed	Generalized theories; interface flux	Measurement and model-selection questions
Inverse characterization	Literature and identifiability not assessed	Relaxation times; kernel parameters	Uniqueness and uncertainty

Table 37: Notation introduced in Section 16.

Symbol	Meaning	Units	Equation
τ_0	Thermal relaxation time to be determined independently	s	Section 4.4
ξ	Nonlocal length scale to be determined independently	m	Section 7.4
$K(t-\xi)$	Memory kernel function	–	Section 4.10
α	Fractional derivative order	–	Section 11.1
λ	Interface impedance or coupling parameter	–	Section 9.3
θ	Temperature increment	K	Section 3.4

16.11 Future directions summary table

16.12 Notation summary for Section 16

16.13 Open questions for Section 16

See OQ10.1–OQ10.8, OQ12.1–OQ12.3, and challenge-specific open questions above.

17 Conclusions and Scientific Outlook

17.1 The shape of the field as this review has presented it

Read across the seventeen chapters of this review, generalized continuum mechanics for multi-physics wave propagation is a field with an unusually clear intellectual structure and an unusually weak evidential one. The intellectual structure is the work of a small number of decisive ideas, each of which the field has absorbed, extended, and preserved in a form that is recognisably the same. The evidential weakness is the condition under which those ideas have been tested.

The main modelling choices surveyed here are finite-rate heat-flux laws, alternative free-energy and entropy structures, additional field couplings, characteristic lengths, and time-history or fractional operators. Their mathematical consequences depend on the precise constitutive equations and parameter restrictions; a family name alone does not establish finite propagation, dissipation, stability, or well-posedness. Comparing these choices requires the same geometry, material data, boundary conditions, frequency convention, and validation target. This revision derives only the classical local energy flux and selected algebraic corrections. It does not adjudicate which generalized family is physically superior.

The repository manuscript reviews analytical wave formulations and interface calculations, but this audit did not reproduce those studies or verify their complete primary texts. One local result was re-derived: for the classical first law and momentum balance, total energy per unit volume is $\mathcal{E} = \frac{1}{2}\rho v_i v_i + \rho e$ and the outward-positive flux is $J_i = q_i - \sigma_{ij} v_j$ for symmetric stress. The modal-scattering equations, benchmark conservation values, and published piezothermoelastic energy ratios remain claims for future claim-level checking; they are not treated here as independently reproduced or verified.

What the field has not achieved is equally clear, and it is the more important half of the assessment. The comparisons between families rest on parameter substitutions rather than on convergence theorems with error estimates. The validation axis of the taxonomy of Section 14 is dominated by self-consistency—a numerical solution agreeing with a closed-form solution of the same equations—and is nearly empty of comparison against experiment or against independently implemented competing formulations. The energy flux of the pure generalized families, which is the quantity on which an interface energy partition depends, is not available in the reviewed corpus in a form that can be independently verified. And the parametric flexibility that the field has cultivated, admirable as it is, has outpaced the ability of the available measurements to constrain it.

17.2 What is established, and what is not

17.2.1 Derived in this revision

- The classical local total-energy density and outward-positive flux, from the displayed momentum and first-law balances, subject to symmetric Cauchy stress and no internal heat supply.

- The high-wavenumber limits of the explicitly stated one-dimensional nonlocal dispersion relation in Section 7.
- The stated-domain distinction between a quasistatic electric potential model and a full Maxwell system: the former has no displacement-current electromagnetic wave branch.

These calculations do not certify the remaining constitutive equations, source citations, scattering benchmarks, or generalized-family fluxes.

17.2.2 Open

- The explicit time-domain energy flux and dissipation function for each of the families listed in Section 10.4, obtainable in principle from the theory-specific constitutive evolution laws but not yet derived in a verifiable form.
- The interface conditions for non-local and memory-dependent media, and the conditions under which the integral and differential formulations give consistent interface tractions.
- Rigorous limit theorems between the theory families, with error estimates in a stated norm, which are the precondition for a quantitative rather than a qualitative cross-family comparison.
- Experimental discrimination among the competing families on a common material, and identification of the relaxation, kernel, and length-scale parameters from data that were not generated by the theory being tested.
- Reproducible numerical benchmarks executed independently, which would allow the validation axis of the taxonomy to be populated with something other than self-consistency.

17.3 Methodological stance: tracing differences to assumptions

Differences between formulations may reflect constitutive choice, material symmetry, geometry, boundary conditions, harmonic convention, normalization, or approximation level; they should not be labelled contradictions without comparing these assumptions. This revision distinguishes results derived directly from displayed equations (DERIVED), conditional results, metadata-level checks, secondary-source statements, unverified claims, and open derivations. A bibliographic match is not a claim-level source check. Legacy labels in the inherited register and appendices are not evidence of verification unless explicitly re-derived or checked in this version.

The stance has a cost, which should be stated. By declining to adjudicate between formulations, this review leaves the reader with a map rather than a recommendation in those places where the literature does not itself contain one. The map is, however, the more durable object: the particular theory that will be found correct in a given application is not yet decidable on the available evidence, and a review that pretended otherwise would be asserting more than the evidence supports. What the map does provide is a statement of what evidence would be needed to decide, and that is the more useful contribution at the present stage of the field.

See the single historical overview in Figure 1.

17.4 The three problems that matter most

The field's open items are not of equal weight, and the review would be less useful if it did not distinguish them. Three of them are, in the reviewer's judgement, the ones on which the scientific standing of generalized continuum mechanics now depends.

Energy accounting for the generalized families. Every physical claim made about an interface—every energy ratio, every absorption coefficient, every statement about how a disturbance is shared between two media—presupposes an energy flux for the theory in which the calculation is performed. For the piezothermoelastic family that flux exists and has been computed independently by several groups; for the pure generalized families it does not exist in a verifiable form. The consequence is that a substantial part of the recent literature on generalized thermoelastic interfaces rests on an energy definition that has not been derived from the theory it accompanies. Closing this gap is a derivation problem, not an experimental one, and it is the one piece of unfinished work that the present review can specify exactly.

The status of the limits. The assertion that the generalized theories reduce to classical thermoelasticity in the appropriate limit is the connective tissue of the entire comparative enterprise. It is currently supported by parameter substitutions, and a parameter substitution is not a convergence theorem. If the limits were established with error estimates, a prediction made with, say, a memory-dependent formulation could be given a quantified statement of how far it may be from the corresponding classical prediction, and the choice of theory could be made on the basis of a stated tolerance rather than on preference. This too is a derivation problem.

Discriminating evidence. Almost every comparison between competing formulations in the reviewed corpus is a comparison between two solutions of the same equations, and none of them is a comparison between a prediction and a measurement. The obstruction is not a shortage of candidate observables; Section 7.3 identifies one—the dispersion signature—that separates two physically different mechanisms, and Section 13 assembles the application scenarios in which such an observable could be measured. What is missing is a protocol: a common material, a common measurement, and a pre-registered prediction from each competing theory. The framework of the taxonomy, if adopted, would make such a protocol easy to specify, since it would identify the cells of the product space in which the competing theories differ.

17.5 Final outlook

Against that assessment, the next decade of research in the field should pursue the following, in the order in which it would most improve the standing of the subject.

1. **Complete the energy flux derivations** for the generalized families, using the theory-specific constitutive evolution laws as the common template, and publish the dissipation function alongside the flux.
2. **Establish rigorous limit theorems** between the theory families, with explicit error estimates and in a stated norm.
3. **Formulate the interface conditions** for non-local and memory-dependent media, and demonstrate their consistency between the integral and differential representations.
4. **Create reproducible numerical benchmarks** that independent groups can execute without sharing code, following the structure of Appendix F.
5. **Validate experimentally**, beginning with the thermal wave velocity and the interface energy partition, and with the dispersion measurement that distinguishes a non-local from a gradient mechanism.

6. **Extend the coupled-field models** to the materials for which the coupling is strongest—two-dimensional systems, metamaterials, and semiconductors—where the additional constitutive complexity is most likely to be repaid by predictive power.

7. **Characterize the parameters inversely**, so that the relaxation times, kernel parameters, and length scales are measured rather than fitted, and so that the number of free parameters in a model is matched to the number of independent measurements available.

The foundation is solid: the classical balance laws are verified, the generalized governing equations are established, and the scattering framework is consistent. The path forward is clear, and it is narrower than the field's recent rate of extension would suggest. The most valuable work available is no longer the construction of additional generalizations—the space of those has been explored extensively—but the derivation, the proof, and the measurement of the ones already in hand. The constructive organizational approach employed here follows the model of comprehensive review articles organized around conceptual themes rather than around individual publications, such as the recent review of generalized thermoelasticity and its modified models Shakeriaski et al. [2021] and the large multi-author collaborative reviews that have defined the format in neighbouring fields Abdalla et al. [2022].

17.6 Notation summary for Section 17

Table 38: Notation introduced in Section 17.

Symbol	Meaning	Units	Equation
J_i	Total energy flux vector	$\mathrm{W\,m^{-2}}$	(109)
σ_{ij}	Cauchy stress tensor	Pa	Section 3
q_i	Heat flux vector	$\mathrm{W\,m^{-2}}$	Section 3
τ_0	Mechanical relaxation time	s	Section 4
c	Thermal wave speed	$\mathrm{m\,s^{-1}}$	Section 6
ω	Angular frequency	$\mathrm{rad\,s^{-1}}$	Section 8
$\Delta \boldsymbol{X}$	Scattering matrix	—	Section 9

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A Equation Verification Register

This appendix records equation claims and their audit status. Legacy status labels from earlier manuscript work are not accepted as verification in version 1.1 unless the present register explicitly identifies a derivation performed here. No comprehensive primary-source equation audit has been completed.

- **DERIVED:** Algebra checked directly from displayed governing equations in this revision.
- **CONDITIONAL:** Valid only under the stated continuum, constitutive, or boundary assumptions.
- **PRIMARY-SOURCE-CHECKED:** Checked against the identified primary source; this does not assert an independent derivation.
- **SECONDARY-SOURCE:** Supported only by a named secondary source; not equivalent to primary verification.
- **UNVERIFIED:** Not checked against a primary source or independently derived in this phase.
- **OPEN:** Required derivation or source evidence is absent; not a canonical result.

Historical statuses and source notes are not evidence of present verification. The scientific-audit report identifies the limited equations derived in this phase.

Table 39: Equation register for Section 3 (Foundations of Continuum Wave Mechanics).

EQ ID	Equation	Status	Source / notes
EQ-001	$\rho\dot{v}_i = \sigma_{ji,j} + \rho f_i$	CONDITIONAL	Classical Cauchy continuum; infinitesimal motion; no couple stresses
EQ-002	$\sigma_{ij} = \sigma_{ji}$	CONDITIONAL	Classical angular momentum balance; no couple stresses or body couples
EQ-003	Mechanical power balance	DERIVED	Product of momentum balance and velocity; classical stress power
EQ-004	$\rho\dot{e} = \sigma_{ij}\dot{\varepsilon}_{ij} - q_{i,i}$	CONDITIONAL	Classical local continuum; no internal heat supply; e specific internal energy
EQ-005	Dissipation inequality ≥ 0	UNVERIFIED	Legacy equation; exact entropy/free-energy assumptions not audited
EQ-006	Coleman–Noll restrictions	UNVERIFIED	Legacy equation; state variables and free-energy convention not audited
EQ-007	$J_i = q_i - \sigma_{ij}v_j$	DERIVED	Section 3.8, outward-positive convention; symmetric stress
EQ-008	$v_i = \dot{u}_i$	UNVERIFIED	Definition of velocity
EQ-009	$\varepsilon_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i})$	UNVERIFIED	Definition of infinitesimal strain
EQ-010	$\omega_{ij} = \frac{1}{2}(u_{i,j} - u_{j,i})$	UNVERIFIED	Definition of rotation tensor

A.1 Layer I — First Principles (Section 3)

A.2 Layer II — Classical and Generalized Thermoelastic Theories (Section 4)

A.3 Layer III — Coupled Multiphysics Continua (Section 5)

A.4 Layer IV — Generalized Transport and Memory (Section 6)

A.5 Layer V — Spatial Nonlocality and Gradient Continua (Section 7)

A.6 Layer VI — Interface Equations and Energy Ratios (Section 9)

A.7 Layer VII — Energy Transport and Dissipation (Section 10)

A.8 Verification Summary

A.9 Open Derivations Requiring Full-Text Verification

The following model families require a dedicated claim-level source and derivation audit before an energy-flux formula is asserted: Lord–Shulman; Green–Lindsay; Green–Naghdi types II and III; dual- and triple-phase-lag; Moore–Gibson–Thompson; memory-dependent; and nonlocal/gradient models. The full texts, exact constitutive conventions, free energies, external-field power terms, boundary conditions, and evolution laws must be checked for each case. The citations and formulae in the retained working appendix are not evidence that such verification has occurred in this phase.

These model-specific flux derivations remain open; legacy Appendix E/Sec. 10 questions identify work items, not canonical formulae.

Table 40: Equation register for Section 4 (Thermoelastic Theories).

EQ ID	Equation	Status	Source / notes
EQ-011	$\rho\ddot{u}_i = (\lambda+\mu)u_{j,ji} + \mu u_{i,jj} - \beta T_{,i}$	UNVERIFIED	Classical coupled thermoelasticity (Biot 1956)
EQ-012	$\rho c_v \dot{T} = k\nabla^2 T - \beta T_0 \dot{u}_{i,i}$	UNVERIFIED	Classical heat equation (Fourier conduction)
EQ-013	$q_i + \tau_0 \dot{q}_i = -kT_{,i}$	UNVERIFIED	Lord–Shulman (1967); primary source
EQ-014	$c_T = \sqrt{k/(\rho c_v \tau_0)}$	UNVERIFIED	LS thermal wave speed; $\tau_0 > 0$
EQ-015	Green–Lindsay Fourier-flux energy equation and temperature-rate stress coupling (Eqs. 4.9–4.10)	PRIMARY-SOURCE-CHECKED	Green–Lindsay (1972), F1; no tau0 term on strain-rate coupling; tau1 greater than or equal to tau0 greater than zero
EQ-016	$\kappa^* \nabla^2 \alpha_{\mathrm{GN}} = \rho c_e \ddot{\alpha}_{\mathrm{GN}} + \beta T_0 \dot{u}_{i,i}$	PRIMARY-SOURCE-CHECKED	GN-II 1993 Eq. (3.16) integrated once; source-free alpha form, zero integration constant; F17
EQ-017	$p_i = \kappa^* (\alpha_{\mathrm{GN}})_{,i}$	UNVERIFIED	Candidate isotropic GN-II entropy flux; coefficient notation aligned with EQ-016; free energy and primary source not checked
EQ-018	$q_i(t+\tau_q) = -kT_{,i}(t+\tau_T)$	UNVERIFIED	DPL constitutive law (Tzou 1995)
EQ-019	First-order three-lag TPL law with thermal-displacement-gradient contribution (Section 4.8)	PRIMARY-SOURCE-CHECKED	Roy Choudhuri (2007), Eqs. 3–4, full-text finding F6; no second-order term
EQ-020	$D_{\omega_M} f(t) = \frac{1}{\omega_M}\int_{t-\omega_M}^{t} \kappa(t-\xi) f'(\xi)\, d\xi$	UNVERIFIED	MDD candidate; finite horizon distinguished from angular frequency; source convention not checked
EQ-021	Inherited specific MDD kernel formula	OPEN	Withheld from the text pending equation-level and dimensional comparison with its cited source; no kernel is asserted
EQ-022	Legacy MGT heat-flux equation withdrawn	UNVERIFIED	Dimensionally inconsistent form and scratch cancellation removed; verify selected MGT variant against primary source
EQ-022F	${}^{C}D_t^{\alpha} T = a_\alpha \nabla^2 T$	CONDITIONAL	Scalar fractional diffusion–wave example only, $0 < \alpha \leq 2$; $[a_\alpha] = \mathrm{m}^2\mathrm{s}^{-\alpha}$; coupled embedding unverified

Table 41: Equation register for Section 5 (Multiphysics Continua).

EQ ID	Equation	Status	Source / notes
EQ-023	$\sigma_{ij} = c_{ijkl}\varepsilon_{kl} - e_{kij}E_k$	UNVERIFIED	Piezoelectric constitutive relation
EQ-024	$D_i = e_{ikl}\varepsilon_{kl} + \kappa_{ij}E_j$	UNVERIFIED	Electric displacement (piezoelectric)
EQ-025	$E_i = -\phi_{,i}$	UNVERIFIED	Electric field from potential
EQ-026	$D_{i,i} = 0$	UNVERIFIED	Charge conservation (no free charge)
EQ-027	$\sigma_{ij} = c_{ijkl}\varepsilon_{kl} - e_{kij}E_k - \beta_{ij}T$	UNVERIFIED	Piezothermoelastic (Barak et al. Barak et al. [2023a], Eq. 3)
EQ-028	$D_i = \kappa_{ij}E_j + e_{ikl}\varepsilon_{kl} + p_iT$	UNVERIFIED	Piezothermoelectric (Barak et al. Barak et al. [2023a], Eq. 6)
EQ-029	Inherited electrical storage formula	OPEN	Withdrawn from the working flux appendix; no universal piezoelectric storage expression is asserted
EQ-5.17	Isotropic micropolar force stress with independent microrotation ϕ_k	UNVERIFIED	Linear isotropic specialization; source equation and coupled thermoelastic moduli require full-text verification

Table 42: Equation register for Section 6.

EQ ID	Equation	Status	Source / notes
EQ-6.1	Schematic constitutive operator $\mathcal{D}[\boldsymbol{q}] = -k\nabla T$	CONDITIONAL	Not a universal representation; only for a specified linear flux law
EQ-6.5	${}^{C}D_t^{\alpha}T = a_{\alpha}\nabla^2 T$	CONDITIONAL	Scalar equation only; $0 < \alpha \leq 2$; dimensional a_{α}; no coupled-law inference
EQ-6.6	Hereditary convolution	CONDITIONAL	Kernel integrable on its stated history interval; model-specific thermodynamic constraints not implied
EQ-6.7	Finite-window MDD operator	CONDITIONAL	$\omega_M > 0$; kernel, normalization, regularity and admissibility must be specified
EQ-6.10	$J_i = q_i - \sigma_{ij}v_j$	CONDITIONAL	Classical local mechanical balance with specified heat-flux vector; not a general MDD total-energy theorem
EQ-6.11	MDD candidate flux law	CONDITIONAL	Only for the displayed kernel/operator convention; stored energy and boundary power remain open

Table 43: Equation register for Section 7.

EQ ID	Equation	Status	Source / notes
EQ-7.1	Integral nonlocal constitutive relation	CONDITIONAL	Kernel and boundary/domain prescription must be specified
EQ-7.2	Differential nonlocal stress equation	CONDITIONAL	Helmholtz differential form only; not identical to every integral kernel/boundary problem
EQ-7.3	Strain-gradient constitutive relation	CONDITIONAL	Positive gradient modulus and extra boundary data as stated in the text
EQ-7.4	$c(k) = c_0/\sqrt{1+\xi^2 k^2}$	CONDITIONAL	One-dimensional bar and the specific Eringen differential model displayed

Table 44: Equation register for Section 9.

EQ ID	Equation	Status	Source / notes
EQ-9.3	Perfect-contact mechanical conditions	CONDITIONAL	Perfectly bonded interface; no slip or separation; common traction normal
EQ-9.4	Perfect-contact thermal conditions	CONDITIONAL	Perfect thermal contact; no Kapitza jump; specified heat-flux law
EQ-9.5	Electrical interface conditions	CONDITIONAL	Quasi-static electric field; no free surface charge for continuous normal displacement
EQ-9.9	Degree-eight secular polynomial	CONDITIONAL	6mm medium; sagittal plane contains c-axis; quasi-static electric closure; specified P–SV–T–ϕ plus SH set; trivial root removed
EQ-9.10	Total-field R, T, A_{int} definitions	CONDITIONAL	Same normal and normalization; include all cross terms; A_{int} is interface-local only
EQ-9.11	$\sum R + \sum T = 1$	CONDITIONAL	Passive lossless interface; zero storage/residual; complete total-field accounting
EQ-9.12	$\sum R + \sum T + A_{\mathrm{int}} = 1$	CONDITIONAL	Total-field interface fluxes include all modes/cross terms; interfacial power/storage accounted; bulk dissipation kept separate

Table 45: Energy-balance register. DERIVED and CONDITIONAL labels reflect this revision's limited calculations; all model-specific OPEN entries require further work.

EQ ID	Equation	Status	Source / notes
EQ-10.1	$\mathcal{E} = \frac{1}{2}\rho v_i v_i + \rho e$	DERIVED	Classical thermoelasticity; e specific internal energy
EQ-10.2	$\partial_t \mathcal{E} + \partial_i J_i = \rho f_i v_i$	DERIVED	No internal heat source; classical model
EQ-10.3	$J_i = q_i - \sigma_{ij} v_j$	DERIVED	Outward-positive flux, symmetric Cauchy stress
EQ-10.4	$\langle J_i \rangle_t = \bar{q}_i - \frac{1}{2}\Re[\hat{\sigma}_{ij}\hat{v}_j^*]$	CONDITIONAL	$e^{-i\omega t}$ convention; linear harmonic fields; mean heat flux evaluated separately
EQ-10.5	Interfacial jump balance	CONDITIONAL	Pillbox limit; jump orientation, storage, and interfacial power specified
EQ-10.6	Passive-interface normal flux continuity	CONDITIONAL	Passive interface; zero interfacial power and storage; same normal orientation
EQ-10.11	$\sum R + \sum T + A_{\text{int}} = 1$	CONDITIONAL	Interface-local total-field fluxes; include all cross terms; interface residual only
EQ-10.12	$\sum R + \sum T = 1$	CONDITIONAL	Lossless interface with zero storage/residual; complete total-field accounting including cross terms
EQ-10.13	Generalized-theory energy fluxes	OPEN	Derive from each model's free energy and boundary power

Table 46: Overall verification status by status class.

Status	Meaning in this revision
DERIVED	Algebra checked from displayed equations under stated assumptions
CONDITIONAL	Relation applies only under the assumptions stated in the register
PRIMARY-SOURCE-CHECKED	Equation statement checked against the named primary text; not necessarily independently re-derived
SECONDARY-SOURCE	Supported only by a named secondary source; not a primary-source verification
UNVERIFIED	Not checked against a primary source or independent calculation in this phase
OPEN	Required derivation or evidence is absent; not a canonical result

B Detailed Balance Law Derivations

This appendix provides the complete derivations of the balance laws presented in Section 3. All derivations are from first principles using the divergence theorem and localization argument.

B.1 Derivation of Linear Momentum Balance (EQ-001)

Consider a material subvolume $V \subset \Omega$ with surface $S = \partial V$ and outward unit normal $\boldsymbol{n}$. The linear momentum balance states:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho v_i \,\mathrm{d}V = \int_S \sigma_{ji} n_j \,\mathrm{d}S + \int_V \rho f_i \,\mathrm{d}V. \tag{124}$$

Applying the divergence theorem to the surface integral:

$$\int_S \sigma_{ji} n_j \,\mathrm{d}S = \int_V \sigma_{ji,j} \,\mathrm{d}V. \tag{125}$$

Substituting (125) into (124):

$$\int_V \left(\rho \dot{v}_i - \sigma_{ji,j} - \rho f_i\right) \mathrm{d}V = 0. \tag{126}$$

Since V is arbitrary, the integrand must vanish pointwise:

$$\rho \dot{v}_i = \sigma_{ji,j} + \rho f_i. \tag{127}$$

B.2 Derivation of Angular Momentum Balance → Symmetric Stress (EQ-002)

The angular momentum balance for an arbitrary subvolume V:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho(\boldsymbol{x} \times \boldsymbol{v})_k \,\mathrm{d}V = \int_S \epsilon_{kij} x_i \sigma_{jm} n_m \,\mathrm{d}S + \int_V \rho \epsilon_{kij} x_i f_j \,\mathrm{d}V. \tag{128}$$

Applying the divergence theorem and using the linear momentum balance (127), the volume terms cancel, leaving:

$$\int_V \epsilon_{kij}(\sigma_{ij} - \sigma_{ji}) \,\mathrm{d}V = 0. \tag{129}$$

Since V and k are arbitrary:

$$\sigma_{ij} = \sigma_{ji}. \tag{130}$$

B.3 Derivation of Mechanical Power Balance (EQ-003)

Multiply the linear momentum equation (127) by v_i:

$$\rho \dot{v}_i v_i = \sigma_{ji,j} v_i + \rho f_i v_i. \tag{131}$$

The left-hand side:

$$\rho \dot{v}_i v_i = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}\rho v_i v_i\right). \tag{132}$$

For the stress term, use the identity:

$$\sigma_{ji,j} v_i = (\sigma_{ji} v_i)_{,j} - \sigma_{ji} v_{i,j}. \tag{133}$$

Since $v_{i,j} = \dot{u}_{i,j} = \dot{\varepsilon}_{ji}$ and $\sigma_{ji} = \sigma_{ij}$:

$$\sigma_{ji} v_{i,j} = \sigma_{ij} \dot{\varepsilon}_{ij}. \tag{134}$$

Combining:

$$\rho \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{1}{2} v_i v_i \right) = -\sigma_{ij} \dot{\varepsilon}_{ij} + (\sigma_{ij} v_j)_{,i} + \rho f_i v_i. \tag{135}$$

B.4 Derivation of Internal Energy Balance (EQ-004)

The first law of thermodynamics for a continuum:

$$\rho \dot{e} = \sigma_{ij} \dot{\varepsilon}_{ij} - q_{i,i} + \rho r. \tag{136}$$

With no external heat supply ($r = 0$):

$$\rho \dot{e} = \sigma_{ij} \dot{\varepsilon}_{ij} - q_{i,i}. \tag{137}$$

B.5 Derivation of Classical Energy Flux (EQ-007)

Add the mechanical power balance (135) and the internal energy balance (137). Define the total energy density:

$$\mathcal{E} = \frac{1}{2} \rho v_i v_i + \rho e. \tag{138}$$

Then:

$$\frac{\partial \mathcal{E}}{\partial t} + \frac{\partial}{\partial x_j} \left(q_j - \sigma_{ji} v_i \right) = \rho f_i v_i. \tag{139}$$

With no body forces ($f_i = 0$):

$$\frac{\partial \mathcal{E}}{\partial t} + \frac{\partial J_i}{\partial x_i} = 0, \qquad J_i = q_i - \sigma_{ij} v_j. \tag{140}$$

This is the classical energy flux (EQ-007).

C Benchmark Specifications and Status

The entries below are proposed reproducibility targets, not reported computational results. No benchmark was executed in this revision. In particular, the inherited "VERIFIED" labels and published numerical values have been removed; a mathematical expectation is not a reproduced benchmark.

C.1 BM-1: Classical elastic interface

Problem: Plane P-wave incident on a perfectly bonded interface between two elastic half-spaces.

Governing equations: Navier equations for the stated isotropic elastic model in each medium. Specify densities, Lamé moduli, incidence angle, normal orientation, and amplitude normalization before computing coefficients.

Expected check: For a lossless passive interface, evaluate the complete time-averaged normal flux and verify conservation after all propagating channels and any relevant cross terms are included. The shorthand $R_P + T_P = 1$ is conditional on a single uncoupled channel and consistent flux normalization.

Status: SPECIFIED, NOT EXECUTED. No numerical value is asserted.

C.2 BM-2: Classical thermoelastic interface

Problem: Coupled thermoelastic wave at a welded interface.

Status: OPEN. Requires model-specific coupled flux, boundary data, and primary-source comparison; no benchmark values supplied.

C.3 BM-3: Piezothermoelastic interface

Problem: Coupled elastic, thermal, and quasistatic electrical fields at an interface.

Status: OPEN. Inherited scattering-matrix equations and energy ratios attributed to the cited literature were not rederived or independently checked in this revision. The cited publication is a source lead only, not a reproduction record.

C.4 BM-4: One-dimensional nonlocal elastic dispersion

Problem: For the specific Helmholtz-type model stated in Section 7, compare the dispersion relation and its long- and short-wavelength limits with the classical bar.

Status: ALGEBRAIC LIMITS DERIVED for the stated law; no numerical benchmark or independent implementation was produced. Boundary-power and interface checks remain open.

C.5 BM-5: Memory-dependent thermoelastic wave

Problem: Select a fully specified memory kernel and initial-boundary-value problem; compare a numerical solution with an independently derived reference solution.

Status: OPEN. No kernel-specific benchmark data or reference solution is supplied in this version.

D Figure Source and Audit Status

The eighteen editable SVG figures have generated PDF renditions. Strict CI run 39771440399 checked the assets and compiled the 149-page v1.1 PDF. All four 100-dpi contact sheets and selected full-page renders were reviewed for gross layout and clipping. This is not an exhaustive final-size typography proof or a scientific/source-attribution audit. Per-figure status remains in `figure-audit.csv`.

Before release, inspect every figure at final manuscript size, verify source-to-figure mappings, and reconcile every numerical or evidentiary claim against its equation/source. The strict build and overview-level visual review passed, but the figure-level scientific/source audit and exhaustive final-size typography proof remain open.

Table 47: Figure source inventory and v1.1 review scope. This table is not a visual proof or source verification.

ID	Source file / topic	Status
F-01	`fig01-theory-evolution.svg`: selected historical milestones	PDF generated; source check and at-size proof open
F-02	`fig02-theory-relationships.svg`: provisional family links; no convergence claims	PDF generated; final-size proof open
F-03	`fig03-literature-landscape.svg`: schematic chronology; links are provisional	PDF generated; final-size proof open
F-04	`fig04-continuum-kinematics.svg`: infinitesimal kinematics	PDF generated; content check open
F-05	`fig05-constitutive-relations.svg`: thermoelastic constitutive response	PDF generated; content check open
F-06	`fig06-classical-waves.svg`: classical elastic waves	PDF generated; content check open
F-07	`fig07-theory-map.svg`: provisional taxonomy; links are not verified limits	PDF generated; final-size proof open
F-08	`fig08-wave-types.svg`: model-dependent modal content	PDF generated; final-size proof open
F-09	`fig09-multiphysics-coupling.svg`: coupled-field map	PDF generated; source check and at-size proof open
F-10	`fig10-mdd-memory.svg`: memory-model audit checklist	PDF generated; final-size proof open
F-11	`fig11-nonlocality.svg`: specified 1D dispersion laws	PDF generated; final-size proof open
F-12	`fig12-solution-workflow.svg`: analysis workflow	PDF generated; source check and at-size proof open
F-13	`fig13-interface-reflection.svg`: interface-wave schematic	PDF generated; source check and at-size proof open
F-14	`fig14-energy-flux-channels.svg`: classical balance and open flux checks	PDF generated; final-size proof open
F-15	`fig15-fractional-order.svg`: fractional transport schematic	PDF generated; source check and at-size proof open
F-16	`fig16-observation-workflow.svg`: computation-to-observation workflow	PDF generated; source check and at-size proof open
F-17	`fig17-application-evidence.svg`: evidence-audit checklist	PDF generated; final-size proof open
F-18	`fig18-taxonomy.svg`: five-axis taxonomy	PDF generated;

E Table Source Inventory and Audit Status

The eight principal comparison tables are editable LaTeX sources in `../tables/source/`. They are included by the master manuscript from that directory. This register records source location and current review status; it does not certify every entry. The energy-transport table was rewritten to avoid presenting unverified generalized flux formulae as canonical. Other tables contain inherited entries whose references, dimensions, counts, and assumptions still require row-by-row audit.

Table 48: Principal comparison-table inventory; source-level audit remains incomplete.

ID	Source file	Primary section	Status
T-01	`T1_historical_development.tex`	4	Rewritten as provisional metadata/source leads; claim-level checks open
T-02	`T2_generalized_thermoelastic.tex`	4	Rewritten as broad model classes; equation-level checks open
T-03	`T3_coupled_multiphysics.tex`	5	Caveated; couplings and cited equations not row-audited
T-04	`T4_wave_types.tex`	3	One universal mode claim narrowed; remaining entries open
T-05	`T5_solution_techniques.tex`	8	Caveated; methods and source claims not independently checked
T-06	`T6_interface_configurations.tex`	9	Caveated; conditions require model-specific review
T-07	`T7_energy_transport.tex`	10	Rewritten; generalized entries explicitly OPEN
T-08	`T8_applications_validation.tex`	13	Rewritten as evidence questions; no validation score assigned

A release audit must check each cell against the cited source, preserve units and tensor conventions, verify inventory counts and labels against the underlying data, and inspect the compiled table for legibility and overflow. The source has not been typeset in this revision, so table-rendering and editorial gates remain OPEN.

F Reproducibility and evidence status

This appendix describes the status of the repository materials accompanying this revision; it is not a claim that every result in the manuscript has been reproduced. The working bibliography is `references_local.bib`. Literature metadata is drawn from the repository inventory and selected DOI checks. DOI resolution is bibliographic verification only. It does not establish that a source was read in full or that a particular equation, interpretation, or numerical result has been checked against that source.

The versioned source bundle contains the LaTeX manuscript, embedded table sources, editable SVG figures, and generated PDF figure assets. The baseline PDF is separately identified as `v1.0-baseline.pdf`; it is not a rendering of the edited source in this directory. The repository copy of that PDF is 161 pages with embedded metadata identifying pdfTeX 1.40.25 and a 2023-11-14 creation date; the separately described 170-page/pdfTeX-1.40.29 attachment is not available in this workspace, so its provenance cannot be reconciled here. A local TeX engine remains unavailable, but strict CI run 39771440399 compiled the v1.1 source, passed the

strict build gates, and published the 149-page PDF and contact sheets; its source commit was ce7bba4550b69dca8d213bf860b153224726f564. The CI report records 483 underfull hbox warnings and zero underfull vbox warnings. Eighteen SVG figures were converted to PDF with the pinned Node renderer; CairoSVG was unavailable because its native Cairo library could not be loaded. Source files, commands, and limitations are recorded in `../README_build.md` and `../reports/build-status.md`.

The engineering and scientific audit registers distinguish the limited calculations checked directly from claims that remain conditional, unverified, or open. Specifically, the classical local energy balance is re-derived in Sections 3 and Appendix B under an outward-positive convention; the other family-specific balance identities and all newly asserted benchmark values remain outside the verified set unless an explicit derivation and source record is provided. The literature search is not a systematic review, and no screening flow diagram or comprehensive database query is claimed.

All tables and figure artwork are editable source, and the manuscript's 18 distinct figure references resolve to generated PDFs in the self-contained master tree. Static XML/PDF checks and the strict manuscript build passed. The four contact sheets and selected full-page renders were reviewed for gross layout and clipping; they do not constitute an exhaustive final-size typographic or scientific proof. No generated numerical data, fit, or independent benchmark dataset is represented as completed evidence.

G Introduction

The classical outward-positive energy flux was re-derived in Section 3 (EQ-007); this appendix is a retained, unverified working collection:

$$J_i = q_i - \sigma_{ij} v_j, \tag{141}$$

derived from the mechanical power balance and internal energy balance (Section 3.8 of the main manuscript).

Some generalized models may introduce additional stored-energy variables, transport channels, dissipation, or boundary-power terms; none of these is universal. This appendix retains selected source leads and open questions. It does not assert a general energy-flux law for any generalized family.

H Working assumptions to check model by model

The following assumptions are provisional bookkeeping items, not a set known to hold simultaneously for every model discussed below:

1. Infinitesimal strain is used where a classical linearized mechanical balance is displayed.
2. Material homogeneity, isotropy, and linearity must be checked for each example.
3. Body force and heat supply are set to zero only where explicitly stated.
4. The harmonic convention $e^{-i\omega t}$ applies only to the displayed frequency-domain substitutions.
5. The classical baseline is $\mathcal{E} = \frac{1}{2}\rho v_i v_i + \rho e$ with specific internal energy e; additional field/internal-variable storage is model-specific and not assigned here.
6. No common balance of the form $\partial_t \mathcal{E} + \partial_i J_i = \mathcal{P} - \mathcal{D}$ is asserted for the generalized models.

I Open audit item 1: Lord–Shulman (LS) Theory

I.1 Governing equations

The LS theory modifies Fourier's law by introducing a flux relaxation time $\tau_0 > 0$:

$$q_i + \tau_0 \dot{q}_i = -kT_{,i}, \tag{142}$$

where q_i is the heat flux, k is the thermal conductivity, T is the temperature increment, and τ_0 is the relaxation time.

The following internal-energy balance is retained as a classical-form candidate only; its validity with the LS thermodynamic state and constitutive equations requires source checking:

$$\rho \dot{e} = \sigma_{ij} \dot{\varepsilon}_{ij} - q_{i,i}. \tag{143}$$

The mechanical power balance (from Section 3.3) remains:

$$\rho \frac{d}{dt} \left(\frac{1}{2} v_i v_i \right) = -\sigma_{ij} \dot{\varepsilon}_{ij} + (\sigma_{ij} v_j)_{,i} + \rho f_i v_i. \tag{144}$$

I.2 Classical flux form

Adding (143) and (144), defining $\mathcal{E} = \frac{1}{2}\rho v_i v_i + \rho e$:

$$\frac{\partial \mathcal{E}}{\partial t} + (q_i - \sigma_{ij} v_j)_{,i} = \rho f_i v_i. \tag{145}$$

The expression $J_i = q_i - \sigma_{ij} v_j$ is the classical local flux under the stated convention. Its compatibility with a complete LS energy balance is not established by the constitutive substitution below and remains open.

I.3 Frequency-domain expression

For time-harmonic fields with $e^{-i\omega t}$ convention, (142) gives:

$$q_i(1 - i\omega\tau_0) = -kT_{,i}, \qquad q_i = \frac{-kT_{,i}}{1 - i\omega\tau_0}. \tag{146}$$

Therefore, the energy flux in the frequency domain is:

$$J_i = -\sigma_{ij} v_j - \frac{kT_{,i}}{1 - i\omega\tau_0}. \tag{147}$$

I.4 Time-domain interpretation

In the time domain, (147) corresponds to:

$$J_i = q_i - \sigma_{ij} v_j, \qquad \text{where } q_i \text{ solves } q_i + \tau_0 \dot{q}_i = -kT_{,i}. \tag{148}$$

There is no unique expression for J_i solely in terms of T in the time domain without q_i, because the relaxation equation introduces memory.

I.5 Status

UNVERIFIED WORKING CLAIM. The flux form $J_i = q_i - \sigma_{ij} v_j$ is valid for LS theory with q_i from (142). The frequency-domain expression (147) is retained as unverified working material. A time-domain expression solely in terms of T (eliminating q_i) would require solving the relaxation ODE, which introduces an integral (memory) term. This remains as an open formulation question.

J Open audit item 2: Green–Lindsay (GL) Theory

J.1 Status of inherited equations

Green–Lindsay is described in the corrected manuscript as a temperature-rate-dependent, single-temperature theory with two relaxation parameters. Earlier equations in this appendix introduced a second temperature and are withdrawn as inaccurate legacy material; they are not reproduced as candidate GL equations. The complete constitutive system and its energy balance remain OPEN pending primary-source checking and derivation. The classical local expression $J_i = q_i - \sigma_{ij} v_j$ is not, by itself, a verified Green–Lindsay energy-flux theorem.

K Open audit item 3: Green–Naghdi Type II (GN-II) Theory

K.1 Governing equations

GN-II theory (thermoelasticity without energy dissipation) introduces the thermal displacement α defined by:

$$\dot{\alpha} = T, \tag{149}$$

where T is the temperature increment. The source-free thermal-displacement form, obtained by integrating the primary temperature equation once with zero integration constant, is Green and Naghdi [1993]:

$$\kappa^* \nabla^2 \alpha = \rho c_e \ddot{\alpha} + \beta T_0 \dot{u}_{i,i}. \tag{150}$$

In the displayed isotropic convention, the candidate entropy flux uses the same scalar coefficient:

$$p_i = \kappa^* \alpha_{,i}, \tag{151}$$

where p_i is the entropy-flux vector. No anisotropic tensor generalization is asserted here.

K.2 Energy flux

For GN-II, there is no dissipation ($\mathcal{D} = 0$). The energy balance is:

$$\frac{\partial \mathcal{E}}{\partial t} + \partial_i J_i = \mathcal{P}. \tag{152}$$

The entropy flux p_i is determined by the same free energy potential as the stress. The relationship between the energy flux J_i and the entropy flux p_i requires the full GN-II thermodynamic formulation.

K.3 Status

OPEN WORKING CLAIM. The candidate isotropic entropy-flux expression $p_i = \kappa^* \alpha_{,i}$ is a source lead, not checked here; its notation is aligned with the scalar coefficient in the displayed GN-II equation. The complete energy flux and its relation to entropy flux require primary-source verification and a model-specific derivation.

L Open audit item 4: Green–Naghdi Type III (GN-III) Theory

L.1 Governing equations

For the source-free, linear isotropic GN-III system transcribed from Green and Naghdi (1992), the temperature equation is:

$$\rho c_e \ddot{T} + \beta T_0 \ddot{u}_{i,i} = \kappa \nabla^2 \dot{T} + \kappa^* \nabla^2 T. \tag{153}$$

L.2 Energy flux

The rate term can contribute dissipation when $\kappa > 0$; the exact energy flux and dissipation identity are not derived here.

L.3 Status

OPEN WORKING CLAIM. The GN-III energy flux and dissipation identity require a derivation from the full 1992 formulation, including its free energy and boundary power. No mixed-flux identity is asserted here.

M Open audit item 5: Dual-Phase-Lag (DPL) Theory

M.1 Governing equations

The DPL constitutive law (Tzou, 1995):

$$q_i(t + \tau_q) = -k T_{,i}(t + \tau_T), \tag{154}$$

where τ_q is the heat flux phase lag and τ_T is the temperature gradient phase lag.
Taylor expansion to first order:

$$q_i + \tau_q \dot{q}_i = -k(T_{,i} + \tau_T \dot{T}_{,i}). \tag{155}$$

M.2 Energy flux

The following internal-energy balance is retained as a classical-form candidate only; its validity with the LS thermodynamic state and constitutive equations requires source checking:

$$\rho \dot{e} = \sigma_{ij} \dot{\varepsilon}_{ij} - q_{i,i}. \tag{156}$$

The expression $J_i = q_i - \sigma_{ij} v_j$ is classical; its role in the displayed generalized balance remains open. However, expressing q_i explicitly in terms of T requires solving the DPL constitutive equation.

M.3 Frequency-domain expression

For time-harmonic fields:

$$q_i(1 - i\omega\tau_q) = -kT_{,i}(1 - i\omega\tau_T), \tag{157}$$

$$J_i = -\sigma_{ij}v_j - kT_{,i}\frac{1 - i\omega\tau_T}{1 - i\omega\tau_q}. \tag{158}$$

M.4 Status

UNVERIFIED WORKING CLAIM. The frequency-domain expression (158) is retained as unverified working material. The time-domain expression solely in terms of T requires solving the DPL constitutive equation, which introduces higher-order time derivatives. The thermodynamic admissibility condition $\tau_q \geq \tau_T$ must be verified for the flux formulation.

N Open audit item 6: Triple-Phase-Lag (TPL) Theory

The inherited TPL constitutive equation and its frequency-domain expansion are withdrawn pending primary-source comparison. Different TPL truncations introduce different derivative or thermal-displacement terms, so the former formula is not retained as a working identity. The exact model, admissibility conditions, and energy balance remain OPEN; no TPL flux is derived here.

O Open audit item 7: Moore–Gibson–Thompson (MGT) Theory

The dimensionally inconsistent legacy equation and its scratch frequency-domain cancellation have been withdrawn. Section 4.9 records a candidate flux/thermal-displacement formulation and marks it UNVERIFIED pending a primary-source comparison. No MGT flux, stored-energy term, dissipation inequality, or frequency-domain reduction is derived here; those tasks remain OPEN.

P Open audit item 8: Memory-Dependent Derivative (MDD) Theory

P.1 Governing equations

The MDD operator (Wang & Li, 2011):

$$D_{\omega_M} f(t) = \frac{1}{\omega_M}\int_{t-\omega_M}^{t} \kappa(t - \xi) f'(\xi)\, d\xi, \tag{159}$$

where $\omega_M > 0$ is the finite memory window and $\kappa(t - \xi)$ is the kernel function.
The MDD heat conduction law:

$$(1 + \tau_0 D_{\omega_M})q_i = -kT_{,i}, \tag{160}$$

or equivalently:

$$q_i + \tau_0 D_{\omega_M} q_i = -kT_{,i}. \tag{161}$$

P.2 Energy flux

The expression $J_i = q_i - \sigma_{ij} v_j$ is classical; its role in the displayed generalized balance remains open. However, q_i is now determined by a convolution-type memory equation.

P.3 Frequency-domain expression

For the finite-window operator in Eq. (159), set $s = t - \xi$ and use the harmonic convention $f(t) = \hat{f} e^{-i\Omega t}$. Its exact multiplier is kernel-dependent:

$$D_{\omega_M} f(t) \longleftrightarrow -\frac{i\Omega}{\omega_M} \left[\int_0^{\omega_M} \kappa(s) e^{i\Omega s} \, ds \right] \hat{f}. \tag{162}$$

This expression is the finite-horizon Fourier multiplier for the displayed kernel and convention; no kernel-independent simplified symbol is asserted.

P.4 Status

OPEN WORKING CLAIM. The MDD energy flux requires the convolution-form flux:

$$J_i = q_i - \sigma_{ij} v_j, \qquad q_i + \tau_0 D_{\omega_M} q_i = -kT_{,i}. \tag{163}$$

The explicit time-domain expression for J_i solely in terms of T requires solving the convolution equation, which depends on the kernel choice. This derivation is open.

Q Open audit item 9: Nonlocal Elasticity (Eringen)

Q.1 Governing equations

Eringen's nonlocal constitutive relation (integral form):

$$\sigma_{ij}(\boldsymbol{x}) = \int_\Omega \alpha(|\boldsymbol{x}' - \boldsymbol{x}|, \tau) \sigma_{ij}^{\text{local}}(\boldsymbol{x}') \, d\Omega_{\boldsymbol{x}'}, \tag{164}$$

or in differential form (Helmholtz kernel):

$$(1 - \xi^2 \nabla^2) \sigma_{ij} = \sigma_{ij}^{\text{local}}, \tag{165}$$

where $\xi = e_0 a$ is the nonlocal length scale parameter.

Q.2 Energy flux

For a specified nonlocal integral law, the local Cauchy-stress product rule need not close the domain power balance. No nonlocal energy-flux formula is asserted in this appendix. Any valid balance must be derived from the selected kernel, domain, constitutive energy, and boundary data, including whatever domain or boundary-power terms follow from those choices.

Q.3 Status

OPEN. The candidate placeholder formula in the inherited working draft is intentionally withheld; it is not a derived universal correction. Derive the domain and boundary power for a specific nonlocal model before stating an energy flux or interface ratio.

R Open audit item 10: Piezoelectric Electrical Energy Transport

R.1 Governing equations

For piezoelectric media, the constitutive relations are:

$$\sigma_{ij} = c_{ijkl}\varepsilon_{kl} - e_{kij}E_k, \tag{166}$$

$$D_i = e_{ikl}\varepsilon_{kl} + \kappa_{ij}E_j, \tag{167}$$

$$E_i = -\phi_{,i}, \qquad D_{i,i} = 0. \tag{168}$$

R.2 Energy flux

No electrical energy density or electrical power-flux expression is asserted here. The coupled energy storage and electrical work depend on the chosen electromechanical potential, field convention, external sources, and boundary conditions; the electrostatic equations displayed above alone do not determine a universal interface flux.

R.3 Status

OPEN. The inherited candidate `-phi dot(D_i)` term and combined piezothermoelastic flux are withheld, not promoted as formulae. Derive the complete balance from a stated electromechanical free energy and field equations, then compare it with the relevant primary source.

S Summary Table

Theory	v1.1 audit status	Required follow-up
Classical	DERIVED in Section 3 for local Cauchy stress and stated convention	Confirm notation consistently across all source files
Lord–Shulman	UNVERIFIED beyond the displayed constitutive law	Check primary-source convention, initial terms, and harmonic flux
Green–Lindsay	OPEN	Check the single-temperature rate-dependent equations and derive the balance
Green–Naghdi II/III	OPEN	Derive free energy, entropy/energy flux, and boundary power for each model
Dual/triple phase-lag	OPEN	Check the exact lag truncation, stability range, and energy identity
Moore–Gibson–Thompson	OPEN	Derive storage, boundary flux, and admissibility for the selected MGT variant
Memory-dependent	OPEN	Specify kernel, initial data, stored energy, and boundary power
Nonlocal / gradient	OPEN	Derive domain- and boundary-condition-specific power terms
Piezoelectric / multiphysics	OPEN	Include external-field power and derive the complete coupled balance

The preceding formulae are retained solely as unverified working material and are not approved generalized-theory identities.

T Recommendations

1. Use only the classical flux derived in Section 3 as the canonical formula; no generalized expression in this appendix is approved.
2. Present frequency-domain forms for LS and DPL as theory-specific expressions with clearly stated assumptions.
3. Present all other formulations as open derivations with explicit assumptions.
4. Do not include any uncertain flux form in comparative tables as a canonical result.
5. Mark each uncertain flux as an open research question (OQ10.1–OQ10.8).